Coding for Vibe Coders, Non-Coders & Everyone Else 🦜

Basic Code Seminar

COURSE READER
Part 1, Version 1

Prof. Dr. Sarah Ciston

CODESHIP Lab
Kunsthochschule für Medien Köln, Winter 2026

Coding for Vibe Coders, Non-Coders and Everyone Else, WI26

🦜 What Is CODESHIP? What Is This Course?

This basic seminar is part of CODESHIP, a supportive community for critical–creative coding that emphasizes co-learning, co-teaching, tinkering, and works in progress. With a welcoming, non-intimidating approach, the course aims to help you gain comfort with the key concepts of computational thinking and to question those concepts through intersectional lenses. You can launch (or troubleshoot) a code-related dream project, ask 'silly' questions and explore new programming skills suited to your artistic practice.

Together we will explore foundational techniques across programming languages, gaining practice with Python and/or Javascript, and you will have the chance to experiment with these and other tools that support your ideas. We will explore how (when, whether) to use AI in your coding practice, as well as better practices of digital care at this pivotal moment of high-risk tech. We will engage the historical, sociotechnical, and material stakes of computation as we create critical-creative projects that treat code as an artistic material and critical aesthetic form. We will test the limits of our own expectations for what code should be and should do.

Who

Prof. Dr. Sarah Ciston (they/them), sarah dot ciston at khm dot de, Sprechstunden: Tues & Thur 14.00–15.50. Make an appointment: https://tinyurl.com/chatwsarah

miguel marañas, Student Assistant, miguel dot rodriguez-bermudez at khm dot de

🏴‍☠️ Where & When

Weekly Seminar Schedule*

Seminars every Thursday 10.00-13.00 CET, 22 Oct–21 Jan @ CODESHIP KHM

* Please keep an eye on the schedule. Assignments and dates will change in order to adapt to the needs of the seminar and students' progress

** See Readings to access the Course Reader and the P4E textbook.

SEMINAR DATE	TOPICS	MATERIALS DUE	WORKSHOPEE
22 Oct	Project Mapping & Pseudocoding	-	-
29 Oct	Variables, Loops & Conditions	Reader: Keeling, Baraka, Blackwell; Python for Everybody (P4E) Chs 1,2,3	
TUE 03 Nov, 11.00 Meet at Computer Lab, Aula Building NOT 05 Nov	Skills & Tools: Critical AI Use	Reader: Ciston et al; P4E Chs 4,5,6,7 *by next week*	-

SEMINAR DATE	TOPICS	MATERIALS DUE	WORKSHOPEE
12 Nov	Strings & Source(ry)	Reader: Chun, Yang, also Yang (2022), Yang 2023a, Yang (2023b); P4E Chs 4,5,6,7	
WED 18 Nov, 14:00-16:00 & Thu 19 Nov	Care & Risk (special seminars for *Wissenschaft gegen Faschismus* Aktionswoche, in collaboration with Dr. Katrin Kämpf's seminar)	Reader: Morrison & Britton; P4E Chs 8,9,10,11 *by next week*	
26 Nov	Materials & (Data)Structures	Course Check-In; Reader: Nakamura; P4E Chs 8,9,10,11	
03 Dec	Databases & Power	Reader: Crawford & Joler, calculatingempires.com, Couldry & Mejias; P4E Chs 12–16 as relevant	
10 Dec	*Open Co-Lab Co-Working*	**Schedule a Project Sprechstunde**	-
17 Dec	Logics & Ideologies	Reader: Steyerl, Amoore	
07 Jan	Project dev & feedback sessions	(TBD based on student interests)	
14 Jan	Project dev & feedback sessions	(TBD based on student interests)	
21 Jan	**Demo Day**: poster presentations & project demos	Final projects & research posters due	-

Special Events Schedule

CODESHIP EVENT	DETAILS
Spooky Screening: *A Girl Walks Home Alone at Night*	Fri 30 Oct, 19-21h
Tech Screening: *Ghost in the Machine* Documentary hosted by KISD	Tue 10 Nov, 17:30, KISD
Tokens Workshop with Ting (email Sarah if interested)	Fri 13 Nov, 13.00-16.00

CODESHIP EVENT	DETAILS
🦜 Stochastic Parrots Social Club	Tue 24 Nov, Time TBD

Why

- because code should not be left to the white tech broligarchy
- because artistic research and experimentation can make tangible shifts in the places most impacted by emerging tech
- because coding is fun and shouldn't be so hidden away with obtuse language

These are some of my (Sarah's) reasons, what are yours?

Goals & Outcomes

You'll understand how artists, engineers, makers, and scholars can embrace coding with a critical eye toward ethics and a creative glint that can help imagine more equitable futures.

You'll gain comfort exploring data and coding, building on any existing skills with new perspectives both technical and critical.

You'll build upon your portfolio of individual and collaborative code projects, as well as your community of practice among critical-creative coders.

Outcomes depend on your own goals and are measured on an individual basis in consultation with the instructor. Start by setting your own goal for what you'd like to get out of the semester and how you'd like to contribute to any group goals. Check in midway and toward the end of the term to see how your goals may have changed after working on your project.

How

Prerequisites

None! No particular experience is required. Any kind of field or background is helpful to get started. Don't worry, no matter your existing level of comfort with code, even if it's none, we meet each other where we are. You don’t have to install or buy anything in advance either; we’ll handle that together. It's great if you have in mind a project or a skill you want to work on. If not, you might start from asking, *"What do I wish were in the world, and why?"*

Seminar Format

Rather than a top-down lecture format, this seminar is organized from the bottom up. It aims for students to direct their own course of learning with the instructor as coach or guide. The purpose of this approach is to provide meta-skills like independent investigation, research intuition, and broader skill-building, as well as a supportive environment for all types of learners to explore and take creative risks.

1. **Opening Check-Ins**: Two minutes per person. *What are you working on, wondering, worrying about, or wowed by?*
2. **Code Workshop**: Group engagement with sample or student code to analyze and offer feedback. Includes review of programming concepts.

3. **Discussion**: Group engagement with the weekly reading or other shared material. Includes review of computing concepts.
4. **Co-working, Troubleshooting, Short Lessons**: Self-directed, based on what came up during check-ins, workshop, and discussion.
5. **Closing Check-In**: Wrap up and set plans for next meeting.

CODE WORKSHOPS

You'll sign up for us to workshop your project on a particular day. Be ready to bring in some code, no matter what stage your project is in.

Each session we will close-read someone's code, troubleshoot any issues, support the project's development, talk through the ideas critically and creatively, but also the code itself. Talk about why something works or how it might work differently. Talk about the style, how it's organized, where it's going next, what tools it uses, and why. Maybe we use AI tools to support or not. We put our heads together as a group with different levels of experience and talk through its sociocultural, aesthetic and technical aspects.

Rather than teach programming in a traditional, linear fashion (you'll have access to that in the sources provided), we will move conceptually based on what is relevant for each project, explaining code skills as we go.

READINGS & DISCUSSIONS

Our seminar is oriented around projects and research you develop, built upon the readings and media we engage with and the conversations we have about them and your own practice. These readings are required and will support your projects and your course progress.

Our primary texts include a READER that you can print or use as a PDF, which combines many selections around the course topics, as well as an introductory text for programming in Python. It also comes as a set of slides or videos if you prefer, and it is translated into multiple languages. If your project is better suited to another programming language, such as Javascript, or if you prefer learning by video or online exercises, we can support your progress in this instead. Please consult with us for equivalent chapters for your assignments.

Requirements for Schein/Certificate:

Please complete all three of the following before requesting the KHM Schein for this seminar:

1. Office Hours (Sprechstunde), one or more visits with Sarah about your course project
2. Successful Research Poster (or other discussed format, e.g. term paper) presented on Demo Day
3. Successful Project (or other discussed format, e.g. curation or collaboration) presented on Demo Day

Lab Guide & Community Commitments

We ask you to read and follow all Community Commitments and the CODESHIP Lab Navigation Guide while you are at the Lab and participating in the seminar. KHM House Rules also apply.

29 Oct – Variables, Loops & Conditionals

Keeling
Baraka
Blackwell (excerpt)
Python for Every (separate) Chs 1,2,3

Cinema Journal 53 | No. 2 | Winter 2014 https://doi.org/10.1353/cj.2014.0004

Queer OS

by Kara Keeling

From new media's eccentric temporalities and reliance on reading codes to their relationships to ephemera, publics, viruses, music, and subcultures, new media intersect with queer theories in a variety of ways. Scholars working at the interfaces of new media, queer theory, and LGBT studies have produced valuable insights into the roles and usages of new media in creating and sustaining forms of LGBT sociality, experiences, and ways of knowing. Vital scholarship on LGBT and queer cybercultures from a variety of perspectives and compelling descriptions and explorations of the role of new media in LGBT, and queer people's lives, have helped scholars understand the centrality and significance of LGBT participation in new media. Important work on representations of LGBT people in, on, and through new media is ongoing.[1]

Within this scholarly milieu, less attention has been dedicated to the interfaces of new media as they have been theorized through conceptualizations of "the digital," "software," "computation," "manufacturing," "information," and "code," and what currently are perceptible as queer ontologies; theories of queer embodiment and materializations; and other issues, logics, and expressions that comprise queer theory, such as, for example, theories of queer temporality, critiques of homonationalism, and investigations into the relationships of queerness, forms of racialization, and contexts of settler colonialism, among others.

Yet as the opening lines of this brief contribution to an evaluation of contemporary intersections of LGBT studies, queer theory, and cinema and media studies suggest, the materiality, rhetorics, forms, and ontologies of new media readily lend themselves to a theoretical encounter with queer theory that might enliven and enrich both film and media studies and queer theory, thus deepening the capacity of each to attend to the sociopolitical registers of contemporary life.

Existing theoretical scholarship on race and new technologies illustrates that new media scholarship that attends to race also might

1 For a helpful, though not exhaustive, gloss on new media and communications scholarship produced at the intersection of queer and cyber, see Kate O'Riordan and David J. Phillips, eds., introduction to *Queer Online: Media Technology & Sexuality* (New York: Peter Lang, 2007). For a consideration of gay participation online, see Ken Hillis, *Online a Lot of the Time: Ritual, Fetish, Sign* (Durham, NC: Duke University Press, 2009). For an ethnography of queer youth using the Internet in rural settings, see Mary L. Gray, *Out in the Country: Youth, Media, and Queer Visibility in Rural America* (New York: New York University Press, 2009).

engage with vital and still-generative scholarship happening in queer theory, but it rarely makes an explicitly queer new media studies or technology studies (or even queering new media and/or technology studies) part of its project. Similarly, compelling work on feminism and the cultural logics of new media technologies is suggestive of a direction amenable to a serious engagement with queer theory, but that work rarely substantively stages such an encounter. With this lacuna in mind, in what follows, I offer preliminary thoughts toward a scholarly political project that I call "Queer OS."[2] As I discuss here, scholarship that might be collected under a rubric of "Queer OS" already exists, and provocative and promising work is currently being produced that might contribute to a project at the interfaces of queer theory, new media studies, and technology studies, such as the one I sketch briefly here.

Queer OS would take historical, sociocultural, conceptual phenomena that currently shape our realities in deep and profound ways, such as race, gender, class, citizenship, and ability (to name those among the most active in the United States today), to be mutually constitutive with sexuality and with media and information technologies, thereby making it impossible to think any of them in isolation. It understands *queer* as naming an orientation toward various and shifting aspects of existing reality and the social norms they govern, such that it makes available pressing questions about, eccentric and/or unexpected relationships in, and possibly alternatives to those social norms.[3]

I have suggested elsewhere, following Antonio Gramsci's work on hegemony, Marcia Landy's reading of Gramsci's work in the context of film studies, and Wahneema Lubiano's work on "common sense" in black nationalism, that common sense is a linchpin in the struggle for hegemony that conditions what is perceptible such that aspects of what is perceptible become generally recognizable only when they work in some way through "common senses." In this context, *queer* offers a way of making perceptible presently uncommon senses in the interest of producing a/new commons and/or of proliferating the senses of a commons already in the making.[4] Such a commons would be hospitable to, perhaps indeed crafted from, just and eccentric orientations within it. Queer OS makes this formulation of *queer* function as an operating system along the lines of what Tara McPherson describes as "operating systems of a larger order" than the operating systems that run on our computers.[5]

Queer OS would take seriously McPherson's suggestion that the cultural logics of the early operating system Unix embed some of the racial logics of the post–World War II era in which Unix (and the modern civil rights movements) were developed.[6]

2 During a Google search for existing formulations of Queer OS, I found a link to an intriguing workshop in Slovenia titled "Workshop Queer OS: Operating System for Fags, Radical Faeries, and Questioning Nerds." See Ljudmila, "Queer OS," http://wiki.ljudmila.org/Queer_OS (accessed June 25, 2013).

3 I develop this formulation of queer in greater detail in my book manuscript in progress, *Queer Times, Black Futures* (New York: New York University Press, forthcoming).

4 For my formulation of "the image of common sense" in the cinematic, see Kara Keeling, *The Witch's Flight: The Cinematic, the Black Femme, and the Image of Common Sense* (Durham, NC: Duke University Press, 2007).

5 Tara McPherson, "US Operating Systems at Mid-Century: The Intertwining of Race and UNIX," in *Race after the Internet*, ed. Lisa Nakamura and Peter Chow-White (New York: Routledge, 2011), 21–37.

6 Ibid., 21.

For McPherson, the logics of US racial formation infuse Unix not because the creators of Unix planned it that way, but because those who developed Unix were working within a sociocultural milieu held together by common senses already saturated by those logics.

Inspired by McPherson's analysis of Unix in the context of US racial formation, Queer OS seeks to make *queer* into the logic of "an operating system of a larger order" that unsettles the common senses that secure those presently hegemonic social relations that can be characterized by domination, exploitation, oppression, and other violences. While it is worth noting here that my references to "the commons" are in critical conversation with existing formulations of the "digital commons," an aim of Queer OS vis-à-vis conceptualizations of commons is to provide a society-level operating system (and perhaps an operating system that can run on computer hardware) to facilitate and support imaginative, unexpected, and ethical relations between and among living beings and the environment, even when they have little, and perhaps nothing, in common.

To begin with, it could be said that in its capacity as a social operating system, Queer OS connects existing distributed areas of scholarly inquiry and activism, thereby producing philosophies and cultures within each of those areas that might unsettle the logics that currently secure them. Here, Queer OS would not be simply interdisciplinary, though because it often evinces a studied promiscuity toward the ideas and methods it assembles, it carries many of interdisciplinarity's risks and promises. Nor is it only transdisciplinary, since it can be relatively indifferent to existing disciplines in an effort to include aspects of the world that have not yet entered the logics of disciplines.[7]

Queer OS names a way of thinking and acting with, about, through, among, and at times even in spite of new media technologies and other phenomena of mediation. It insists upon forging and facilitating uncommon, irrational, imaginative, and/or unpredictable relationships between and among what currently are perceptible as living beings and the environment in the interest of creating value(s) that facilitate just relations.[8] Because Queer OS ideally functions to transform material relations, it is at odds with the logics embedded in the operating systems McPherson discusses. Because it seeks to undermine the relationships secured through those logics, even as, like McPherson does when she points out that she is using a computer and word-processing software that shape her own intellectual work in specific ways, it acknowledges its own imbrication with and reliance on those logics while still striving to forge new relationships and connections.

7 In this regard, Queer OS has affinities with Katie King's formulation of "networked reenactments." See Katie King, *Networked Reenactments: Stories Transdisciplinary Knowledges Tell* (Durham, NC: Duke University Press, 2012); King "A Naturalcultural Collection of Affections: Transdisciplinary Stories of Transmedia Ecologies Learning," *S&F Online*, http://sfonline.barnard.edu/feminist-media-theory/a-naturalcultural-collection-of-affections-transdisciplinary-stories-of-transmedia-ecologies-learning/.

8 It is worth noting here that, although they are of different orders, this description of Queer OS resonates with what I described as "the black femme function" within the cinematic in my book *The Witch's Flight*. See also Franco "BIFO" Berardi, "Precariousness, Catastrophe and Challenging the Blackmail of the Imagination," *Affinities: A Journal of Radical Theory, Culture, and Action* 4, no. 2 (November 23, 2010), http://www.affinitiesjournal.org/affinities/index.php/affinities/article/view/58.

From my own position, it is possible to detect exciting contributions that have already been made, as well as ones on the horizon. Among the early precedents for Queer OS are projects such as Allucquére Rosanne Stone's (Sandy Stone's) *The War of Desire and Technology at the Close of the Mechanical Age*, a book in which Stone is "seeking social structures in circumstances in which the technological is the nature, in which social space is computer code."[9] Fifteen years after the publication of *The War of Desire and Technology*, Margaret Rhee and Amanda Philips introduced their 2010 Humanities, Arts, Science, and Technology Advanced Collaboratory (HASTAC) forum "Gender, Sexuality, and Queerness" by announcing their "hope for dialogues that traverse disciplinary boundaries, borders, and fictive territories."[10] As they described it, the forum invited discussions of questions such as "How does queer theory intersect with technology [and/or] technologies?" "How do issues of gender, sexuality and identity play out in digital media, digital arts, and the Internet?" "How does the body function as a theme within theory and art, emerging from queer, ethnic, and feminist, studies and other related disciplines?" and "Is technology historically closely entangled with sexuality?"[11]

The questions that Rhee and Phillips invited their participants to discuss remain compelling ones to explore. Some of those who have been engaged in their exploration also have participated in the conversations about scholarly technology that have come to characterize the digital humanities. This year, Phillips coauthored an article with Alexis Lothian that seeks to make an intervention into the contested category of "the digital humanities." That article, "Can Digital Humanities Mean Transformative Critique?," builds on the premise that, "if humanities scholars in critical media and cultural studies, queer studies, ethnic studies, disability studies, and related areas are doing work in and with the digital, we should lay claim to our place within digital humanities."[12] In that spirit, Lothian and Phillips offer "a curated list of projects, people, and collaborations that suggest the possibilities of a transformative digital humanities: one where neither the digital nor the humanities will be terms taken for granted."[13]

Picking up on a trajectory of inquiry into technology, gender, and sexuality offered by Jack Halberstam's 1991 essay "Automating Gender: Postmodern Feminisms in the Age of the Intelligent Machine," scholars also are working to queer the histories we tell about computing. Homay King's most recent project centers on pioneering computer

9 Allucquére Rosanne Stone, *The War of Desire and Technology at the Close of the Mechanical Age* (Cambridge, MA: MIT Press, 1995), 38. Jack Halberstam's 1991 essay also might be retrospectively understood as precedent for the later work I am collecting under the rubric of "Queer OS." See Judith Halberstam, "Automating Gender: Postmodern Feminism in the Age of the Intelligent Machine," *Feminist Studies* 17, no. 3 (1991): 439–460.

10 "Queer & Feminist New Media Spaces—HASTAC," http://hastac.org/forums/hastac-scholars-discussions/queer-feminist-new-media-spaces.

11 The citations in Rhee and Phillips's introduction to the forum offer an archive of scholarship in new media studies, technology studies, and gender and sexuality studies on which they invite forum participants to draw. See "Queer & Feminist New Media Spaces—HASTAC," http://hastac.org/forums/hastac-scholars-discussions/queer-feminist-new-media-spaces.

12 Alexis Lothian and Amanda Phillips, "Can Digital Humanities Mean Transformative Critique?," *Journal of e-Media Studies* 3, no. 1 (2013), doi:10.1349/PS1.1938-6060.A.425.

13 See ibid.

scientist Alan Turing's homosexuality in an effort to, as she put it, "queer the computer just slightly."[14] In a similar vein, Jacob Gaboury is compiling "Queer History of Computing," which can be accessed online.[15]

What all of these efforts have in common is an interest in bringing the considerable insights of queer theory and LGBT studies to bear on discussions and studies of new media and their technologies and vice versa. They offer ways of thinking about new media that disrupt what we think we know about it, and they demonstrate what queer theory can gain from an interested consideration of media and technology. Along these lines, in their solo and collaborative performance art work, Zach Blas and Micha Cárdenas have made contributions to our ways of thinking about transgender embodiments, queer sexualities, new media technologies, and other aspects of mediation that might be considered under the rubric of "Queer OS." By innovating things such as "transcoder," which is "a queer programming anti-language," or instructing people on how to build a gay bomb, Blas's work prompts us to question our assumptions about what technology is and what it can do. Both Blas and Cardenas are producing work that strives to forge new relationships between living beings and the environment by working with, through, and at times in spite of technology.[16]

Blas, Cárdenas, and others working at the theory-practice nexus of queer theory, trans and gender studies, and technology can help nuance understandings of *queer*, *gender*, and *technology* because their work points to ways of embracing *queer* and *gender* as technologies. In this regard, a Queer OS project also could involve reading their oeuvres (which can be grasped as Queer OS), as well as those of other artists working to (re)forge queerness within new media and technology, alongside existing scholarship on "race and/as technology" and the artistic expressions and rhetorics that make that formulation perceptible.

A Queer OS project might notice, for example, that Wendy Hui Kyong Chun's essay "Race and/as Technology or How to Do Things to Race" becomes just a bit queerer through the revisions that accompanied its transformation from serving as the introduction to the special issue "Race and/as Technology" of *Camera Obscura* that Chun coedited with Lynne Joyrich in 2007 to a stand-alone essay in the 2012 collection *Race after the Internet*, coedited by Lisa Nakamura and Peter A. Chow-White, and build upon that observation. It is only in the latter version of the essay that Chun turns to Greg Pak's 2003 feature film *Robot Stories* to rethink arguments she made in the past regarding "high tech Orientalism—the high tech abjection of the Asian/Asian American other."[17]

14 See, e.g., "Lecture: 'Keys to Turing' by Homay King Nov. 8, Visual Arts Center, Beam Classroom, English, Calendar (Bowdoin)," http://www.bowdoin.edu/calendar/event.jsp?bid=601095&rid=68848.

15 Jacob Gaboury, "A Queer History of Computing: Part Four," Rhizome.org, http://rhizome.org/editorial/2013/may/6/queer-history-computing-part-four/.

16 See "Queer Technologies—Automating Perverse Possibilities," http://www.queertechnologies.info; Micha Cárdenas, "Micha Cárdenas—Movement, Technology, Politics," http://transreal.org.

17 Wendy Hui Kong Chun, "Race and/as Technology or How to Do Things to Race," in *Race after the Internet*, ed. Lisa Nakamura and Peter Chow-White (New York: Routledge, 2011), 49; Chun, "Introduction: Race and/as Technology; or, How to Do Things to Race," *Camera Obscura* 24, no. 1 70 (2009): 7–35.

Stating that her reevaluation of high-tech Orientalism is inspired by Beth Coleman's essay "Race as Technology," Chun presents Pak's film as an exploration of "the extent to which high tech Orientalism might be the ground from which some other future can be created; the ground from which dreams can be made to fly, flower, in freaky, queer unexpected ways."[18] Chun's discussion of *Robot Stories* attends to the meaningful ways that technology, race, gender, and sexuality work together in the film. She claims that "what is remarkable" by the end of *Robot Stories* is that "the invisibility and universality usually granted to whiteness has disappeared, not to be taken up seamlessly by Asian Americans and African Americans, but rather to be reworked to displace both what is considered to be technological and what is considered to be human."[19]

Though Chun does not pursue an evaluation of the work that "queer" does in *Robot Stories*, it is clear from her discussion (as it is in the film) that something queer persists (even after her brief discussions of the queer sexual acts in the film) in her reading of how *Robot Stories* makes race do things within high-tech Orientalism other than reproduce its logics. It could be argued that what Chun calls Pak's methodology is presented in the film as a Queer OS. Chun describes it in this way:

> The opening credits of *Robot Stories*, which begins with the now stereotypical stream of 1s and 0s, encapsulates Pak's methodology nicely. Rather than these 1s and 0s combining to produce the name of the actors, etc. (as in *Ghosts in the Shell* and *The Matrix*), the credits interrupt this diagonal stream. . . . As the sequence proceeds, little robots are revealed to be the source of the 1s and 0s. Shortly after they are revealed, one malfunctions, turning a different color, and produces a 2. . . . Soon, all the robots follow, turn various colors and produce all sorts of colorful base-10 numbers. Thus, robots turn out in the end to be colorful and operate in the same manner—and in the same numerical base—as humans. The soundtrack features a Country and Western song telling Mama to let herself go free. The 1s and 0s, rather than being readable, are made to soar, to color the robots that are ourselves.[20]

In Chun's description of Pak's methodology, Queer OS can be grasped as a malfunction within technologies that secure "robot" and "human," a malfunction with a capacity to reorder things that can, perhaps, "make race do different things," tell "Mama to let herself go free," and make what was legible soar into unpredictable relations. ✻

Tara McPherson directed me to several of the scholars and essays discussed in this piece. Chandra Ford, Patty Ahn, Damon Young, Julia Himberg, and the editors at Cinema Journal *offered helpful suggestions at different moments in the writing of this piece. All faults are mine.*

18 Chun, "Race and/as Technology," 49. Chun refers to Beth Coleman, "Race as Technology," *Camera Obscura* 24, no. 1 70 (2009): 177–207.

19 Chun, "Race and/as Technology," 56.

20 Ibid., 56.

“Hunting Is Not Those Heads On The Wall” in Home: Social Essays (1966) Amiri Baraka

Thought is more important than art. Without thought, art could certainly not exist. Art is one of many products of thought. An impressive one, perhaps the most impressive one, but to revere art, and have no understanding of the process that forces it into existence, is finally not even to understand what art is.

The artist is cursed with his artifact, which exists without and despite him. And even though the process, in good art, is everywhere perceptible, the risk of perfection corrupts the lazy public into accepting the material *in place of* what it is only the remains of.

The academic Western mind is the best example of the substitution of artifact worship for the lightning awareness of the art process. Even the artist is more valuable than his artifact, because the art process goes on in his mind. But the process itself is the most important quality because it can transform and create, and its only form is possibility. The artifact, because it assumes one form, is only that particular quality or idea. It is, in this sense, after the fact, and is only important because it remarks on its source.

The academician, the aesthete, are like deists whose specific corruption of mysticism is to worship things, thinking that they are God (thought, the process) too. But art is not capable of thought. Just as things are not capable of God, but the reverse is what we mean by the God function, the process I am talking about.

The Supermaker, is what the Greeks identified as "Gods." But here the emphasis is still muddled, since it is what the God can do that is really important, not the fact that he is the God. I speak of the *verb process*, the doing, the coming into being, the at-the-time-of. Which is why we think there is particular value in live music, contemplating the artifact as it arrives, listening to it emerge. *There* it is. And *There*.

But even this is after the fact. Music, the most valuable of artifacts, because it is the most abstract, is still not the activity that makes itself possible. Music is what is left after what? *That* is important.

A museum is a curious graveyard of thinking. But we can go through one hoping to get some inkling of what those various creators who made the creations were thinking. What was he thinking when he did That? is a common question. The answer is obvious, though: That.

Formal art, that is, artifacts made to cohere to preconceived forms, is almost devoid of this verb value. Usually a man playing Bach is only demonstrating his music lessons;

the contemporary sonneteer, his ability to organize intellectual materials. But nothing that already exists is *that* valuable. The most valuable quality in life is the will to existence, the unconnected zoom, which finally becomes in anyone's hands whatever part of it he could collect. Like dipping cups of water from the falls. Which is what the artist does. Fools want to dictate what kind of dipper he uses.

Art is like speech, for instance, in that it is at the end, and a shadowy replica, of another operation, thought. And even to name something, is to wait for it in the place you think it will pass. Thought, "I've written" - understanding even this process is recording. Art-ing is what makes art, and is thereby more valuable. But we speak of the Muse, to make even the verb a thing.

If we describe a man by his life we are making him a verb, which is the only valid method since everything else is too arbitrary. The clearest description of now is the present participle, which if the activity described continues is always correct. Walking is not past or future. Be-ing, the most complex, since it goes on as itself, as adjective-verb, and at the moment of. Art is not a be-ing, but a Being, the simple noun. It is not the verb, but its product. Worship the verb, if you need something. Then even God is after the fact, since He is the leavings of God-ing. The verb-God, is where it is, the container of all possibility. Art, like time, is the

measurement of. Make no mistake.

Even "sense" is clearly a use some energy is put to. No one should fool around with art who is only trying to "make sense." We are all full of meaning and content, but to make that wild grab for more! To make words surprise themselves. Some more of the zoom trembling in its cage, where some fool will be impressed by its "perfection." This is what should be meant by a "primitive" mind, that which is satisfied

with simple order. But "using" words denies the full possibility of expression, which is, we must suppose, impossible, since it could not be stopped and identified. Art is identification, and the slowing-down for it. But hunting is *not* those heads on the wall.

The imitator is the most pitiful phenomenon since he is like a man who eats garbage. A saxophonist who continues to "play like" Charlie Parker cannot understand that Charlie Parker wasn't certain that what had happened had to sound like that. But if a man tries to understand *why* Parker sounded like he did, the real value of his music begins to be understood.

Form is simply *how* a thing exists (or what a thing exists as). When we speak of man, we ask, "How does he make it?" Content is *why* a thing exists. Every thing has both these qualities; otherwise it could not (does not) exist. The art object has a special relationship between these two qualities, but they are not separable in any object.

The recent concern in the West for the found object and chance composition is an attempt to get closer to the non-Western concept of natural expression as an Art object, since of course such an object has form and content in special relationship like any thing a man made. Because a man cannot make a thing that is in this sense unnatural. The unnatural aspect of the man-made object is that it seems to exist only as a result of man, with no other real connection with the nonhuman world. *Artificial*, in this sense, is simply *made*. "Bad art" is usually unnatural, *i. e.*, it seems as if it could not exist without being made by a man. It is strictly artificial.

Western men have always been more concerned with the artifact, the made thing, as "an art" separated from some natural use. Art as a separate category of concern is first

seen when? Functional art is as old as man was when he made *anything*. To posit the idea that you will make a thing whose sole value and function is that you will make it, is a different emphasis.

God (which is separate, and before, A God) is in one sense an art object and was probably the first. In the secular West, God is a nonfunctional (literally) art object. But earlier, God was simply the naming of force, in the same sense I meant earlier of naming a thing by its life. God was "the force out of which the world (and life) issued." The *naming*, nominalization, of that force is finally a step at making it artificial. The arbitrary assignment of content (which means nothing in a strictly local context, *i. e.*, Who will object?) based very likely on need, is the beginning of God as an art object.

But think about this, "the force out of which the world (and life) issued" exists everywhere, as we can see, and this is the basic form (and content) of God. Everything else is most likely to be nationalism. Nothing else exists. But again the confusing of process with artifact, or rather the substitution of process for artifact. When God gets to be a thing, it is an artifact. When the lightning was "the force out of which the world issued," the emphasis was on natural evidence, the natural thing. And lightning is curiously apt, since in its natural form, it is a process, a happening, as well as an artifact. Duplicating God signs was simple education.

When God started to *look like* a human being, men had gotten very sure of themselves. (That is, once the dog, the wolf, the fish, the bear, the leopard, etc., had all been God, and the fallacy of this reasoning, in whatever turn of environmental circumstance, became traditional. Some of the things we have seen are animals,

possibly one of them is God. Men next.) But *naming* is the first appropriation, the earliest humanist trend. Jane Harrison says the Greeks took

the fear away by not only making all the various qualities known in the world men, but understandable, knowable, men. They began to make lives for their Gods, so those Gods could only exist in that certain way. From the unknown verb, to the familiar artifact. Greek Gods are beautiful Greeks, which finally in social/political terms is the beginning of modern nationalism. What the Western white man calls the beginning of democracy was the positing of the sovereign state, wherein everyone was free. The rest of the world could be exploited. *Logos*, then, is not merely thought, but belief. Greeks were Greeks because they had the same beliefs. A Greek was a man who believed the Gods were Greeks.

Humanism is good in this sense, that it puts the emphasis on what we actually have, but there is a *loss* with the loss of the unspecific imagination because knowing man was all there was enabled the less imaginative to show up fully armed. Man's mind is revered and, in the ugliest emphasis, man's inventions. Again the hideous artifact, to replace the valuable process. The most stupid man ought to know he is more important than what he can make. But he will never understand that what moves him is even more important, because of the contemporary (post-Renaissance) loss of prestige for the unseen. When God had a rep, his curious "workings" were given deference. But now that everything is grounded or lodged in the sweaty palm, men only believe in what, as Auden inferred, "takes up space."

Thinking, in the most exalted humanist terms, is God, the force out of which the world issued. Nature, we make a "natural" process. Darwinian determinism provided the frame. From the Renaissance, the boost of the industrial revolution, and man surrounded by his artifacts. Machines, which are completely knowable.

Blackwell, Moral Codes: Designing Alternatives to AI (2024) MIT Press.
(Excerpt of Ch 10 “The Craft of Coding”)

The Abstract Craft of Making Code

To many people, it seems odd to describe software as “material,” when it is also in many ways immaterial.[8] But although code is just bits sent down a cable or stored on a disk drive, the day-to-day experience of a programmer is that code seems to resist what you want to do with it. Writing software involves constant small adjustments—as you type one thing, you find out that it doesn’t work, try something else, realize that the whole idea was wrong, take a different approach, and so on. Sometimes coding seems more like carving a piece of wood—the chisel cuts smoothly for a while before suddenly hitting a hidden knot or a change in the grain, jerking and splitting in an unexpected direction. Traditional craft skills involve recognizing and responding to the characteristics of the material they work with, in a way described by design philosopher Donald Schön as a “conversation with the material,” and reflecting observations of scientific and technical work practices by thinkers including Richard Sennett and Andrew Pickering.[9]

When writing code, the “material” speaks back to the programmer through various kinds of testing, as software development tools test the consistency of one part of the code against another or run specific pieces of local code to confirm that they have the intended effects. The process of testing often results in surprises and subsequent changes to the original plan, especially when working on a problem (such as a research problem) that has not been fully specified in advance.

Much of the craft of programming relates to testing, and professional programmers spend far more time testing and adjusting than they do

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writing new code. But perhaps the most interesting implication of testing is its relationship to the specification process, where the programmer thinks about and abstractly describes what they would like the software to do. That is a central concern of this book.

An essential element of abstract coding is deciding what to call things. Day-to-day programming involves continually inventing names for local pieces of data storage, screen windows, network messages, and even parts of the algorithm itself. Steve McConnell dedicates a whole chapter of *Code Complete* to choosing good names, and emphasizes the importance of names at many stages of the coding process. Inventing a name for something is an exercise in philosophical abstraction. Two or three words might summarize hundreds of lines of code. When chosen well, this name becomes the definition of what that code should do.

Programmers constantly reuse libraries of code created by other programmers, where every element has a function determined by its name. The set of names becomes a language in itself, a way of thinking about the purpose and potential of the library. In a sense, every programmer of complex systems is defining their own language, a vocabulary that will allow the eventual solution to be elegantly expressed.

When building complex systems, a new set of names might be needed as building blocks to construct more complex functions, which can then be combined with others, and so on. At each stage, testing must confirm that every function does what the name says, and that sentences combining these words work as expected. When programming in an exploratory way, it is not unusual to realize that the name initially chosen has not properly captured the abstraction intended, and a different name would have been better. Expert programmers spend much of their time "refactoring" to change earlier names and redefine the logical relationships between them.[10]

The craft practices I have described here apply to all conventional programming languages and will be instantly recognizable to expert programmers. It is interesting to note that both these craft elements—naming and testing—are not foregrounded by the end-user tools I have described so far, such as spreadsheets, GUIs, or automated macro facilities. As a result, many ordinary people use programmable tools without encountering the most basic requirements for serious coding.[11]

The question for the agenda of this book is whether those capabilities ever *could* be provided for use by people who are not programmers? It

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doesn't seem like this should be forbidden on the grounds that ordinary people would be too stupid to use them properly. Ordinary people are able to create names when necessary, whether for a music playlist, an online chat group, or a document folder. Ordinary people are also perfectly accustomed to trying things out when doing routine craft work—whether tugging on a button that was just sewed on or placing the first few books on a shelf recently screwed to the wall. It is the essential interaction between naming and testing that is not well supported by current end-user software tools, and not at all supported by AI systems. Today's "intelligent" systems based on LLMs do not allow users to create new words within the model or test the consequences of doing so, despite the fact that those things were routine in programming languages.

Attention Investment in the Crafts of Naming and Testing

Previous chapters have emphasized the central concept of attention investment—the trade-off that simple automation tasks might save mental effort in the long term, but require some initial thought to get right. According to individual preferences, some people might prefer never to think in advance, while others might do it too much, overthinking and wasting time automating a task they will never need to do again. Effective tools can improve people's ability to make this decision, and mixed-initiative strategies can use machine learning methods to suggest where further investment of attention would be beneficial.

The same principles of attention investment that are central to programming in the small apply to the craft practices of naming and testing. A name is an abstraction that requires thinking in advance to get it right, but with the payoff that a single word can then be used in place of lengthy explanations. Testing involves abstract thinking about the ways that a named component might behave, and expressing these possible behaviors in terms of test cases that relate to possible interpretations of the chosen name. Testing, as every programmer understands, is an insurance investment against the work that would have to be done fixing things up if the program later goes wrong in ways they hadn't anticipated.

Margaret Burnett at Oregon State University leads a program of research into *end-user software engineering* that applies the principles of attention investment to help reduce the frequency of errors in spreadsheets. Their

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TUE! 3 Nov!!!!

Ciston et al Inventing ELIZA (excerpt)
P4E Ch 4,5,6,7 (by next week)

12.00 Conclusion

ELIZA CODE:
SGO ON
PLEASE ‘S

12.01 The AI Effect: The Persistent Allure of Artificial Intelligence?

12.02 Resisting the AI Effect

12.03 ELIZA’s Multiple Futures

After its initial impact had faded, ELIZA was sometimes considered to be more a curiosity in computing history than a serious contribution to computer science. Yet its dismissal misunderstands both ELIZA's historical significance and its continued relevance. In this book, we have undertaken a systematic reconstruction of ELIZA: recovering the program's source code, understanding its cultural context, analyzing its design choices, exploring its multiple identities, and tracing its influence. This code excavation represents a comprehensive study of ELIZA as a software artifact, challenging conventional narratives about early AI development while providing a critical foil for today's algorithmic urgency.

Our media archaeological approach to ELIZA has revealed it to be more than an early, limited chatbot. In fact, ELIZA marks a key juncture of computing and culture. ELIZA emerged from a specific historical moment at MIT in the 1960s, shaped by Cold War exigencies, institutional frameworks, and Weizenbaum's personal history as a Jewish refugee from Nazi Germany. Because Weizenbaum published only the DOCTOR script rather than ELIZA's source code, Lisp and BASIC versions of DOCTOR proliferated, while the original MAD-SLIP code lay hidden away. But the program's technical innovations, from its pattern-matching algorithms to its script architecture, established techniques that are still influential today. The multiple identities of ELIZA beyond the famous DOCTOR script reveal a system designed for adaptability and experimentation rather than merely simulating a Rogerian therapist. Such discoveries conjure up counterhistories of ELIZA that ask us to imagine: What if its complexities had been a bigger part of its influential story all along? By reconstructing the many entangled aspects of ELIZA, this research project shows how software can also reveal complex interconnections across technology, culture, and power.

The book also serves as a useful case study for critical code studies, suggesting ways of approaching code that show the value of interdisciplinary scholarship. Our collaboration has combined technical expertise with cultural analysis, historical context, feminist critique, artistic research practice, and theoretical reflection. It shows how examining source code (even on seemingly simple systems) reveals how the choices, constraints, and ideologies embedded in historical moments can become embedded in technical systems. By drawing on computing archives and oral histories, we have also been able to reconstruct and analyze a version of ELIZA that works as it would have at the time, giving a better sense of the materiality of this software as a media object. Working across methods of critical analysis and creative practice in this way supports a deeper understanding of many other aspects of the project, as it helps bring to life the situated context of ELIZA that is essential to grasping its development and reception.

Crucially, our analysis of ELIZA highlights the need for careful attention to the historical context and archival traces of computational systems. It is all too easy for key narratives or technical aspects of a system to be recontextualized, misremembered, or altogether lost; the adaptability

of software can be an asset, but its metamorphoses should remain part of its story. Studying what a program does, or what people have claimed about it, can be valuable; but it is also worthwhile to understand the why, where, who, when, and how of a software artifact. That can require critically examining the code itself, through a close reading of the instructions that constitute its logic and by connecting these to its social and technical context. As AI systems increasingly shape public discourse, social relationships, and economic opportunities, this form of critical analysis becomes not merely academically interesting but also socially necessary.

Yet the field of critical code studies faces significant challenges ahead. The proprietary nature of most contemporary AI systems makes much source code inaccessible. The scale and complexity of software development often exceeds traditional methods of code analysis. Even when source code is available, its operation may depend on data, infrastructure, and organizational contexts that keep it obfuscated or allow it to be destroyed. These challenges demand new methods, institutional arrangements, and regulatory frameworks.[1] As computation increasingly mediates political discourse, economic opportunity, and social relationships, citizens need the tools to understand, critique, and shape these systems in meaningful ways.

The trajectory from ELIZA to today's large language models is neither straight nor inevitable. Rather than viewing AI history as a steady march of progress, we have shown examples of how certain paths were privileged while others were abandoned. Weizenbaum's original vision of ELIZA as a system for studying human-computer communication, not a simulation of human intelligence, has been largely forgotten as commercial interests and military funding shaped AI development. His early ethical concerns about delegating human judgment to machines, articulated decades before today's debates about algorithmic decision-making, have been similarly marginalized. These historical trajectories have profound implications for how we understand and engage with contemporary AI systems.

While Weizenbaum warned that incomprehensibility is not a necessary property of even huge computer systems, today's AI landscape is defined by increasingly opaque systems.[2] Large language models with billions of parameters trained on vast corpora of text operate at scales that resist traditional forms of analysis. The Eliza effect—that is, the tendency to anthropomorphize and overestimate machine intelligence—has expanded into what we might call the "AI effect": a broader cultural willingness to accept computational systems as authoritative, objective, and even sentient (despite the evidence to the contrary).[3] Although commercial developers continue to create the kinds of chatbots Weizenbaum would have opposed, we can draw different lessons from his work, using ELIZA as inspiration for exploring computational personas and questioning cultural assumptions.

Despite Weizenbaum's prescient warnings, society continues its headlong rush toward algorithmic solutions and automated systems,

particularly through machine learning applications that further alienate users from understanding how these systems work. We need to apply approaches like critical code studies, critical AI studies, and critical data studies that demystify rather than abstract these technologies.

Additionally, we suggest that the relationship between computation and thinking must be examined through social philosophy, particularly regarding what society considers acceptable to become "computable." Rather than focusing on technical limitations—what computers can or cannot calculate—we should ask what problems should and should not be solved computationally. This reframes the question of what is "uncomputable" from a technical constraint to an ethical choice: What tasks should society place off-limits to computation, even if they are technically feasible? This ethical framing raises fundamental questions, including whether we should aim to create thinking machines or even machines that appear to think.[4]

12.01 The AI Effect: The Persistent Allure of Artificial Intelligence?

Weizenbaum's concerns about human overreliance on computational systems have only grown more relevant in our contemporary AI landscape. What he identified with ELIZA's reception has expanded from a localized effect to a broader societal phenomenon with significant implications for social justice and human agency. Critics like Joy Buolamwini and Ruha Benjamin highlight how these systems perpetuate systemic racism through their design and implementation.[5] Artificial intelligence has been described as "not a monolithic paradigm of rationality but a spurious architecture made of adapting techniques and tricks."[6] Weizenbaum expressed his fear in relation to ELIZA that "understanding" mimicked by a computer program could become normalized, either using tricks or misdirection built on human perceptual vulnerabilities.

It is crucial to understand that computation in society is not neutral; it imposes categorical, selective processes on knowledge and can distort how data is created and transmitted. Computation often brings together formalist and metaphysical thinking, elevating mathematization and formalization of knowledge as superior approaches. This seems to repeat the mistakes of a belief in indubitable knowledge, critiqued here as an ideology, that reduces thought and understanding to instrumental rationality.

Hannah Arendt's critique of how rationality had been reduced to calculation, particularly in Pentagon decision-making, deeply influenced Weizenbaum's thinking.[7] As Arendt observes, policymakers "were not just intelligent, but prided themselves on being 'rational.' … [They] did not *judge*; they calculated … an utterly irrational confidence in the calculability of reality [became] the leitmotif of the decision making."[8] Weizenbaum developed this critique further, arguing against ceding human judgment

to machines: "Computers can make judicial decisions, computers can make psychiatric judgments. They can flip coins in much more sophisticated ways than can the most patient human being. The point is that they ought not be given such tasks. They may even be able to arrive at 'correct' decisions in some cases—but always and necessarily on bases no human being should be willing to accept."[9]

For Weizenbaum, ELIZA revealed a dangerous pattern of human dependence on computers that pointed to a deeper problem. As he argued, "Science promised [humans] power. But, as so often happens when people are seduced by promises of power, the price exacted in advance and all along the path, and the price actually paid, is servitude and impotence."[10] True power, he insisted, lies in the ability to choose, not merely to decide: "Power is nothing if it is not the power to choose. Instrumental reason can make decisions, but there is all the difference between deciding and choosing."[11]

Today's digital technologies, with their computational and calculative logic, pose even greater challenges to human reasoning.[12] What began as simple software applications designed to aid human judgment have evolved into complex, interconnected platforms that often exceed their users' understanding while becoming indispensable to daily life. These systems privilege instrumental logic over human cognitive processes, leading to what Weizenbaum described as "the feeling of powerlessness so ubiquitous among individuals in our society ... the widespread alienation of people from one another and from their work ... the perception of ordinary people that they are living in the interstices of a gigantic system."[13] His warnings about technology's powers of persuasion, deception, and mediation remain relevant to artificial intelligence's continuing development.

Like the steam engine of 1776, AI systems are viewed as transformative "general purpose" technologies with wide-ranging applications. They already permeate daily life through email filtering, content recommendations, navigation, payment processing, and countless other uses. AI is branching into new forms, including agentic systems, multimodal models, AI-driven browsers, and enabling protocols like Model Context Protocol (MCP). The question remains: What are the trade-offs? Although AI's impact on individuals and society is undeniable, understanding its full implications remains challenging—particularly as developers tend to celebrate AI's potential while remaining vague about specifics. Rather than emphasizing creator responsibility, regulatory frameworks often treat AI systems as autonomous agents. These systems are now being deployed by people across social media to influence politics and culture, including elections, through misinformation,[14] raising critical questions about safety, liability, fairness, and intellectual property. Addressing these challenges requires careful consideration of transparency, bias, infrastructure, design, and the digital literacy needed to create, use, and critique these tools. These concerns echo Weizenbaum's original warnings, now amplified by the scale and pervasiveness of contemporary AI systems that shape nearly every aspect of social, political, and economic life—and creating new forms of synthetic media and social interaction.[15]

12.02 Resisting the AI Effect

What is to be done, when in late capitalist societies the interweaving of economic chaos with rationalization and technology reduces opportunities for reflective thought? This environment privileges instrumental reason, conflating calculation with rational thinking and dismissing anything that cannot be quantified. Such conditions threaten society's capacity for critical thinking. As lecturer and activist Dan McQuillan argues, "The pseudo-rational ideology of artificial intelligence, with its racist and supremacist undertones, makes it an attractive prospect for the already existing authoritarian and fascist tendencies in political movements around the world."[16] Weizenbaum recognized these tendencies early, noting how technical language often serves to obscure, rather than clarify, writing,

> The various systems and programs we have been discussing share some very significant characteristics: they are all, in a certain sense, simple; they all distort and abuse language; and they all, while disclaiming normative content, advocate an authoritarianism based on expertise. Their advocacy is, of course, disguised by their use of rhetoric couched in apparently neutral, jargon-laden, factual language (that is, by what the common man calls "bullshit"). These shared characteristics are, to some extent, separable, but they are not independent of one another.[17]

Weizenbaum began warning about the potential harms of these advancements, which he witnessed from his particular perspective—both as a technologist and also as a Jewish German-American who saw the uses and impacts of technology in World War II, in the Vietnam War, and in American culture. Many of the questions he asked continue to be posed today, and his works offer reflections and reminders for the future—in particular for working with contemporary and emergent systems like LLMs and chatbots. Drawing from our archival research, code analysis, and historical contextualization, we have identified several crucial lessons from ELIZA that remain vitally relevant today. Here is some of what we've learned.

12.02.01 LOOK FOR WHAT'S MISSING.

The first and perhaps most important of these lessons concerns our tendency to overlook our own contributions to seemingly intelligent computer interactions. ELIZA reveals how little complexity is required to create the impression of understanding. Users supply much of what makes the interaction seem meaningful, going to great lengths to fill information gaps. Often these gaps are intentionally created, as with the persona of the Rogerian therapist that Weizenbaum chose for its plausible, even wise, unknowingness. Even a pseudorandomized "yes" or "no" response can feel like personal exchange given the right framing.[18] As Weizenbaum notes, "The human speaker will contribute much to clothe ELIZA's responses in vestments of plausibility."[19]

This insight remains crucial for evaluating today's more sophisticated systems. Don't assume complexity or dynamic growth behind an interface. Just because a chatbot interacts well or provides some correct information doesn't mean its other outputs are accurate or trustworthy. To quote MrMind, "I AM A BOT ALL BOTS ARE LIARS." The bot is not your friend (nor your enemy)—it's just logic and math, a lot of it moving very fast, putting one word after another.

Critical analysis must examine not just what these systems do but what they omit, who they exclude, and whose labor they invisibly incorporate. Ask: What aren't they telling or showing? What isn't being offered? Where are we doing most of the work of creating an intellectual or emotional connection? Beyond the interface itself, we should also ask: Who is left out of the stories? Who is excluded from the design and decision-making process? Whose data is included without consent? Whose experiences are flattened or misrepresented?

12.02.02 CHALLENGE UNDERSTANDING.

Programs are theories. Weizenbaum insisted that computational models always embody theories about the world. These theories are necessarily partial, selective, and ideologically shaped. Rather than accepting computational representations as objective, we must interrogate their underlying assumptions and recognize what they systematically exclude or distort. Weizenbaum also insisted on distinguishing between how models are designed often to "just work" and how models necessarily map a theory of the world. [20] These objectives often blur, with the results reflecting back to culture a computational representation of the world in which only its most "machine-readable" parts are validated. Weizenbaum argues,

> To those fully in the grip of the computer metaphor, to understand X is to be able to write a computer program that realizes X. ... In other words, the computer metaphor has become another lamppost under whose light, and only under whose light, men seek answers to burning questions.[21]

In settling for programs that "just work," we risk assuming that because a program works it can also provide an adequate or accurate depiction of the world or the problem at hand. This is how we end up with language models that generate reactionary, reductive versions of the world, because that is the world that they have been trained on. If "programming is theory building," as Peter Naur suggests,[22] then the theory of the world that ELIZA maps is one in which understanding is always partial. In writing the ELIZA program, Weizenbaum theorized a form of conversation that can be exchanged through a set of predetermined templates, rules, and syntax swaps—without the need for understanding. In the DOCTOR script, a therapist persona is an incomplete reflection and a transformative echo. This recognition that incompleteness and misunderstanding are inherent to computational systems offers a profound counterpoint to contemporary AI rhetoric that promises ever-increasing accuracy and comprehension.

Weizenbaum mapped a theory of communication built on assumptions of understanding itself, and he was interested in what could be made productive from the communications gaps between people and machines.[23] He continually returned focus to questions of (the impossibility of) true human understanding rather than computational understanding: "the very asking of the question, 'Has the computer captured the essence of man?' is a diversion and, in that sense, a trap. For the real question, 'Does man understand the essence of man?' cannot be answered by any technological instrument."[24] Moving beyond critique, our investigation of ELIZA also invites us to consider paths not taken and possibilities not yet realized in computational design.

12.02.03 IMAGINE ALTERNATIVE FUTURES.

If ELIZA had been known beyond the DOCTOR script, as a poetry teacher, a math tutor, or a "translating processor," perhaps its legacy and influence might have followed very different paths. Similarly, we need not accept current trajectories of AI development as inevitable or determined. Different configurations, priorities, and relationships between humans and machines remain possible. Moving beyond conventional thinking about AI capabilities requires reimagining the relationship between humans and machines. Weizenbaum resisted anthropomorphizing computer systems and warned about their illusory effects. In building ELIZA to use other scripts, he also mapped the potential for software tools that are not beholden to any single worldview. In the SPACKS script, ELIZA (1968+) discusses the poem "The Freshmen" by Barry Spacks (see chapter 5.2). The resulting dialogue unpacks the meaning of a line referencing a World War II German concentration camp: "Is any heart in order after Belsen?" What alternative scripts and alternative futures can we imagine for AI systems now? What might we ask of computational systems if we weren't constrained by existing paradigms? Might the chatbot interface paradigm, while initially democratizing access to complex LLM technologies, ultimately constrain AI development and limit the kinds of qualitative leaps that are needed to move AI forward—a kind of innovators dilemma for the AI age?[25] Imagine what causal models, and computation in general, could offer if we were not fixed into the models we have been given that focus on imitating intelligence in one particular optimized, machine-readable representation.

These alternative futures provide pivot points, suggesting we reframe the questions we ask about computational systems. Weizenbaum himself advocated for this reorientation. One of his most profound insights was to argue that the question "Can machines think?" distracts from the more important question, "How should humans think about machines?" The value of computational systems lies not in their capacity to mimic human cognition but in their ability to complement, challenge, and extend human understanding. Computers are most valuable not when they provide answers but when they help us ask better questions about ourselves, our societies, and our relationships with technology.

Rather than pursuing the impossible goal of complete machine understanding, Weizenbaum invites us to embrace partiality and limitation. We might hold computational agents to a different standard, not expecting them to know but to help us recognize what we don't know. As he notes, "Indeed, one of the most cogent reasons for using computers is to expose holes in our thinking. Computers are merciless critics in this respect."[26] These alternative possibilities require us to recognize the multifaceted nature of computational artifacts and the diverse approaches needed to fully understand them.

12.03 ELIZA's Multiple Futures

ELIZA can be "read" through many lenses: A media archeologist might be thrilled by the hardware it runs on, a programmer by the discovery of its code's operations, an interaction designer by how its scripts are composed, a psychologist by how people respond to it, an artist by its creative spinoffs, a philosopher by its gender implications, or a critical code studies researcher by how these aspects fit together and are enacted by the code itself. In the end, we found no final version of ELIZA—no single story, no definitive version of the code, no simple story to sum up how ELIZA came to be and what it became.

Weizenbaum's creation is so much more than a chatbot. We could not have made many of the discoveries in this book without the code in hand; a description of the algorithm was not enough. It required a team of scholars brought together by the code and ongoing conversation about the code, in order to develop this deeper understanding of ELIZA. The code became an entry point for explorations of the culture of the time, its technological limitations and innovations, Weizenbaum's life and opinions, as well as the programs that followed.

This case study of ELIZA demonstrates how to trace some of the many meandering threads of artificial intelligence and human–computer interaction (HCI). It is an object lesson in how much can get missed, and how quickly a singular history of code gets codified. In fact, code proliferates and travels, like seeds carried by wind, creating new growth in unexpected places. Concepts from ELIZA reached PARRY and BASIC ELIZA, and extended to Hal and *Her*, in trajectories that even its developer could not control. These versions, Weizenbaum's responses, and the gaps in the story, all make up part of the ongoing conversations that ELIZA has produced.

ELIZA also shows the role that critical examinations of source code can play in emerging systems and the need for more transparent, explainable systems—particularly as artificial intelligence scales. This transparency does not mean mere access to code, but the demand that corporations cooperate with users for more agency. It requires the capacity for deep, interdisciplinary literacy with the latest, largest AI systems in order to understand them as complex sociotechnical systems and public problems—as agents of inequality, homogenization, and oppression.[27]

This book represents a conversation between Weizenbaum and ELIZA, between programmers and cultural critics, between artists and philosophers. We invite you to join this ongoing dialogue, to approach computational systems with both technical understanding and critical awareness, and to participate in shaping more humane technological futures. As we experience an era of increasingly powerful and pervasive computational systems, ELIZA stands as both warning and inspiration. It warns against the seduction of apparent machine intelligence and the delegation of human judgment to algorithmic processes. It inspires us to imagine computational systems differently, for expanding human and non-human flourishing rather than replacing human capacities.

When running the DOCTOR script, ELIZA asked users to tell it their problems. As we face increasingly powerful and opaque AI systems, we might now pose our own questions: What problems do these systems create? What values do they encode? Whose interests do they serve? ELIZA reminds us that computational systems are never neutral technical objects but rather culturally situated artifacts embedded with ideologies, assumptions, and power dynamics. By recovering and analyzing their code, context, and consequences, critical code studies offers essential tools for democratic engagement with technologies that increasingly shape our collective future. This work has only begun, but ELIZA, in its multiplicity and lasting influence, shows us how to proceed: by asking the right questions.

12 Nov – Strings & Source(ry)

Yang, Yang 2022, Yang 2023a, Yang 2023b
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P4E Chs 4,5,6,7

Pleiades Journal Oct 22, 2024

Katherine Yang

A World Encoded

I had never thought so obsessively about poetry as I did when I was learning about syntactic linguistics.

There were different kinds of verbs, my professor had told me. If John threw a ball, that throwing was a doing. But if John loved Mary, my professor stressed that, within the structure of the sentence, John served as what was called an “experiencer” of the emotion of love. Loving was not a doing — syntactically speaking, of course. He drew for us two diagrammatic trees that demonstrated how the nature of a doing or experiencing verb influenced how the sentence was modeled in our internal understanding of language.

There is poetry in constellations, and poetry in patterns of birdsong, but this class was how I learned about the poetry that inhabits the formal structures of language. Was the professor aware of the symphony he was plotting out on the blackboard? He was composing out loud a striking thesis on what it meant to perform the act of love, or, indeed, to simply be the experiencer of some cosmic feeling. I was desperate to learn more, especially if the professor continued to explain these concepts in a way that inadvertently sounded like a poetic proclamation.

Over the course of the semester, I learned a litany of beautiful linguistics. I also learned that, as humans, we had the capacity to understand and forgive a lot that our models couldn’t.

Elsewhere in the textbook, we found that “John likes herself” was an ungrammatical sentence. The reflexive pronoun “herself” couldn’t be used to stand in for John in that sentence because John was not a girl; the sentence didn’t compute.

What the twenty-year-old introductory textbook didn’t satisfactorily explain was that the conflicting pieces of information arose not from the words themselves, but rather from our expectations of who “John” was likely to be — namely, a boy who used the reflexive pronoun “himself.” We knew nothing about “John” but that the models in our prototypical grammar had categorically precluded John from being anything other than a textbook example of a boy.

If John strayed from expectations, the expression would be struck out as a grammatical error.

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I can’t help but think about words and their power, everywhere I look.

In the computer code that I write and read every day, the word “let” is strewn everywhere. Each instance is a wish or desire — a mundane one, certainly, but powerful. With a word, I can make space in memory, create a variable called *x*, and give it a value of ten.

There’s something magical and intoxicating about that ability. It calls to mind the arcane magic of rituals and ceremony: the magic words “I name you” have the power to bring life to an inanimate object, bind a spirit, or set the ball rolling on a great prophecy.

In speech act theory, certain kinds of sentences serve as both expressions and actions. For those three magic words, “I name you” may appear to be simply a verbal expression, but in our verbal culture, words have the potential to carry the great weight and power of context. By vocalizing an

intention, we can actualize that unspoken life into the physical world. Put simply, we can speak ideas into existence.

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Much like the field of syntactic linguistics, the field of computer science is rich with systems of abstraction. A computer is, at its core, a highly sophisticated calculator, which we have creatively wrangled into the ultimate tool to solve any and all of our most complex interpersonal challenges.

Take, for example, some tens of thousands of people in the employment of a large corporation. Placed in front of this contingent and handed a sophisticated calculator, what could you proceed to do but enumerate them? The pieces then fall into place accordingly. This is how we begin to make sense of people, to understand and manage them. We collect pieces of data about people so we can store them in a big, centralized database. We compress human beings into a linear array, so we can aggregate them, categorize them, and delineate them.

These are the systems that keep our world running in its infinite complexity. For better or for worse, they're built upon millions of human judgments about how we see the world.

Abstraction runs down the design of the computer to such fundamental concepts like the binary. Even at the microscopic level of a bit flipping on and off, there are big questions to be asked. In a system built upon millions of combinations of on or off, one or zero, true or false — what does it mean to occupy space on a spectrum? What does it mean to be messy and to color outside the lines?

We built these systems. We built these worlds. I was starting to understand how they worked, and I was starting to imagine we could build others, too. I wanted to burst open these models and examine their pumping hearts.

I was beginning to form a nascent question. If these were the trappings that held the languages we speak and the languages we equip, what could a language of my own look like? What could it say — not just in its lexicon, but in its syntax and function? What alternative values could I encode; what new models could I construct?

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Over the course of a year, I undertook a project of designing and writing my own small programming language.

I knew how to program, but up until now, I had no sense of how the gears turned under the hood. With a little bit of research and a lot of trial and error, I soon found that making a programming language is not unlike teaching a child how to read.

In broad strokes, the engine that powers a programming language works by taking a program written in the designated syntax and translating that text into executable instructions.

It starts by reading in each character one by one, piecing them together to form words that it can compare against its lexicon. If it encounters the word "let," for example, it knows, from how the word has been encoded, that the word is a cue for the declaration of a new variable. It can now expect the name and value of a new variable to follow — information that it will store and use to understand future references to that variable name. (If what follows doesn't match the expectation, it stutters to a stop from the resulting syntax error, much like our assumed "John likes herself" example.) Word by word, statement by statement, and function by function, it thus walks through the entirety of the program to assemble a structured representation of the actions to be taken.

At this point, I was feeling something like a spark of life under my fingertips. With this malleable clay in my hands, I set about trying to mold it into something interesting.

Instead of having it munge through numbers and executing complex algorithms like a traditional programming language might, I wanted my programming language to be engaged with written language. I wanted it to be a vehicle for examining and reflecting on language.

I wanted to retain the computing practice of storing and manipulating values in memory, but beyond that, I started thinking about ways I could bring in ideas from the literary and poetic world.

The question of the binary lingered with me. It is a fundamental model that drives so much of the computing world, determining logical pathways and producing lightning-fast calculations. But life rarely falls so neatly into these labeled buckets. Do concepts such as ambiguity and plurality have any place in a system?

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In *Babel* by R. F. Kuang, the world runs on a type of magic called silver — more specifically, magical artifacts called silver bars. These bars are stuck to the bottom of carriages to make them run more safely quickly, for example, or embedded in gardens to induce greater tranquility. The bars are characterized by engraving two words of a chosen match pair on either side, each representing the same concept in different languages. For example, a match pair might consist of the words “speed,” from English, against its Latin root, “spēs,” whose meanings include the idea of hope. The two words are closely tied, yes, but translation is an ambitious and tricky endeavor. More often than not, we aren’t hoping to communicate every nuanced facet of a word and its place in a language, but we’re settling instead for the closest approximation or an emulation of the original idea. Indeed, it’s this elusive translational distance in context and connotation that serves as the rich source of power in this world’s magic system.

I find poetry in the shape of a system and in the awkward spaces where it can’t quite fit. To a layperson, to a mind new to the intricacies of interpretation — to a computational system, perhaps — the machine of translation seems so straightforward until you consider the sheer scope of its objective.

Words occupy amorphous spaces in our galaxy of language. Poets, who work as craftspeople in the medium of language, are immensely talented in understanding how to wield the raw material of words, both in isolation and in combination, to tell a specific story or evoke a specific emotion. And still — or perhaps, as a result — a million readers will inevitably have a million different interpretations of the same poem and the words used within.

In contrast, computers don’t traffic in anything so human and changeable. If one system communicates to another that *x* is equal to ten, no shred of nuance can possibly be lost in that exchange.

And yet, if pressed, I might make the case that, in the computing world, a device called the regular expression is one that seems to play on a lot of the same questions that human language does.

Let’s say that you wanted to find every North American phone number in a spreadsheet. Instead of typing in random sequences of ten-digit numbers until you found every one, you could write one magical expression into the search box that would encapsulate every phone number that could possibly exist. Assuming a consistent format, the expression could look like this:

```
\d{3}-\d{3}-\d{4}
```

The backslash followed by a “d” is a special code that covers any digit from zero to nine. The dash means you want a literal dash, and the “3” or “4” in curly brackets means you want three or four of those numbers. In one concise expression, you would be able to find the phone numbers “123-456-7890” or “555-555-5555” or “206-555-0100.”

Handy for doing a batch operation on a spreadsheet, of course, but that’s not why we’re here. Even in the weeds of formal computational grammars, there’s something in the act of searching and the descriptive language of representation that rings of poetry to me.

Instead of dealing with phone numbers, let’s say that you have a habit of confusing the words “word” and “world” — that pesky “l” sneaks in and out whenever you’re not paying attention. You want to find every use of those two words, perhaps to mark the places where you’ll need to reread the passage and confirm the spelling of the word. You could search for each word individually, but here’s a regular expression you might write instead:

```
worl?d
```

Here, most of the bare letters are requests for those literal letters, but if the letter is followed by a question mark, that letter is marked as optional. When the system uses this query to search through the text, it’ll look for every word that starts with the letters “wor,” then either does or doesn’t have the letter “l,” and finally ends with the letter “d.” In other words, here is an expression that searches for, or perhaps represents, both “word” and “world.” It encapsulates the orthographic, and perhaps semantic, span of the two words; it, in a way, *is* both “word” and “world.” Once the syntax of the system is decoded, the concealed layers of meaning begin to reveal themselves.

The question mark is a special character that represents either absence or presence, and thus represents both simultaneously, leading to the creation of something like Schrodinger’s letters. Its cloaking ability isn’t limited to just letters, either.

Take this expression:

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(will )?(have )?miss(ed)?
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The parentheses group letters or words together, so that they can be operated upon as a unit by the following question mark. This expression marks the words “will” and “have” as optional; same for the past tense “-ed” postfix on the verb “miss.” It encapsulates multiple possible tenses and aspects of the act and feeling of missing: we “miss,” “missed,” “will miss,” “have missed,” and “will have missed.” It is past, present, future, and more, in one fell swoop. Each symbolic marking unlocks another layer of possibility. In its most expanded form, the expression only consists of three words, but the journey of its decoding starts to feel like a poem.

This is where I started to find a home for my language.

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Coem (“co-uhm”) is a programming language, in its bones, but it has grown into a broader experiment in the contemplation of words and their relationships.

Variable names are built with regular expressions: with the question mark at your disposal, you can speak about words and worlds always in the same breath; a hearth can always carry a heart. With another special operator, representing the choice between multiple options between groups of letters, you could consider the holy union of poetry and pottery, or gaze upon a moon that always abuts the night.

It's a two-way relationship between the definition and the reference. When I write worl?d, I mean to say something about both "word" and "world"; when I later ask for "word," the system returns to me that same joint definition. Like two halves of an imperfect match pair, the words occupy the same space in memory, giving rise to something new from their juxtaposition, like overlapping shards of colored glass.

On top of this foundation, I've been exploring more approaches to examining and interacting with words. One is a palimpsest mode, which allows for variables to grow throughout the piece. In the traditional paradigm, redefining a variable replaces the old value with the new one, but in a palimpsest mode, variables can expand with each new definition, taking on traces of time like layers of writing on parchment. When it's asked to report its value, it returns not only its latest definition, but every definition it's taken on throughout the course of the piece.

The idea of the "output" has also been an area of exploration. In programming, there is the most visible output: the result of the program, or the file or webpage or software that is produced by the program. There is also the programmer's internal kind of output. In an area of the window called the console, the programmer can "print" messages at certain points in the program to help identify problems in the code.

The focus of Coem is on the writing and the text. The output is worth reading, but so is the source code, and so, too, is the internal output. These are all elements of "the text" to me. In Coem, then, printing the result of a variable is the output, and the output is fed back into the source code as a comment or annotation, in a feedback loop of definition, reference, and affirmation.

Piece by piece, the language starts to feel like a tiny looking glass through to makings of a little world. It's a world where we make space for plurality, where we platform intentional expression over algorithmic detritus, where we pull the curtain back, where we acknowledge growth and complexity, where we feel our way forward through language and poetry.

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On "Sourcery," or Code as Fetish

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Abstract

This essay offers a sympathetic interrogation of the move within new media studies toward "software studies." Arguing against theoretical conceptions of programming languages as the ultimate performative utterance, it contends that source code is never simply the source of any action; rather, source code is only source code after the fact: its effectiveness depends on a whole imagined network of machines and humans. This does not mean that source code does nothing, but rather that it serves as a kind of fetish, and that the notion of the user as super agent, buttressed by real-time computation, is the obverse, not the opposite of this "sourcery."

Debates over new media often resonate with the story of the six blind men and the elephant. Each man seizes a portion of the animal and offers a different analogy: the elephant is like a wall, a spear, a snake, a tree, a palm, a rope. Refusing to back down from their positions—based as they are on personal experiences—the wise men then engage in an unending dispute with each "in his own opinion / Exceeding stiff and strong / Though each was partly in the right, / And all were in the wrong!" The moral, according to John Godfrey Saxe's version of this tale, is: "So oft in theologic wars, / The disputants, I ween, / Rail on in utter ignorance / Of what each other mean, / And prate about an Elephant / Not one of them has seen!"[1] It is perhaps

1. John Godfrey Saxe, "The Blind Men and the Elephant." http://www.wordinfo.info/words/index/info/view_unit/1/?letter=B&spage=3.

Configurations, 2008, 16:299–324 © 2010 by The Johns Hopkins University Press and the Society for Literature and Science.

profane to compare a poem on the incomprehensibility of the divine to arguments over new media, but the invisibility, ubiquity, and alleged power of new media (and technology more generally) lend themselves nicely to this analogy. It seems impossible to know the extent, content, and effects of new media. Who knows the entire contents of the WWW or the real extent of the Internet or of mobile networks? How can one see and know all time-based, online interactions? Who can expertly move from analyzing social-networking sites to Japanese cell-phone novels to *World or Warcraft* to hardware algorithms to ephemeral art installations? Is a global picture of new media possible?

In response to these difficulties, an important strain of new media has moved away from content and from specific technologies to what seems to be common to all new media objects and moments: software. All new media allegedly rely on—or, most strongly, can be reduced to—software, a "visibly invisible" essence. Software seems to allow one to grasp the entire elephant because it is the invisible whole that generates the sensuous parts. Based on and yet exceeding our sense of touch—based on our ability to manipulate virtual objects we cannot entirely see—it is a magical source that promises to bring together the fractured field of new media studies and to encapsulate the difference this field makes.

But, what is software? What does it mean to know software and, most importantly, what does positing software as the essence of new media do? This essay responds to these questions, arguing that software as source relies on a profound logic of "sourcery"—a fetishism that obfuscates the vicissitudes of execution and makes our machines demonic. Further, this sourcery is the obverse rather than the opposite of the other dominant trend in new media studies: the valorization of the user as agent. These sourceries create a causal relationship among one's actions, one's code, and one's interface. The relationship among code and interface, action and result, however, is always contingent and always to some extent imagined. The reduction of computer to source code, combined with the belief that users run our computers, makes us vulnerable to fantastic tales of the power of computing. To break free of this sourcery, we need to interrogate, rather than venerate or even accept, the grounding or logic of software. Crucially, though, closely engaging software will not let us escape fictions and arrive at a true understanding of our machines, but rather make our interfaces more productively spectral. As a fetish, source code can provide surprising "deviant" pleasures that do not end where they should. Framed as a *re*-source, it can help us think through the machinic and human rituals that help us imagine our technologies and their executions.

The Logos of Software

To exaggerate slightly, software has recently been posited as the essence of new media, and knowing software as a form of enlightenment. Software is allegedly the truth, the base layer, the logic of new media. Lev Manovich in his groundbreaking *The Language of New Media* for instance asserts that

> new media may look like old media, but this is only the surface . . . to understand the logic of new media, we need to turn to computer science. It is there that we may expect to find the new terms, categories, and operations that characterize media that become programmable. *From media studies we move to something that can be called "software studies"—from media theory to software theory*.[2]

his turn to software—to the logic of what lies beneath—has offered a solid ground to new media studies, allowing it, as Manovich argues, to engage presently existing technologies and to banish so-called "vapor theory," theory that fails to distinguish between demo and product, fiction and reality, to the margins.[3] This call to banish vapor theory, made by Geert Lovink and Alexander Galloway amongst others, has been crucial to the rigorous study of new media, but, this rush away from what is vapory—undefined, set in motion—is also troubling because vaporiness is not accidental, but rather essential to, new media and, more broadly, to software. Software, after all, is ephemeral, information ghostly, and new media projects that have never, or barely, materialized are among the most valorized and cited. (Also, if you take the technical definition of information seriously, information increases with vapor, with entropy). This turn to computer science also threatens to reify knowing software as truth, an experience that is arguably impossible: we all know some software, some programming languages, but does anyone really "know" software? What could this knowing even mean? Regardless, from myths of all-powerful hackers who "speak the language of computers as one does a mother tongue"[4] or who produce abstractions that re-

2. Lev Manovich, *The Language of New Media* (Cambridge, MA: MIT Press, 2001), p. 48, emphasis in original.

3. Vapor theory is a term coined by Peter Lunenfeld and used by Geert Lovink to designate theory so removed from actual engagement with digital media that it treats fiction as fact. This term, however, can take on a more positive resonance, if on takes the non-materiality of software seriously. (Geert Lovink, "Enemy of Nostalgia, Victim of the Present, Critic of the Future Interview with Peter Lunenfeld," 31 Jul 2000 <http://www.nettime.org/Lists-Archives/nettime-l-0008/msg00008.html> accessed 2/1/2007).

4. Alexander R. Galloway, *Protocol: How Power Exists after Decentralization* (Cambridge, MA: MIT Press, 2004), p. 164.

lease the virtual[5] to perhaps more mundane claims made about the radicality of open source, knowing (or using) the right software has been made analogous to man's release from his self-incurred tutelage.[6] As advocates of free and open source software make clear, this critique aims at political, as well as epistemological, emancipation: as a form of enlightenment, it is a stance of how not to be governed like that—an assertion of an essential freedom that can only be curtailed at great cost.[7]

To be clear, I am not dismissing the political potential of free or open source software, or the importance of studying or engaging software; rather, I am arguing that we need to interrogate how knowing (or using free or open source) software does not simply enable us to fight domination or rescue software from evil-doers such as Microsoft, but rather is embedded in—mediates between, is part of—structures of knowledge-power. For instance, using free software does not mean escaping from power, but rather engaging it differently, for free and open source software profoundly privatizes the public domain: GNU *copyleft* does not seek to reform *copyright*, but rather to spread it everywhere.[8] It is thus symptomatic of the move in contemporary society away from the public/private

5. McKenzie Wark, "A Hacker Manifesto." version 4.0 (n.d.). http://subsol.c3.hu/subsol_2/contributors0/warktext.html.

6. See Richard Stallman, "The Free Software Movement and the Future of Freedom; March 9th 2006." http://fsfeurope.org/documents/rms-fs-2006-03-09.en.html. Immanuel Kant famously described enlightenment as "mankind's exit from its self-incurred immaturity" ("An Answer to the Question: What Is Enlightenment," *What Is Enlightenment? Eighteenth-Century Answers and Twentieth-Century Questions*, Ed. James Schmidt (Berkeley: California UP, 1996), 58).

7. For more on enlightenment as a stance of how not to be governed like that, see Michel Foucault, "What Is Critique?" in *What Is Enlightenment?,* ed. James Schmidt (Berkeley: University of California Press, 1996), pp. 382–398.

8. As Niva Elkin-Koren notes in "Creative Commons: A Skeptical View of a Worthy Project," the Creative Commons strategy "does not aim at creating a public domain, at least not in the strict legal sense of a regime that is free of any exclusive proprietary rights. The strategy is entirely dependent upon a proprietary regime and drives its legal force from its existence. The normative framework assumes that it is possible to replace existing practices of producing and distributing informational works by relying on the existing proprietary regime. The underlying assumption is that if intellectual-property rights remain the same, but rights are exercised differently by their owners, free culture would emerge" (http://www.hewlett.org/NR/rdonlyres/6D4BFD1E-09BB-4F89-9208-7C1E4B141F2A/0/Creative_Commons_Amsterdam_final2006.pdf). Although Elkin-Koren is writing about Creative Commons in this passage, she makes it clear that this strategy of extending and revising intellectual-property rights is drawn from the Free Software Movement's GPL.

dichotomy to that of open/closed.[9] More subtly, the free software movement, in insisting that freedom stems from free software—from freely accessible source code—amplifies the power of source code, erasing the vicissitudes of execution and the structures that ensure the coincidence of code and its execution. It buttresses the logic of software—that is, software as *logos*.

Software as we now know it (importantly, software was not always software) conflates word with result, logos with action. The goal of software is to conflate an event with a written command. Software blurs the difference among human-readable code (readable because of another program), its machine-readable interpretation, and its execution by turning the word "program" from a verb to a noun, by turning process in time into process in space, by turning execution into inscription—or at least attempting to do so. An example I've used elsewhere, Edsger Dijkstra's famous condemnation of "go to" statements, encapsulates this nicely.[10] In "Go to Statement Considered Harmful," Dijkstra argues, "the quality of programmers is a decreasing function of the density of go to statements in the programs they produce," because go to statements work against the fundamental tenant of what Dijkstra considered to be good programming: namely, the necessity to "shorten the conceptual gap between static program and dynamic process, to make the correspondence between the program (spread out in text space) and the process (spread out in time) as trivial as possible."[11]

This is important, since if a program suddenly halts because of a bug, go tos make it difficult to find the place in the program that corresponds to the buggy code. Go tos make difficult the conflation of instruction with its product—the reduction of process to command—that grounds the emergence of software as a concrete entity and commodity; that is, go tos make it difficult for the source program to act as a legible source.[12] As I've argued elsewhere, this conflation of

9. For more on this, see Wendy Hui Kyong Chun, *Control and Freedom: Power and Paranoia in the Age of Fiber Optics* (Cambridge, MA: MIT Press, 2006).

10. See Wendy Hui Kyong Chun, "On Software, or the Persistence of Visual Knowledge," *grey room* 18 (winter 2005): 27–52.

11. Edsger Dijkstra, "Go to Statement Considered Harmful," in *Software Pioneers: Contributions to Software Engineering*, ed. Manfred Broy and Ernst Denert (Berlin: Springer, 2002), p. 342.

12. Structured programming was introduced as a way to make programs, rather than "programmer priest," the source, although the term programmer priest complicates the notion of source: Is the source the programmer, or is it some mythic power she mediates?

instruction or command with its product is also linked to software's gendered, military history: in the military, there is supposed be no difference between a command given and a command completed, especially to a "girl."[13] The implication here is: execution does not matter—as in conceptual art, execution is a perfunctory affair; what really matters is the source.[14]

This drive to forget or trivialize execution is not limited to programmers confounded with the task of debugging errant programs; it also extends to many critical analyses of code. These theorizations, which importantly question the reduction of new media with screen, emphasize code as performative or executable. For instance, Alexander Galloway, in *Protocol: How Control Exists after Decentralization,* claims: "code draws a line between what is material and what is *active*, in essence saying that writing (hardware) cannot *do* anything, but must be transformed into code (software) to be effective. . . . Code is language, but a very special kind of language. *Code is the only language that is executable*."[15] Drawing in part from Galloway, N. Katherine Hayles, in *My Mother Was a Computer: Digital Subjects and Literary Texts,* distinguishes between the linguistic performative and the machinic performative by arguing that

> [c]ode that runs on a machine is performative in a much stronger sense than that attributed to language. When language is said to be performative, the kinds of actions it "performs" happen in the minds of humans, as when someone says, "I declare this legislative session open" or "I pronounce you husband and wife." Granted, these changes in minds can and do reach in behavioral effects, but the performative force of language is nonetheless tied to external changes through complex chains of mediation. Code running in a digital computer causes changes in machine behavior and, through networked ports and other interfaces, may initiate other changes, all implemented through transmission and execution of code.[16]

13. See Chun, "On Software" (above, n. 11).

14. For more on software art as conceptual art, see Florian Cramer, "Concepts, Notations, Software, Art." 2002. http://userpage.fu-berlin.de/~cantsin/homepage/writings/software_art/concept_notations/concepts_notations_software_art.html.

15. Galloway, *Protocol* (above, n. 5), p. 165, emphasis in original.

16. N. Katherine Hayles, *My Mother Was a Computer: Digital Subjects and Literary Texts* (Chicago: University of Chicago Press, 2005), p. 50. Hayles's argument immediately poses the question: What counts as internal versus external to the machine, especially given that, in John von Neumann's foundational description of stored program computing, the input and output (the outside world to the machine) was a form of memory?

The independence of machine action—this autonomy, or automatic executability of code—is, according to Galloway, its material essence:

> the material substrate of code, which must always exist as an amalgam of electrical signals and logical operations in silicon, however large or small, demonstrates that code exists first and foremost as commands issued to a machine. Code essentially has no other reason for being that instructing some machine in how to act. One cannot say the same for the natural languages.[17]

Galloway thus strongly states, in "Language Wants to Be Overlooked: Software and Ideology," that "to see code as subjectively performative or enunciative is to anthropomorphize it, to project it onto the rubric of psychology, rather than to understand it through its own logic of 'calculation' or 'command.'"[18]

To what extent, however, can source code be understood outside of anthropomorphization? Does understanding voltages stored in memory as commands/code not already anthropomorphize the machine? (The inevitability of this anthropormorphization is arguably evident in the title of Galloway's article: "Language *Wants* to Be Overlooked" [emphasis added].) How is it that code "causes" changes in machine behavior? What mediations sustain the notion of code as inherently executable?

To make the argument that code is automatically executable, the process of execution itself must not only be erased, but source code also must be conflated with its executable version. This is possible because, it is argued, the two "layers" of code can be reduced to each other. Indeed, in *Protocol*, Galloway argues that

> uncompiled source code is *logically* equivalent to the same code compiled into assembly language and/or linked into machine code. For example, it is absurd to claim that a certain value expressed as a hexadecimal (base 16) number is more or less fundamental than that same value expressed as [a] binary (base 2) number. They are simply two expressions of the same value.[19]

He later elaborates on this point by drawing an analogy between quadratic equations and logical layers:

> One should never understand this "higher" symbolic machine as anything empirically different from the "lower" symbolic interactions of voltages

17. Alexander R. Galloway, "Language Wants to Be Overlooked: Software and Ideology," *Journal of Visual Culture* 5:3 (2006): 315–331.

18. Ibid., 321.

19. Galloway, *Protocol* (above, n. 5), p. 167, emphasis in original.

> through logic gates. They are complex aggregates yes, but it is foolish to think that writing an "if/then" control structure in eight lines of assembly code is any more or less machinic than doing it in one line of C, just as the same quadratic equation may swell with any number of multipliers and still remain balanced. The relationship between the two is *technical*.[20]

According to Galloway's quadratic equation analogy, the difference between a compact line of higher-level programming code and eight lines of assembler equals the difference between two equations, in which one contains coefficients that are multiples of the other. The solution to both equations is the same: one equation can be reduced to the other. This reduction, however, does not capture the difference between the various instantiations of code, let alone the empirical difference between the higher symbolic machine and the lower interactions of voltages (the question here is: Where does one make the empirical observation?); that is, to push Galloway's conception further, a technical relation is far more complex than a numerical one.

Significantly, a technical relation engages art or craft. According to the *Oxford English Dictionary* (*OED*), a technical person is one "skilled in or practically conversant with some particular art or subject." Code does not always nor automatically do what it says, but does so in a crafty manner. To state the obvious, one cannot run source code: it must be compiled or interpreted. This compilation or interpretation—this making code executable—is not a trivial action; the compilation of code is not the same as translating a decimal number into a binary one; rather, it involves instruction explosion and the translation of symbolic into real addresses. Consider, for example, the instructions needed for adding two numbers in PowerPC assembly language, which is one level higher than machine language:

```
li      r3,1          *load the number 1 into register 3
li      r4,2          *load the number 2 into register 4
add     r5,r4,r3      *add r3 to r4 and store the result in r5
stw     r5,sum(rtoc)  *store the contents of r5 (i.e. 3)into the
                        memory location
                      *called "sum"s (where sum is defined elsewhere)
blr                   *end of this snippet of code[21]
```

20. Galloway, "Language Wants to Be Overlooked" (above, n. 18), p. 321, emphasis in original.

21. This example draws from the *PowerPC Assembly Language Beginners Guide*. http://www.lightsoft.co.uk/Fantasm/Beginners/Chapt1.html.

This explosion is not equivalent to multiplying both sides of a quadratic equation by the same coefficient or to the difference between E and 15. It is, rather, a breakdown of the steps needed to perform what seems a simple arithmetic calculation. It is the difference between a mathematical identity and a logical equivalence, which depends on a leap of faith. This is most clear in the use of numerical methods to turn integration—a function performed fluidly in analog computers—into a series of simpler, repetitive arithmetical steps. This translation from source code to executable is arguably as involved as the execution of any command. More importantly, it depends on the action (human or otherwise) of compiling/interpreting and executing. Also, some programs may be executable, but not all compiled code within that program is executed; rather, lines are read in as necessary. *So, source code thus only becomes source after the fact.* Source code is more accurately a *re-source,* rather than a source. Source code becomes a source when it becomes integrated with logic gates (and at an even lower level, with the transistors that comprise these gates); when it expands to include software libraries, when it merges with code burned into silicon chips; and when all these signals are carefully monitored, timed, and rectified.

Code is also not always the source, because hardware does not need software to "do something." One can build algorithms using hardware, for all hardware cannot be reduced to stored program memory. Figure 1, for instance, is the logical statement: "if notB and notA, do CMD1 (state P); if notB and notA and notZ OR B and A (state Q) then command 2." To be clear, this is not a valorization of hardware over software, as though hardware necessarily escapes this drive to conflate space with time. Crucially, this schematic is itself an abstraction. Logic gates can only operate "logically" if they are carefully timed. As Philip Agre has emphasized, the digital abstraction erases the fact that gates have "directionality in both space (listening to its inputs, driving its outputs) and in time (always moving toward a logically consistent relation between these inputs and outputs)."[22] When a value suddenly changes, there is a brief period in which a gate will give a false value. In addition, because signals propagate in time over space, they produce a magnetic field that can corrupt other signals nearby ("crosstalk"). This schematic erases all these various time-based effects. Thus hardware schematics, rather than escaping from the logic of sourcery, are also embedded within this structure. Indeed, John von Neumann, the alleged

22. Philip E. Agre, *Computation and Human Experience* (Cambridge: Cambridge University Press, 1997), p. 92.

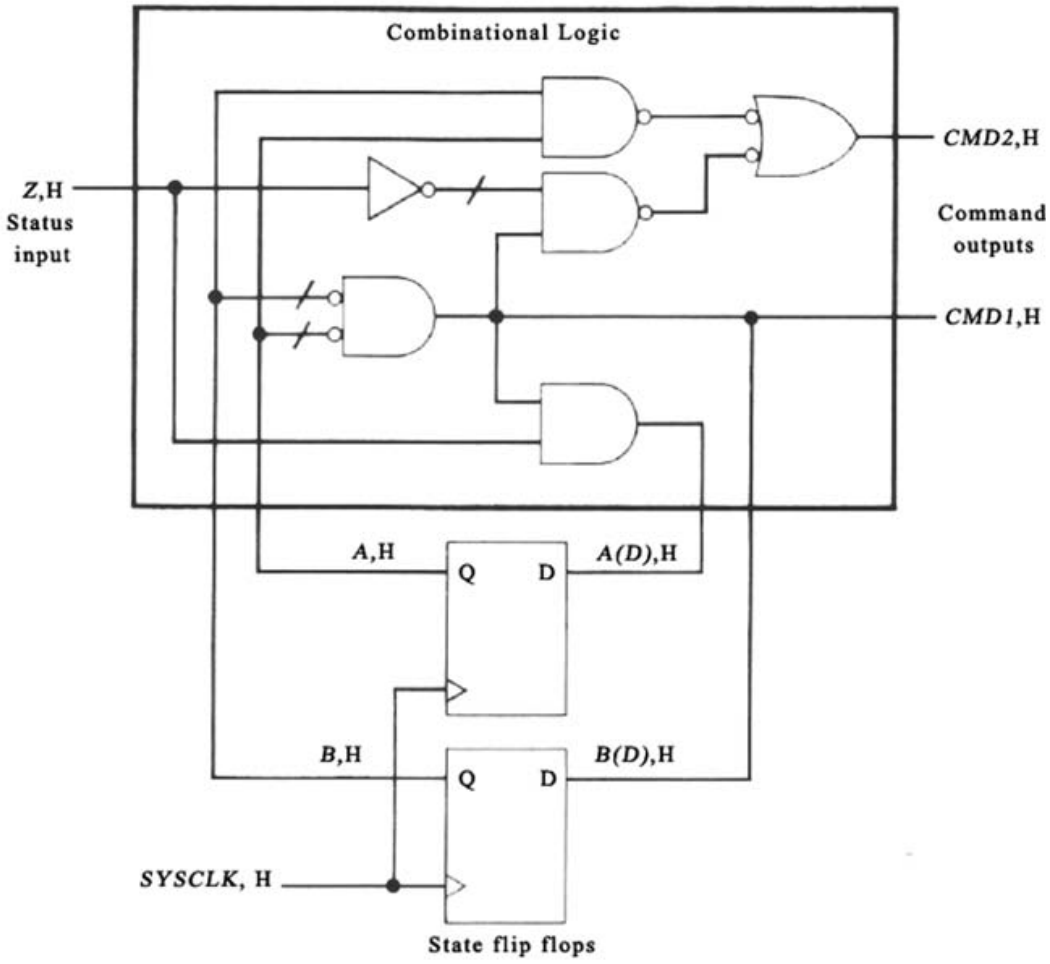


Figure 1. If notB and notA, do CMD1 (state P); if notB and notA and notZ OR B and A (state Q) then command.

architect of the stored-memory digital computer, drew from Warren McCulloch and Walter Pitts's conflation of neuronal activity with its inscription in order to conceptualize our modern computers. According to McCulloch and Pitts (who themselves drew from Alan Turing's work), "the response of any neuron [is] factually equivalent to a proposition which proposed its adequate stimulus."[23] That is, the response of a neuron can be reduced to the circumstances that make it possible: instruction can substitute for result. It is perhaps appropriate then that von Neumann spent the last days of his life—dying from a cancer most probably stemming from his work at Los Alamos—reciting from memory *Faust,* Part 1.[24] At the heart

23. Warren McCulloch and Walter Pitts, "A Logical Calculus of the Ideas Immanent in Nervous Activity," in *Embodiments of Mind* (Cambridge, MA: MIT Press, 1965), p. 21.

24. Norman McRae, *John von Neumann* (New York: Pantheon, 1992), p. 378.

of stored-program computing lies the Faustian substitution of word for action.

Not surprisingly, this notion of source code as source coincides with the introduction of alpha-numeric languages. With them, human-written, nonexecutable code becomes source code and the compiled code, the object code. Source code thus is arguably symptomatic of human language's tendency to attribute a sovereign source to an action, a subject to a verb.[25] By converting action into language, source code emerges. Thus Galloway's statement "to see code as subjectively performative or enunciative is to anthropomorphize it, to project it onto the rubric of psychology, rather than to understand it through its own logic of 'calculation' or 'command'" overlooks the fact that to use higher-level alphanumeric languages is already to anthropomorphize the machine, and to reduce all machinic actions to the commands that supposedly drive them. In other words, the fact that "code is law"—something Lawrence Lessig pronounces with great aplomb—is hardly profound.[26] After all, code is, according to the *OED*, "a systematic collection or digest of the laws of a country, or of those relating to a particular subject." What is surprising is the fact that *software is code*, that code is—has been made to be—executable, and that this executability makes code not law, but rather every lawyer's dream of what law should be, automatically enabling and disabling certain actions and functioning at the level of everyday practice.

Source Code as Fetish

Source code as source means that software functions as an axiom, as "a self-evident proposition requiring no formal demonstration to prove its truth, but received and assented to as soon as it is mentioned."[27] In other words, whether or not source code is only source code after the fact or whether or not software can be physically separated from hardware,[28] software is always posited as already existing, as the self-evident ground or source of our interfaces. Software is axiomatic. As a first principle, it fastens in place a certain

25. According to Friedrich Nietzsche in *The Genealogy of Morals*, "there is no 'being' behind the doing, effecting, becoming: 'the doer' is merely a fiction added to the deed—the deed is everything"; see Nietzsche, *The Birth of Tragedy & Genealogy of Morals*, trans. Francis Golffing (New York: Doubleday, 1956), pp. 178–179.

26. Lawrence Lessig, *Code and Other Laws of Cyberspace* (New York: Basic Books, 1999).

27. *Oxford English Dictionary Online*. http://www.oed.com.

28. See Friedrich Kittler, "There Is No Software," *Ctheory*. 1995. http://www.ctheory.net/articles.aspx?id=74.

logic of cause and effect, based on the erasure of execution and the privileging of programming that bleeds elsewhere and stems from elsewhere as well.[29] As an axiomatic, it, as Gilles Deleuze and Félix Guattari argue, artificially limits decodings.[30] It temporarily limits what can be decoded, put into motion, by setting up an artificial limit—the artificial limit of programmability—that seeks to separate information from entropy by designating some entropy information and other "nonintentional" entropy noise. Programmability, discrete computation, depends on the disciplining of hardware and the desire for a programmable axiomatic. Code is a medium in the full sense of the word. As a medium, it channels the ghost that we imagine runs the machine—that we see as we don't see—when we gaze at our screen's ghostly images.

Understood this way, source code is a fetish. According to the *OED*, a fetish was originally an ornament or charm worshipped by "primitive peoples . . . on account of its supposed inherent magical powers."[31] The term *fetisso* stemmed from the trade of small wares and magic charms between the Portuguese merchants and West Africans; Charles de Brosses, in 1757, coined the term fetishism to describe "primitive religions." According to William Pietz, Enlightenment thinkers viewed fetishism as a

> false causal reasoning about physical nature [that became] the definitive mistake of the pre-enlightened mind: it superstitiously attributed intentional purpose and desire to material entities of the natural world, while allowing social action to be determined by the . . . wills of contingently personified things, which were, in truth, merely the externalized material sites fixing people's own capricious libidinal imaginings.[32]

That is, fetishism, as "primitive causal thinking," derived causality from desire rather than reason:

> Failing to distinguish the intentionless natural world known to scientific reason and motivated by practical material concerns, the savage (so it was argued) superstitiously assumed the existence of a unified causal field for personal actions and physical events, thereby positing reality as subject to animate powers whose purposes could be divined and influenced. Specifically, humanity's

29. Namely, twentieth-century genetics.

30. Gilles Deleuze and Félix Guattari, "Capitalism: A Very Special Delirium." 1995. http://www.generation-online.org/p/fpdeleuze7.htm.

31. *Oxford English Dictionary Online* (above, n. 28).

32. William Pietz, "Fetishism and Materialism," in *Fetishism as Cultural Discourse*, ed. Emily Apter and William Pietz (Ithaca, NY: Cornell University Press, 1993), pp. 138, 139.

> belief in gods and supernatural powers (that is, humanity's unenlightenment) was theorized in terms of prescientific peoples' substitution of imaginary personifications for the unknown physical causes of future events over which people had no control and which they regarded with fear and anxiety.[33]

Fetishes thus allow the human mind too much and not enough control by establishing a "unified causal field" that encompasses both personal actions and physical events. Fetishes enable a semblance of control over future events—a possibility of influence, if not an airtight programmability—that itself relies on distorting real social relations into material givens.

This notion of fetish as false causality has been most important to Karl Marx's diagnosis of capital as fetish. He famously argued that

> the commodity-form . . . is nothing but the determined social relation between humans themselves which assumes here, for them, the phantasmagoric form of a relation between things. In order, therefore, to find an analogy we must take a flight into the misty realm of religion. There the products of the human head appear as autonomous figures endowed with a life of their own, which enter into relations both with each other and with the human race. So it is in the world of commodities with the products of men's hands. This I call . . . fetishism.[34]

The capitalist thus confuses social relations and the labor activities of real individuals with capital and its seeming magical ability to reproduce, for "it is in interest-bearing capital . . . that capital finds its most objectified form, its pure fetish form. . . . Capital—as an entity—appears here as an independent source of value; a something that creates value in the same way as land [produces] rent, and labor wages."[35]

The parallel to source code seems obvious: we "primitive folk" worship source code as a magical entity—as a source of causality—when in truth the power lies elsewhere, most importantly in social and machinic relations. If code is performative, its effectiveness relies on human and machinic rituals. Intriguingly though, in this parallel, Enlightenment thinking—a belief that knowing leads to control and freedom, a release from tutelage—is not the "solution" to the fetish, but rather what grounds it, for source code historically has been portrayed as the solution to wizards and other myths of

33. Ibid., p. 137.

34. Karl Marx, *Capital: A Critique of Political Economy*, vol. 1, trans. Ben Foskes (New York: Penguin, 1976), p. 165.

35. Marx, as quoted by Pietz, "Fetishism and Materialism" (above, n. 33), p. 149.

programming. According to this popular narrative, machine code provokes mystery and submission; source code enables understanding and thus institutes rational thought and freedom, enabling us to move from being blind to sighted.

John Backus, writing about programming in the 1950s, contends that "programming in the early 1950s was a black art, a private arcane matter."[36] These programmers formed a "priesthood guarding skills and mysteries far too complex for ordinary mortals."[37] Opposing even the use of decimal numbers, these machine programmers were sometimes deliberate purveyors of their own fetishes or "snake oil," systems that allegedly had "mysterious, almost human abilities to understand the language and needs of the user; closer inspection was likely to reveal a complex, exception-ridden performer of tedious clerical tasks that substituted its own idiosyncrasies for those of the computer."[38] Automatic programming languages such as FORTRAN, developed by Backus, allegedly exorcised the profession of the priesthood by making programs more readable and thus making it easier to discern the difference between snake oil and the real thing.

Similarly, Richard Stallman, in his critique of nonfree software, has argued that machine-executable programs create mystery rather than enlightenment. Stallman contends that an executable "is a mysterious bunch of numbers. What it does is secret."[39] As I argue below, this notion of executables and executing code as magic is buttressed by real-time operating systems that posit the user as the source. Against this magical execution, source code supposedly enables pure understanding and freedom—the ability to map and understand the workings of the machine, but again only through a magical erasure of the gap between source and execution, an erasure of execution itself. Tellingly, this move to source code has hardly deprived programmers of their priest-like/wizard status. If anything, the notion of programmers as super-human has been disseminated ever more, and the history of computing—from direct manipulation to hypertext—has been marked by various "liberations."

But clearly, source code can do things: it can be interpreted or compiled, and it can be rendered into machine-readable commands

36. John Backus, "Programming in America in the 1950s—Some Personal Impressions," in *A History of Computing in the Twentieth Century*, ed. N. Metropolis et al. (New York: Academic Press, 1980), p. 126.

37. Ibid., p. 127.

38. Ibid.

39. Richard Stallman, "Copyright and Globalization in the Age of Computer Networks." 2001. http://www.gnu.org/philosophy/copyright-and-globalization.html.

that are then executed. Source code is also read by humans and is written by humans for humans: it is thus the source of some understanding. Although Ellen Ullman and many others have argued that "a computer program has only one meaning: what it does. It isn't a text for an academic to read. Its entire meaning is its function," source code must be able to function, even if it does not function—that is, even if it is never executed.[40] Source code's readability is not simply due to comments that are embedded in the code, but also due to English-based commands and programming styles designed for comprehensibility. This readability is not just for "other programmers"; when programming, one must be able to read one's own program—to follow its logic and predict its outcome, whether or not this outcome coincides with one's prediction.

This notion of source code as readable—as creating some outcome regardless of its machinic execution—underlies "codework" and other creative projects. The Internet artist Mez, for instance, has created a language, "mezangelle," that incorporates formal code and informal speech. Mez's poetry deliberately plays with programming syntax, producing language that cannot be executed but nonetheless draws on the conventions of programming language to signify.[41] Codework, however, can also work entirely within an existing programming language. Graham Harwood's perl poem, for example, translates William Blake's late-eighteenth-century poem "London" into London.pl, a script that contains within it an algorithm to "find and calculate the gross lung-capacity of the children screaming from 1792 to the present."[42] Regardless of whether or not it can execute, code can be, must be, worked into something meaningful. Source code, in other words, may be sources of things other than the machine execution it is "supposed" to engender.

Source code as fetish, understood psychoanalytically, embraces this nonteleological potential of source code, for the fetish is a genital substitute that gives the fetishist nonreproductive pleasure. It is a deviation that does not "end" where it should, a deviation taken on so that the child may combat castration for both himself and his mother, while at the same time accommodating to his world's larger Oedipal structure. It both represses and acknowledges paternal symbolic authority. According to Freud, "the fetish is a substitute for the woman's (mother's) phallus which the little boy once believed in

40. Ellen Ullman, interviewed by Scott Rosenberg, "21st: Elegance and Entropy." 1997. http://dir.salon.com/story/tech/feature/1997/10/09/interview/print.html.

41. See mez's site at http://www.hotkey.net.au/~netwurker/.

42. See http://www.scotoma.org/notes/index.cgi?LondonPL.

and does not wish to forego,"[43] but the fetish, formed the moment the little boy sees his mother's phallus, also transforms the phallus:

> It is not true that the child emerges from his experience of seeing the female parts with an unchanged belief in the woman having a phallus. He retains this belief but he also gives it up; during the conflict between the deadweight of the unwelcome perception and the force of the opposite wish, a compromise is constructed such as is only possible in the realm of unconscious thought—by the primary processes. In the world of psychical reality the woman still has a penis in spite of it all, but this penis is no longer the same as it once was. Something else has taken its place, has been appointed as its successor, so to speak, and now absorbs all the interest which formerly belonged to the penis.[44]

The fetish is a transformation of the mother's phallus, whose magical power "remains a token of triumph over the threat of castration and a safeguard against it."[45] As such, it both fixes a singular event—turning time into space—and enables a logic of repetition that constantly enables this safeguarding. As Pietz argues

> the fetish is always a meaningful fixation of a singular event; it is above all a "historical" object, the enduring material form and force of an unrepeatable event. This object is "territorialized" in material space (an earthly matrix), whether in the form of a geographical locality, a marked site on the surface of the human body, or a medium of inscription or configuration defined by some portable or wearable thing.[46]

Although it fixes a singular event, the fetish works only because it can be repeated, but again, what is repeated is both denial and acknowledgment, since the fetish can be "the vehicle both of denying and asseverating the fact of castration."[47]

Slavoj Zizek draws on this insight to explain the persistence of the Marxist fetish:

> When individuals use money, they know very well that there is nothing magical about it—that money, in its materiality, is simply an expression of social relations. . . . on an everyday level, the individuals know very well that there are relations between people behind the relations between things. The problem is that in their social activity itself, in what they are *doing*, they are *acting*

43. Sigmund Freud, "Fetishism," in *Sexuality and the Psychology of Love* (New York: Macmillan, 1963), p. 205.

44. Ibid., p. 206.

45. Ibid.

46. William Pietz, "The Problem of the Fetish," pt. 1. *Res* 9 (spring 1985): 12.

47. Freud, "Fetishism" (above, n. 44), p. 208.

> as if money, in its material reality is the immediate embodiment of wealth as such. They are fetishists in practice, not in theory. What they "do not know," what they misrecognize, is the fact that in their social reality itself—in the act of commodity exchange—they are guided by the fetishistic illusion.[48]

Fetishists importantly know what they are doing—knowledge is not an answer to fetishism. The knowledge that source code offers is therefore no cure for source-code fetishism: if anything, this knowledge sustains it.

To make explicit the parallels, source code, like the fetish, is a conversion of event into location—time into space—that does affect things, but not necessarily in the manner prescribed. Its effects can be both productive and nonexecutable. Also, in terms of denial and acknowledgment, we know very well that source code is not executable, yet we persist in treating it as so. And it is this glossing over that makes possible the ideological belief in programmability.

Code as fetish means that computer execution deviates from the so-called source, as source program does from programmer. Alan Turing, in response to the objection that computers cannot think because they merely follow human instructions, argued:

> Machines take me by surprise with great frequency. . . . The view that machines cannot give rise to surprises is due, I believe, to a fallacy to which philosophers and mathematicians are particularly subject. This is the assumption that as soon as a fact is presented to a mind, all consequences of that fact spring into the mind simultaneously with it. It is a very useful assumption under many circumstances, but one too easily forgets that it is false. A natural consequence of doing so is that one then assumes that there is no virtue in the mere working out of consequences from data and general principles.[49]

This erasure of the vicissitudes of execution coincides with the conflation of data with information, information with knowledge—the assumption that what's most difficult is the capture, rather than the analysis, of data. This erasure of execution through source code as source creates an intentional authorial subject: the computer, the program, or the user, and this source is treated as the source of meaning. The fact that there is an algorithm, a meaning intended by code (and thus in some way knowable), sometimes structures our experience with programs. When we play a game, we arguably try to reverse-engineer its algorithm, or at the very least to link

48. Slavoj Zizek, *The Sublime Object of Ideology* (New York: Verso, 1989), p. 31, emphasis in original.

49. Alan Turing, "Computing Machinery and Intelligence," *Mind* 59. 1950. http://www.loebner.net/Prizef/TuringArticle.html.

its actions to its programming, which is why all design books warn against coincidence or random mapping, since it can induce paranoia in its users. That is, because an interface is programmed, most users treat coincidence as meaningful. To the user, as with the paranoid schizophrenic, there is always meaning: whether or not the user knows the meaning, she or he knows that it regards her or him. Whether or not she or he cares, however, is another matter. Regardless, the fact that the code allegedly driving an interface can be revealed, the fact that source code exists means that the truth is out there. To know the code is to have a form of "X-ray" vision that makes the inside and outside coincide, and the act of revealing sources or connections becomes a critical act in and of itself.[50]

Real Time, This Time

This discussion of the user brings to light an intriguing fact: the notion of user as source, introduced though real-time operating systems, is the obverse, rather than the antidote, to code as source. According to the *OED*, real time is "the actual time during which a process or event occurs, especially one analyzed by a computer, in contrast to time subsequent to it when computer processing may be done, a recording replayed, or the like." Crucially, hard and soft real-time systems are subject to a "real-time constraint"—that is, they need to respond, in a forced duration, to actions predefined as events. The measure of real time, in computer systems, is its reaction to the live—its liveness.

The notion of real time always points elsewhere—to "real-world" events, to user's actions—thereby introducing indexicality to this supposedly nonindexical medium; that is, whether or not digital images are supposed to be "real," real time posits the existence of a source—coded or not—that renders our computers transparent. Real-time operating systems create an "abstraction layer" that hides the hardware details of the processor from application software; real-time images portray computers as un-mediated connectivity. As Realplayer reveals, the notion of real time is bleeding into all electronic moving images, not because all recordings are live, but because grainy moving images have become a marker of the real.[51] What is

50. N. Katherine Hayles develops this theme of revealing codes in *My Mother Was a Computer* (above, n. 17), pp. 54–61. Importantly, some software art projects also complicate and frustrate code as X-ray vision and connection as meaning, such as Golan Levin's *Axis Aplet*.

51. See Thomas Levin, "Rhetoric of the Temporal Index: Surveillant Narration and the Cinema of 'Real Time,'" *CTRL Space: Rhetorics of Surveillance from Bentham to Big Brother*, ed. Thomas Levin et al. (Cambridge, MA: MIT Press, 2002), pp. 578–593.

authentic or real is what transpires in real time, but real time is real not only because of this indexicality—this pointing to elsewhere—but also because its quick reactions to user's inputs.

Dynamic changes to webpages in real time, seemingly at the bequest of users' desires or inputs, create what Tara McPherson has called "volitional mobility." Creating "Tara's phenomenology of websurfing," McPherson argues:

> When I explore the web, I follow the cursor, a tangible sign of presence: implying movement. This motion structures a sense of liveness, immediacy, of the now . . . yet this is not the old liveness of television: this is liveness with a difference. This liveness foregrounds volition and mobility, creating a liveness on demand. Thus, unlike television which parades its presence before us, the web structures *a sense of causality* in relation to liveness, a liveness we navigate and move through, often structuring a feeling that our own desire drives the movement. The web is about presence but an unstable presence: it's in process, in motion.[52]

This liveness, McPherson carefully notes, is more the illusion—the feel or sensation—of liveness, rather than the fact of liveness; the choice yoked to this liveness is similarly a sensation rather than the real thing (although one might ask: What is the difference between the feel of choice and choice? Is choice itself not a limitation of agency?). The real-time moving cursor and the unfolding of an unstable present through our digital (finger) manipulations make us crane our necks forward, rather than sit back on our couches, causing back and neck pain. The extent to which computers turn the most boring activities into incredibly time-consuming and even enjoyable ones is remarkable: one of the most popular computer game to date, *The Sims*, focuses on the mundane; action and adventure games reduce adventure to formulaic, motion-restricted activities. This volitional mobility, McPherson argues, reveals that the "hype" surrounding the Internet does have some phenomenological backing. This does not necessarily make the Internet an empowering medium, but at the very least means that it can provoke a desire for something better: true volitional mobility, true change.[53]

52. Tara McPherson, "Reload: Liveness, Mobility and the Web," in *The Visual Culture Reader*, 2nd ed., ed. Nicholas Mirzoeff (New York: Routledge, 2002), p. 462, emphasis in original.

53. Coming from film rather than television studies and focusing more on applications than phenomenology, Galloway in *Protocol* (above, n. 5) similarly argues that continuity makes websurfing "a compelling, intuitive experience for the user": "On the Web, the browser's movement is experienced as the user's movement. The mouse movement is substituted for the user's movement. The user looks through the screen into an imaginary world, and it makes sense. The act of "surfing the web," which, phenomenologically,

As noted earlier, however, the user is not the only source of change in real time. Real-time images, such as those provided by webcams, make our computer screen seem, for however brief a moment, alive. Refreshing on their own and pointing elsewhere, they make our networks seem transparent and thus fulfill the promise of fiber-optic networks to connect us to the world, as do real-time stock quotes and running news banners.[54] As a whole, these moments of "interactivity" buttress the notion of transparency at a larger level. The source of a computer's actions always stems from elsewhere, because real time makes it appear as though only outside events—user mouse-clicks, streaming video—cause the computer actions. These real-time interactions, which were initially introduced to make computation more efficient, have almost erased computation altogether. Once again, the movements within the computer—the constant regeneration, the difference between the textual representation of a program, a compiled program, a program stored on the hard drive, and the program read-in instruction by instruction into the processor—are all erased. These movements create our spectral interface, without which our machines (we erroneously believe) could not work, and it is these spectral interfaces that allow us to include computers within the broader field of "visual-culture" studies, or screen media. Viewed as the alpha and omega of our computers, interfaces stand in, more often than not, for the computer itself, erasing the medium as it proliferates its specters, making our machines transparent producers of unreal visions—sometimes terrifying, but usually banal imitations or hallucinations of elsewhere.

Demonic Media

Not accidentally, this spectrality of digital media makes our media demonic; that is, inhabited by invisible processes that, perhaps like Socrates' *daimonion* (mystical inner voice), help us in our time of need. They make executables magic. UNIX—that operating system seemingly behind our happy spectral Macs—runs daemons. Daemons run our e-mail, our webservers. Macs thus not only proudly display that symbol of Judeo-Christian man's seduction and fall from grace—that sanitized but nonetheless telling bitten apple—it also inhabits its operating systems with daemons that make it a veritable "paradise lost" (see Fig. 2).

should be an unnerving experience of radical dislocation—passing from a server in one city to a server in another city—could not be more pleasurable for the user. Legions of computer users live and play online with no sense of radical dislocation" (p. 64).

54. For more, see chapter 5 of Chun's *Control and Freedom* (above, n. 10).



Figure 2. FreeBSD mascot.

So why are these daemons called "send mail" and not Satan? Most simply, a daemon is a process that runs in the background without intervention by the user (usually initiated at boot time). They can run continuously, or in response to a particular event or condition (for instance, network traffic), or at a scheduled time (e.g., every five minutes, or at 05:00 every day). More technically, UNIX daemons are parentless—that is, orphaned—processes that run in the root directory. You can create a UNIX demon by forking a child process and then having the parent process exit, so that INIT (the daemons of daemons) takes over as the parent process.[55]

55. The following PERL program, for instance, says hello every five minutes (from http://www.webreference.com/perl/tutorial/9/3.html):

```
use POSIX qw(setsid);
#turns the process into a session leader, group leader, and ensures that it doesn't
#have a controlling terminal
chdir '/' or die "Can't chdir to /: $!";
umask 0;
open STDIN, '/dev/null' or die "Can't read /dev/null: $!";
#open STDOUT, '>/dev/null' or die "Can't write to /dev/null: $!";
open STDERR, '>/dev/null' or die "Can't write to /dev/null: $!";
defined(my $pid = fork) or die "Can't fork: $!";
exit if $pid;
setsid or die "Can't start a new session: $!";
while(1) {
sleep(5);
print "Hello...\n";
}
```

UNIX daemons supposedly stem from the Greek word *daemon,* meaning, according to the *OED*, "a supernatural being of a nature intermediate between that of gods and men; an inferior divinity, spirit, genius (including the souls or ghosts of deceased persons, esp. deified heroes)." A daemon is thus already a medium, an intermediate value, albeit one that is not often seen. The most famous daemon is perhaps Socrates' *daimonion*—that mystical inner voice that assisted Socrates in time of crisis by forbidding him to do something rash. The other famous daemon, more directly related to those spawning UNIX processes, is Maxwell's demon. According to Fernando Corbato, one of the original members of the Project MAC group in 1963:

> Our use of the word daemon was inspired by the Maxwell's daemon of physics and thermodynamics. (My background is Physics.) Maxwell's daemon was an imaginary agent which helped sort molecules of different speeds and worked tirelessly in the background. We fancifully began to use the word daemon to describe background processes which worked tirelessly to perform system chores.[56]

Daemonic processes are slaves that work tirelessly and, like all slaves, define and challenge the position of the master.

The introduction of multiuser, command line processing—real-time operating systems—necessitates the mystification of processes that seem to operate automatically without user input, breaking the interfaces' "diegesis." What is *not seen* becomes daemonic rather than what is normal, because the user is supposed to be the cause and end of any process. Real-time operating systems, such as UNIX, transform the computer from a machine run by human operators in batch-mode to "alive" personal machines, which respond to users' commands. Real-time content—stock quotes, breaking news, and streaming video—similarly transforms personal computers into personal media machines. These moments of "interactivity" buttress the notion of our computers as transparent. Real-time processes, in other words, make the user the "source" of the action, but only by orphaning those processes without which there could be no user. By making the interface transparent or "rational," one creates demons.

It is not perhaps surprising then that Nietzsche condemned Socrates so roundly for his daemon (and similarly language of its attribution of subject to verb). According to Nietzsche, Socrates was

56. "The Origin of the word Daemon." http://ei.cs.vt.edu/~history/Daemon.html. This is why Neal Stephenson, in *Snow Crash* (New York: Bantam, 1992), describes robots or servants in the *Metaverse* as daemons.

himself a daemon because he insisted on the transparency of knowledge, because he insisted that what is most beautiful is also most sensible. Crucially, Socrates' divine inner voice only spoke to dissuade. Socrates introduced order and reified conscious perception, making instinct the critic and consciousness the creator. Perhaps as a sign of the desire for the transparency of knowledge—the reigning of rationality—daemon is also a backronym. Since the first daemon was apparently a program that automatically made tape backups of the file system, it has been assumed that daemon stands for "Disk And Executive MONitor." This first daemon is appropriately about memory: an automated process, stored in memory, that transfers data between secondary and tertiary forms of memory, and that stores the code so that it can be viewed as source. Memory is what makes possible daemons, makes our media daemonic.[57] The questions that remain are: How to deal with these daemons and their alleged source codes? Should these daemons be exorcised, or is this spectral relationship not central to the very ghostly concept of information and the commodity itself?[58]

Code as Re-Source

To answer these questions, let me return to *code as re-source*, for code as re-source enables us to think in terms of the gap between source and execution, and makes an interface a process rather than a stable thing. This gap complicates any analysis of user determination by software: as Matthew Fuller points out, the "gap between a model of a function and its actualities . . . in some cases describes a degree of freedom, and . . . in others puts into place a paralyzing incapacity to act."[59] Reading Microsoft Word's mammoth feature mountain, Fuller compelling argues that the more features offered in an anxious attempt to program the user—the more codes provided—the more ways the user can go astray.[60] Thinking in terms of this gap also means thinking of how information is undead; that is, how it returns over and over again.[61] Second, code as re-source allows us to take seriously

57. See Wendy Chun, "The Enduring Ephemeral, or the Future Is a Memory," *Critical Inquiry* 35:1 (2008): 148–171.

58. See Thomas Keenan, "The Point Is the (Ex)Change It: Reading *Capital*, Rhetorically," in *Fetishism as Cultural Discourse*, ed. Emily Apter and William Pietz (Ithaca, NY: Cornell University Press, 1993), pp. 152–185.

59. Matthew Fuller, *Behind the Blip* (New York: Autonomedia, 2003), p. 107.

60. See Matthew Fuller, "It looks as though you're writing a letter," *Telepolis*. 2001. http://www.heise.de/tp/r4/artikel/7/7073/1.html.

61. See Chun, "Enduring Ephemeral" (above, n. 58).

the entropy, noise, and decay that code as source renders invisible. By taking decay seriously, we can move away from the conflation of storage with memory that grounds current understandings of digital media. Finally, understanding code as re-source links its effectiveness to history and context. If code is performative, it is because of the community (human and otherwise) that enables such utterances to be repeated and executed, that one joins through such citation.

This larger view of the performative has been developed by Judith Butler, who argues against that the felicity of a performative utterance does not depend on the sovereign subject who speaks it. Instead, she argues that what is crucial to a performative utterance's success or failure is its *iterability*, where iterability is "*the operation of that metalepsis by which the subject who 'cites' the performative is temporarily produced as the belated and fictive origin of the performative*."[62] In other words, when a speaker executes a performative utterance, she or he cites an utterance that makes "linguistic community with a history of speakers."[63] What is crucial here is: first, code that succeeds must be citations—and extremely exact citations at that. There is no room for syntax errors; second, that this iterability precedes the so-called subject (or machine) that is supposedly the source of the code; and third, and most importantly, an entire structure must be in place in order for a command to be executed. This structure is as institutional and political as it is machinic.

To make this point, I'll conclude by directly addressing an example that has "haunted" this essay: the current struggle between free software and open source software. At the material level, this disagreement makes no sense, for what is the difference between free and open source software, between linux and gnu linux? Nothing and everything—an imagined yet crucial difference. According to Richard Stallman, the difference lies in their values:

> The fundamental difference between the two movements is in their values, their ways of looking at the world. For the Open Source movement, the issue of whether software should be open source is a practical question, not an ethical one. As one person put it, "Open source is a development methodology; free software is a social movement." For the Open Source movement, non-free software is a suboptimal solution. For the Free Software movement, non-free software is a social problem and free software is the solution.[64]

62. Judith Butler, *Excitable Speech: A Politics of the Performative* (New York: Routledge, 1997), p. 51, emphasis in original.

63. Ibid., p. 52.

64. Richard Stallman, "Why 'Free Software' is better than 'Open Source.'" http://www.gnu.org/philosophy/free-software-for-freedom.html.

The fact that code automatically does what it says is hardly central. The difference between open source and free software lies in the network that is imagined when one codes, releases, and uses software—the type of community one joins and builds when one codes. The community being cited here, worked through, is one committed to this notion of freedom, to coding as a form of free speech. Open source and free software movements are aligned, however, in their validation of "open": freedom is open access.[65]

This emphasis on imagined networks I hope makes it clear that I'm not interested in simply exorcising the spectral or the visual, but am rather trying to understand how its spectrality lies elsewhere. Capturing ghosts often entails looking beyond what we "really" see to what we see without seeing, and arguably, digital media's biggest impact on our lives is not through its interface, but through its algorithmic procedures. Agre has emphasized its mode of capture rather than surveillance—that is, the ways in which computers capture our routine tasks as data to be analyzed and optimized, from shopping to driving a car.[66] Capture re-works the notion and importance of access: one does not need a "personal computer" in order to be captured—all one needs is a RFID tag or to work in a factory or to simply be in debt. Capture also emphasizes tactility and the mundane rather than the spectacular—but also the political and theoretical importance of imagining these invisible networks and technologies that envelope us in their dense thicket of passive and active signals, which can only be imagined.

This necessary imagining, this visually spectral other, underscores the fact that we do not experience technology directly, although to what extent human sensory experience or affect is the end all and be all of computation remains to be seen, for emphasizing human perception can be a way of clinging to a retrograde humanism. This necessary imagining also means that software can only be understood in media res—in the middle of things. "In media res" is a style of narrative that starts in the middle as the action unfolds. Rather than offering a smooth chronology, the past is introduced through flashbacks—interruptions of memory. To return to the parable of the six blind men relayed much earlier, this means that the position of the blind men who know without knowing is not one that can be superseded, but rather the exact position from which we can inter-

65. The question that remains, which is the subject of another paper, is: What does this meaning of open close down?

66. Philip E. Agre, "Surveillance and Capture: Two Models of Privacy," *Information Society* 10:2 (1994): 101–127.

vene and know. Software in media res also means that we can only begin with things—things that we grasp and touch without fully grasping, things that unfold in time—things that can only be rendered "sources" or objects (if they can) after the fact.

Wed 18 Nov!! & 19 Nov – Care & Risk

Morrison & Britton
P4E Chs 8,9

CATALYST
feminism, theory, technoscience

How Can You Spell *Care* with Only 1s and 0s?

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Keywords
practice, feel, mutual aid, scale, noticing, disability, notation, archive, ritual, computation, programmability, quilts, preparation, Black, measurability, materiality, relationship

Ritual for Reading
Be with this text on your computer, smartphone, tablet, piece of paper or whatever other device and pause for a moment. Take the chance to feel the humming of your processor, the ink on the page, the brightness of the screen and the contrast (or not) from this text with your surrounding environment. Choose how you want to enter into the text, will you sit? Lie down? Read out loud? Have the text read to you? Consider if you need a sip of something wet, check in with your body and when ready - begin.

What follows is a conversation between artists and scholars Romi Ron Morrison and Loren Britton who have become friends since they met right before the Covid-19 pandemic. During this time of distance in space and time they have found themselves in multiple conversations over questions of who gets to have a life in the computational presents we are enmeshed within. Their shared interests lie in moments of everyday noticing, attention, and community processes that

Morrison, Romi Ron and Loren Britton. 2021. "How can you spell *care* with only 1s and 0s?" *Catalyst: Feminism, Theory, Technoscience* 7 (2): 1–14.

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form the capacity for greater resistance against algorithmic normalizations that continually foreclose life chances for those most vulnerable.

Recently they have collaborated on the web plugin *Soundings* that is an invitation to willfully interrupt web browsing, and slow down. Both Morrison and Britton are members in the collaborative oracle(s) with Helen Pritchard and Eric Snodgrass, which is a practice of developing poetic technologies that might free ourselves from the computational practices of measurement, categorization, extraction and optimization.

In this conversation there are interruptions by different emojis that pop into the text to introduce references or extra associations for our readers. These interruptions use emojis from Mutant.Tech that work towards an open source and more diverse emoji future.

 ("Infinity". Emoji description: A black and white emoji that is like the number '8' turned sideways to signal "infinity".) This comes up when we reference an idea directly from a scholar or artist.

 ("Moon Viewing Ceremony". Emoji description: A moon in the top right part of the emoji shines over a triangle and some plants.) This comes up when we have a little extra association or lagniappe.

 Dzuk. Mutant Standard. 2017 - 2020. Accessed 29 July 2021. https://mutant.tech/.

 Britton, L., Morrison, R., Pritchard, H., and Snodgrass, E. 2020. *oracle(s) - becoming undone together*. Accessed 29 July 2021. https://vimeo.com/441452535.

 Morrison, Romi Ron and Loren Britton. 2020. S*oundings.* Accessed 29 July 2021. https://altcph.dk/soundings-romi-ron-morrison-loren-britton/.

Loren Britton (LB): Maybe there's no right way to start so I'll pick up and begin. In other writings of yours you have spoken about a re-imagination of programmability that would remake computing towards Black Computational Thought, a term you are developing that refuses computing that stems from white western modernity as its sole archive. In this writing you talk about social formations that exceed measurability and separable difference and stay with proximity, shared risk, accountability and receptivity. I wanted to ask you about what examples you work with that might speak to these computational moments that exceed measurability? I thought this might be a chance to talk about your

project engaging the Freedom Quilts as well as the work you are currently developing for Transmediale that is called Noticing the Preconditions for _____.

Silva, Denise Ferreira da. 2016. "On Difference Without Separability." *32nd Bienal De São Paulo Art Biennial: Incerteza viva*: 57-65.

Chen, Nancy N. 1992. "'Speaking nearby': A conversation with Trinh T. Minh-ha." *Visual Anthropology Review* 8 (1): 82–91. https://doi.org/10.1525/var.1992.8.1.82.

Accountability reminds me of the entry, Dreaming Accountability, from Mia Mingus on her blog, Leaving Evidence, that reads "*What if accountability wasn't scary? It will never be easy or comfortable, but what if it wasn't scary? What if our own accountability wasn't something we ran from, but something we ran towards and desired, appreciated, held as sacred?"* In the context of our conversation here we discuss amongst other things where our sacred accountabilities lie, and how we can develop ways of caring for ourselves and these practices as we make and unmake technoscience.

Mingus, Mia. Leaving Evidence. "Dreaming Accountability," 5 May 2019. Accessed 29 July 2021. https://leavingevidence.wordpress.com/2019/05/05/dreaming-accountability-dreaming-a-returning-to-ourselves-and-each-other/.

Morrison, Romi Ron & Mimi Onuoha. 2021. *Noticing the Preconditions for _____*. Transmediale. Accessed 20 July 2021. https://transmediale.de/almanac/noticing-the-preconditions-for_.

Romi Ron Morrison (RRM): I want to start by acknowledging that I appreciate the ways that you're attending to my interest in programmability particularly as a way to get towards this concept of Black Computational Thought, which feels like a practice of recovering these instances that just often aren't seen as being relevant to computation. I'm really trying to push beyond an acceptance of computation, programmability, software, data, measurement—these things that have become so standard within the lexicon of computation, as self-evident and transparent. Instead, I want to understand the much longer, historical, and residual baggage these practices carry due to the violences they inherit from Post-Enlightenment scientific thought.

Here I am using transparent akin to the *transparent I* evoked in Silva, Denise Ferreira da. 2007. *Toward a Global Idea of Race*. Minneapolis, MI: University of Minnesota Press.

 For a deeper dive into the ways that colonial violences structure scientific thought see Wynter, Sylvia. 2003. "Unsettling the coloniality of being/power/truth/freedom: Towards the human, after man, its overrepresentation—an argument." *New Centennial Review* 3 (3): 257–337. https://doi.org/10.1353/ncr.2004.0015.

So for me, the practice of returning to certain social practices, formations and social technical infrastructures that arise out of Black diasporic movement is an attempt to think about these terms very differently. I believe that these instantiations of Black technical practice begin to illustrate the contours for what a space might look like that is adverse to but subtends the totalizing residue of colonization and Post-Enlightenment scientific discourse.

 I am thinking about residue alongside Lisa Lowe's work on the residual, borrowing from Raymond Williams in Lowe, Lisa. 2015. *The Intimacies of Four Continents*. Durham, NC: Duke University Press.

I'm looking for examples in which people have contended with concepts like measurability or data on very different grounds and so the Negro Motorist Green Book, as a collection of safe places for traveling Black people to access, is an example of a kind of cybernetic system that doesn't mirror the automated control that can be found in Norbert Wiener's imaginary about how a city should function.

 Green, Victor H., 1936 - 1967. *The Green Book*. New York, NY: Victor H. Green & Company. Accessed 29 July 2021. https://digitalcollections.nypl.org/collections/the-green-book#/?tab=about&scroll=10.

In some of the unpublished writings that I shared with you there's a whole essay about Jennifer Light's brilliant work charting the use of military technologies for municipal use. She helps me understand how immensely rewritten American cities became within pervasive cybernetic logics of management in the face of Cold War inspired threat, particularly Black urban rebellions from 1965-1969. So, for me I'm looking at a dialectic for how Blackness produces other types of technologies, not just those that are about capture and control, and The Negro Motorist Green Book as a site of cybernetic operation opens itself for there to be dialogue about trust, and emphasizes the face to face encounters of people making decisions together within a system that still circulates, and is about information flow. Something gets lost or foreclosed when the emphasis is only on optimization and automation. What's fascinating to me about the Green Book are the ways in which the information is sourced, compiled, and then circulated through Victor and Alma Green's social relationships within the U.S. Postal

Workers Union. By having postal workers literally walking their routes, engaging with neighbors, and having relationships with business owners, they would note safe places for Black people, send them to Victor and Alma Green who would compile them and circulate them back out as destinations in the Green Book. Even at the end of every printing of the Green Book there's always a call for more destinations to be added, so it works as a crowdsourced directory that is explicitly about the preservation of Black life across space that is not a neutral container, but is overtly hostile.

RRM: And I think there's also something interesting about this context because the operations are not transparent. The Negro Motorist Green Book isn't written as a survival guide for Jim Crow or white supremacy it's written innocuously as a travel companion. This is a form that engages with different types of opacity that can be productive and necessary. It also complicates these cybernetic operations by foregrounding how people are engaged in shared risk in a way that doesn't remove the flesh and allows for a kind of presence with each other. This emphasis on shared risk and flesh expands the ways that I want to think about programmability, as an engagement, albeit often unequal, that can hold different power relations allowing people to step into other ways of becoming together. I'm trying to think about programmability in that sense, rather than accept the master slave logic within computation, or the determinism of programmability that Wendy Chun has traced to Mendelian genetics and American eugenicists.

 These thoughts on programmability and shared risk are informed by Zakiyyah Iman Jackson's work on symbiosis in *Becoming Human: Matter and Meaning in an Antiblack World*.

 Chun, Wendy Hui Kyong. 2011. *Programmed Visions: Software and Memory*. Cambridge, MA: MIT Press.

 Jackson, Zakiyyah Iman. 2020. *Becoming Human: Matter and Meaning in an Antiblack World.* New York, NY: NYU Press.

LB: In your work on the Freedom Quilts which were quilts carefully constructed and stitched to direct fugitive Black people escaping slavery in the US South towards freedom; you have written about the quilting bees on plantations where the language, pattern, stitching and constant reformulation / modulation of these quilts would be re-inscribed through their community production. In this way these quilts are highly specific, contextual land cues, a culturally situated practice towards freedom. I wonder if you want to say more about this, also perhaps on the relation of the materiality of these quilts to the land that they were present on?

 These practices of language, patterning, contextual specificity and slippage relate to a project I have worked on with my collaborator Isabel Paehr called Hacking Concrete. This work engages material instability to make concrete 'open' even though it is often a material that can foreclose access to spaces and colloquially is understood to be solidified.

 MELT (Loren Britton & Isabel Paehr). 2021. Meltionary. "*Hacking Concrete*." Accessed 29 July 2021. http://meltionary.com/hackingconcrete/start.html.

RRM: Yeah, so this is really becoming important and crucial for me, this practice of trying to hold the productive tension of seemingly disparate things together—which reads like an autobiography at this point. My relation to scholarship begins within Black Studies and Gender Studies, centering Queer Black Feminist Theory in particular, and then extended to experimental practices of design, architecture, and geography. And I now carry those practices through to consider the effects of computation on the production of space while still reading from a Queer Black Feminist lens. This allows me to interrogate and denaturalize certain computational practices as self-evident or already fully understood.

I'm trying to better understand the ways that theories of computation, which are never fully deracinated or separate from Post Enlightenment thoughts of empiricism, measurability, or calculation, have a huge effect on how we understand land and space. Both in the managing and organizing of space but more importantly in the types of spatial imaginaries that computational logics allow. Jennifer Light's book *From Warfare to Welfare: Defense Intellectuals and Urban Problems in Cold War America* does a great job of this. It shows the ways that military actors like the RAND Corporation, essentially trying to diversify their bottom line, introduce certain computational informatics and cybernetic architectures into municipal governments. There is a clear way in which the city as a whole is being reimagined from the 1950s to the 1970s through large scale logistics management, urban planning, data capture, and predictive analytics.

 Light, Jennifer. *From Warfare to Welfare: Defense Intellectuals and Urban Problems in Cold War America*. Baltimore, MD: John Hopkins University Press.

So I want to pay attention to how modern ideas about measurability, control, and conquest inform cybernetics as a new science of management, and produce specific typologies of urban space. Alternately, The Negro Motorist Green Book and the Freedom Quilts as Black technological practices produce different spatial and computational logics. So I'm deeply curious about the Black production of

space and technology that come from things like The Negro Motorist Green Book and the Freedom Quilts. Coming back to your question about the Freedom Quilts, the locations of the quilting bees on plantations are essential sources for information gathering that have a great deal of geographical and topographical specificity because people are literally having to navigate the landscape in order to then tell others about what resources they might find and encode that data through detailed stitching patterns. These patterns and ties were also used to plot mile markers so that people can plan how far they can travel within a day. Maintaining this contextual basis for place expands computation beyond the glossy and frictionless descriptions often assigned to it where we simply input and output information, but stays with the troubled materiality in which computation is co-constituted through the specificity of place, social engagements and shared agreements. This approach to computation is sociogenic and gives a broader understanding for how computation is constituted otherwise at different times, by different people, in different places. And so I try to hold on to these instantiations because I think they're particularly helpful now, as we're trying to engage in rich critical discourse about computation and violence, and rather than trying to speak to the violences within the same language that has given birth to them, I'm trying to find these other lexicons or dialogues because I think they are a lot more expansive.


Wynter, Sylvia. 2013. "Towards the Sociogenic Principle: Fanon, identity, the puzzle of conscious experience, and what it is like to be 'black'." *National Identities and Socio-Political Changes in Latin America,* 46-82. https://doi.org/10.4324/9781315052717-8.

LB: Totally. I am taking note of the stitching patterns and the materiality of these quilts as socio-technical objects, the patterns or ties for example that would signal distance. These questions of materiality, spatial relationships and the capacity for negotiation towards movements towards freedom reminds me of my work as MELT with Isabel Paehr on a project called Rituals Against Barriers. In this project we have been developing ritual and material based experiments that are inspired by practices of survival in oppressive contexts and take the form of ritual practice. We think about how rituals can invite movements away from normativities and can allow entry into practices like refusing to ignore a feeling, refusing ableist spaces, and refusing to speed up even if that is the normalized tempo. In that work, rituals are moments of joy, of refusal, collective practice, of uncomfort and unlearning.


Although in this work rituals are upheld as a space for practicing towards liberation and pushing back against ritualized oppressions, we also attend to the ways in which oppression becomes normalized by dominant and oppressive culture. In her book *Dark Matters: On the*

Surveillance of Blackness, Simone Browne points to "the ritualized practices and trauma of white supremacy" (105) which is another kind of structural violence which must be refused.

Browne, Simone. 2015. *Dark Matters: On the Surveillance of Blackness*. Durham, NC: Duke University Press.

MELT (Loren Britton & Isabel Paehr). 2021. Meltionary. "*Rituals Against Barriers*." Accessed 29 July 2021. http://www.meltionary.com/meltries/a.html.

We have been looking into the history, present, possibilities and devastating failures of disability justice/activism within the United States and in Germany. We have thought a lot about these entanglements of place, space, access and technology nearby Aimi Hamraie in their book *Building Access: Universal Design and the Politics of Disability.* In the chapter, Sloped Technoscience, they tell the story of how disabled people and their accomplices drove around in Berkeley, California in the 1970s and took sledgehammers to inaccessible streets in order to crumble curbs, repour cement and physically make the streets more accessible. Reflecting on this history, they ask: "If we take a sledgehammer to the seemingly concretized sidewalks of disability rights history, what layered sedimentations of resistance do we find below? (Hamraie, 103)". They share images of concrete pieces that are artifacts or evidence of hacking the city and making material interventions towards other more accessible worlds. It's an example from a different archive, but it resonates with how the materials of socio-technical systems say so much about their context, like you were talking about.

 A Ritual from *Rituals Against Barriers*: Ritual for tending to the "not perceivable": From *Undrowned* by Alexis Pauline Gumbs, spend time with the question: "What becomes possible when we are immersed in the queerness of forms of life that dominant systems cannot chart, reward or even understand?"

Hamraie, Aimi. 2017. *Building Access: Universal Design and the Politics of Disability*. Minneapolis, MI: University of Minnesota Press.

Gumbs, Alexis Pauline. 2020. *Undrowned: Black Feminist Lessons from Marine Mammals.* Chico, CA: AK Press.

RRM: Yeah, absolutely. It makes me think about a few things. There's a way in which Ruth Wilson Gilmore describes abolition as a radical presence, and I think that has a lot of relevancy here. Especially the power and devastation that certain imaginaries concretize. Which is why the hack is super enticing and provocative, and also attuned to a close proximity for understanding needs and desire. Urban

planning and programming don't often come out of histories in which they're anticipating that people will imbue new uses within their creations, the master plan is not something that's very adaptable because it's setting 10-year goals linearly into the future that are self-contained. What is imagined in the abstract as ideal or commonsensical becomes written, codified, and then materialized. And practices of urban design and software design both tend to imagine ideal universal users. What's so lovely about the materiality in your example is that it breaks this imagined standard and forces a kind of focus around quotidian daily practices, even rituals, that are about what it means to notice your own desires and needs, and then to be poised to take the sledgehammer and edit the curb. This is not arbitrary, it's not a random act of destruction, it is considered and is indeed a relationship to place. When I'm trying to search for examples that practice computation on different grounds they almost always exhibit that kind of everyday attentiveness to refute the universalizing claims of computation as a transcendent cloud.

Gilmore, Ruth Wilson. 2007. *Golden Gulag: Prisons, Surplus, Crisis, and Opposition in Globalizing California*. Oakland, CA: University of California Press.

Instead I'm trying to stay in the dirt and the soil, walking the ground. So thinking about the curb edits, the postal workers walking their daily routes, the plantation grapevine that then feeds information to quilting bees, they are practices that engage the materiality of shared risk to intervene in everyday desires in excess of the present.

LB: Yes exactly, it reminds me of a quote from Remi M. Yergeau that "Bodies are not for hacking, bigotry is.!" So, I understand your work on Rituals of Black Fugitivity: directly picks up on practices of ritual as they meet Black diasporic practices and I wonder if you want to speak more to this? As I've thought about them, rituals have been about accessibility creating practices, so rituals are something that can be enacted in community or alone that enact the possibility of a specific sense of space within uninhabitable conditions.

Yergeau, Remi M. 2014. "Disability Hacktivism." *Hacking the Classroom: Eight Perspectives,* edited by Jentery Seyers & Mary Hocks. Last accessed 29 July 2021. http://web.uvic.ca/~jentery/hacking/.

Morrison, Romi Ron. 2018. "Rituals of Black Fugitivity:." Accessed 29 July 2021. https://elegantcollisions.com/rituals-of-black-fugativity-p.

RRM: For me ritual is a very spatial-temporal relationship. Before sitting in front of my altar, talking to ancestors, or writing I need to prepare myself and my space. I can't sit in front of my altar if my room is messy, or I haven't washed my face yet. The preparation and the making of space is part of the ritual. It becomes vital to prepare the space to be ready for the ritual to show me what it needs to show me, otherwise I'm not engaged in ritual. So, there's a kind of preparation for me, that's a part of ritual that in turn produces its own space-time.

LB: I mean, yeah. I really appreciate the way that you come back to preparation. The distinction between habit and ritual is something that I've been considering as well, when is something pathologized and when not, who decides? What are the stakes of the amount of preparation that must be enacted in order to withstand great precarity? And who has access to that time, space, resources, and community practice to be well prepared?

These practices that affirm agency remind me of Rosemarie Garland-Thomson's concept of misfitting as a shifting spatial and temporal relationship that affirms the agency and value of disabled peoples' experiences. She discusses how misfitting avoids genericizing disabled experience and undoes myths around dependency/ interdependency. Misfitting time-space-experience could then be a way to move within and care for our brilliant knowledges and challenge worlds that were not built for us. Thinking about sensory experiences, there are quite some preparations needed to withstand, or to find ways to be okay in shifting contexts that are specific to my body-mind. I very much relate to what you're saying.

> Bodymind is a materialist feminist disability studies concept from Margaret Price and has been taken up by multiple disability studies scholars to discuss the enmeshment of the body and mind which have been constructed as separate due to Cartesian dualisms of western philosophy. Theorists I follow who engage this concept include: Margaret Price, Sami Schalk and Aimi Hamraie.
>
> Price, Margaret. 2015. "The Bodymind Problem and the Possibilities of Pain". *Hypatia*, 30 (1): 268–284. https://doi.org/10.1111/hypa.12127; Schalk, Sami. 2020. *Bodyminds Reimagined: (Dis)ability, Race, and Gender in Black Women's Speculative Fiction.* Durham: NC: Duke University Press.; Garland-Thomson, Rosemarie. 2011. "Misfits: A feminist materialist disability concept". *Hypatia* 26 (3): 591–609. http://doi.org/10.1111/j.1527-2001.2011.01206.x.

RRM: Yeah I definitely feel that, even in things that we understand to be really phenomenological or stochastic, like improvisation within jazz music or

polyrhythmic West African drumming ensembles. There is this great conversation between Arthur Jafa and Steve Coleman and they're speaking about jazz and this misnomer in which people assume improvisation is just pure unbridled creation and that you start from nothing, and it's like no there is so much preparation that goes into constantly playing a scale over and over again so that your approach to the scale becomes instinctual. You're trying to build an instinctual muscle memory for having a lot of notes at your disposal to play with. But the actual improvisation isn't just an act of randomization; it's an act of preparation that allows for expansive forms of emergence to happen. That feels very akin to me when thinking about ritual to respond to precarity that is not about prediction. Because prediction is about trying to eliminate flux, change, and what can't be conceived at the time. It is about the limiting of options and limiting the ability of other options to arrive. I think Kara Keeling speaks to that really well in *Queer Times Black Futures*. Specifically, in corporate futures planning…

 Keeling, Kara. 2019. *Queer Times Black Futures.* New York, NY: NYU Press.

 Jafa, Arthur. 2018. "Love is the Message: An Evening with Arthur Jafa." Accessed 30 July 2021. https://www.youtube.com/watch?v=yOYd_IAPIeo.

LB: Talking about Shell I think.

RRM: Right! And I love the ways that she is bringing in thoughts on antifragility and practices within Black life that are made to be inherently antifragile, the ways that she talks about James Snead's work on repetition, modulation, coverage, and the cut, as ways to build accidents into the process and repetition as that kind of preparation, the ability to return to the altar to make space for feeling composed enough to meet the precarity or unpredictability of the day rather than attempting to map out any potential threat at the beginning of the day and then avoid them as much as possible. There is a very different gesture there. So I think there is a lot here to consider towards Black Computational Thought that invites ritual practice rather than prediction when trying to mitigate risk by making space for it as a relation.

 James Snead, "Repetition as a Figure of Black Culture," in *Black Literature and Literary Theory*, ed. Henry Louis Gates and Sunday Ogbonna Anozie (New York; Routledge, 1984), 59.

LB: So, relatedly, we read together many great texts in prep for our conversation, like Michelle Murphy who talks about phantasmagrams. Phantasmagrams as Murphy introduces them are about the affective or emotional attachment to

enumeration or the emotional quality of numeracy and all of the things that fall outside of numbers themselves that are relatable. In her text she talks about Gross Domestic Product (GDP) and care labor that is excluded from the GDP. I'm curious about your interest in phantasmagrams as expressions of the excess of numeracy as they sit alongside Black Computational Thought? Do you see these as being in relation?

 Murphy, Michelle. 2017. *The Economization of Life*. Durham, NC: Duke University Press.

RRM: Yeah, so I think phatasmagrams allows for me to bring in certain forms of Black Studies scholarship that would maybe otherwise be seen as really outside the bounds of any theory of computation or data. I'm hesitant to make a parallel between phantasmagrams and Black Computational Thought even though I think there are some shared components. This relationship to numeracy and what lives in excess of the digit as well as our attachments to them, are both components where there is a shared investment between how Michelle Murphy is talking about phantasmagrams and how I am trying to think about Black Computational Thought. So what phantasmagrams do for me is they speak to the often removed affective attachment to a metric. Often critiques of big data or algorithmic bias is about "oh these just aren't the right numbers, or these aren't the right inputs and that's why we're getting bad junk data out" it still treats the numeral, the integer, as if it is just a cold quantifiable thing. It's just computed, it's not something that we also have this libidinal attachment to. So what I like about Michelle Murphy's text is that, she's looking at things like the rise of population and its relation to GPD in this post Malthisian natural scarcity world and is really explicitly talking about the forms of violence that then become justified because of this affective attachment to GDP as an indicator of the literal health or potential progress of the nation. So, it is a really provocative example to bring up the affective attachment to numbers or enumeration, particularly at a large scale when talking about population or GDP as nascent concepts for nation building.

So, for me that is really helpful when looking back at Jennifer Light's work. Light details how so many cybernetic interventions that happened within municipalities failed in terms of making cities more efficient. She notes how technicians acknowledged their failures, but the affective attachment to the imaginary of control over large urban processes still overwrote that recognition of failure. So phantasmagrams are really important because they speak to the affective attachment to enumeration as fact or truth. That is really useful for intervening in critiques of technology and opens the space for me to bring in the work of scholars like Sylvia Wynter and talk about sociogeny and the ways in which storytelling, language, and the imaginary are not separate from what we think of as empirical, scientific, or measurable. It is a hybrid. This situates computation so that we can tell different stories about it.

LB: I had a question about forms of attention and sensemaking, that might be related to other stories we could tell about computing, we talked already about modulation or vibration. And I'm curious to think with you about other sorts of sense-based practices like smell or things that exceed the written, which seems like something that you might be interested in, but we didn't directly talk about. It's also something I'm quite interested in. And then I also had a question about scale, which is maybe a different direction.

RRM: This project that I've been working on with Mimi Onuoha for Transmediale is pretty explicitly about a kind of sensemaking, which we've been talking about as noticing. For us, noticing is about giving a lot of attention to small, very ordinary things that hold a lot of potential. And that can be instructive for recognizing a desire for how we might want to live or be together, that's not always allowable in the present.

>  Here I am thinking with José Muñoz's work in *Cruising Utopia: The Then and There of Queer Futurity* in which desire is essential for reaching towards the horizon of queer futurity. Desire becomes a manner of knowledge to seek what may not be allowable now but needs to exist. It is a practice of beginning to conjure.

And so the content is essentially all about a particular form of sensemaking. And a sensemaking for paying attention to what otherwise might fall out of memory. In particular, Mimi and I both share a kind of investment in Blackness as a political analytic to frame the things that we notice. Repetition and ritual return here to be able to meet risk from a different vantage, it's not about prediction, but it's about building up a practice of making space for what can be accounted for. So in this piece we each separately notice an expression of care from our daily lives, document it, and make an interpretation of it in the form of a short media work. Then through the act of correspondence we share those works with each other to iterate upon them, and then send it back to repeat again. So that, that original thing is occluded, but what we get instead are different permutations based on interpretation. It makes interpretation an explicit part of data collection and also includes my relationship with Mimi as an important part of that interpretation. It is relational. This kind of shared correspondence reminds us of the things that a single person can't hold within their attention and is an attempt to recognize the value in building relational forms of sensemaking.

LB: Yes, definitely. I like how you and I in our practices make sensemaking at different scales. There must be structural change but also there are ways in which change is made every day. I don't know if you know the collective called SINS INVALID? They're based in California. I'm a huge fan of their work, they are a BIPOC led Disability Justice collective, and they center disabled people focusing

as they say on "An Unashamed Claim to Beauty in the Face of Invisibility". They share a lot of resources about disability justice on their website and have many performances also online to view. They have this amazing podcast titled "Into the Crip Universe" that explores the intersection of disability justice and climate chaos. In this podcast Patty Berne, one of the founders of Disability Justice, talks about all the ways that in the midst of climate chaos the kind of support that might be needed. In the episode, "We Love like Barnacles", Berne poses the question "When climate disaster strikes, will able-bodied allies be there for us?". She goes on to consider what kinds of small mutual aid we might need to practice like: having a text message group to check in, sharing fridge space and making sure that people have access to their meds. These small scale mutual aid initiatives really create access for people right because when we're talking about accessibility, we know that we can't always rely on larger structures to care for us so we practice caring for each other.

 SINS INVALID: Into the Crip Universe Podcast Episode 1. "We Love like Barnacles." 2020. Accessed 29 July 2021. https://www.sinsinvalid.org/podcast/2020/10/16/episode-1-foundations-of-climate-justice-and-disability-justice.

Ritual for Closing

Allow these words to wash over you. Release the pressure to capture them and stow them away. Instead allow yourself to be a sieve. Give attention to what bubbles up. Those errant turbidities that collect in patches. These are the beginnings for new ground. Moving takes practice and we begin with a question. What if we cared for each other with an ease that holds shared risk?

Author Bios

Romi Ron Morrison is a Black queer non-binary artist, researcher, and educator. Their work investigates the personal, political, ideological, and spatial boundaries of race, ethics, and social infrastructure within digital technologies. Using maps, data, sound, performance, and video, their installations center Black Feminist technologies that challenge the demands of an increasingly quantified world—reducing land into property, people into digits, and knowledge into data.

Loren Britton is a white trans* interdisciplinary artist and researcher tuning with practices of Critical Pedagogy, Trans*FeministTechnoScience and Disability Justice. Engaging the queer potential of undoing norms they practice joyful accountability to matters of collaboration, computing, trans*gender politics, Black Feminisms, and transformations. They love slowness, reading out loud, following non-linear processes and experimenting towards greater accessibility.

26 Nov – Materials & (Data)Structures

Course Check-In
Nakamura
P4E Chs 10,11

CODESHIP Course Check-In

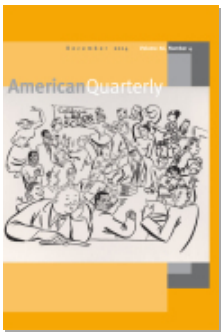
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Indigenous Circuits: Navajo Women and the Racialization of Early Electronic Manufacture



Lisa Nakamura

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Indigenous Circuits: Navajo Women and the Racialization of Early Electronic Manufacture

Lisa Nakamura

Donna Haraway's foundational cyberfeminist essay "A Cyborg Manifesto" is followed by an evocative subtitle: "An Ironic Dream of a Common Language for Women in the Integrated Circuit." She writes, "The nimble fingers of 'Oriental' women, the old fascination of little Anglo-Saxon Victorian girls with doll's houses, women's enforced attention to the small take on quite new dimensions in this world. There might be a cyborg Alice taking account of these new dimensions. Ironically, it might be the unnatural cyborg women making chips in Asia and spiral dancing in Santa Rita jail whose constructed unities will guide effective oppositional strategies."[1] In this passage Haraway draws our attention to the irony that some must labor invisibly for others of us to feel, if not actually *be*, free and empowered through technology use: technoscience is, indeed, an integrated circuit, one that both separates and connects laborers and users, and while both genders benefit from cheap computers, it is the flexible labor of women of color, either outsourced or insourced, that made and continue to make this possible.[2]

Haraway's critical perspective on digital technology's possibilities and opportunities for intersectional feminism expresses itself in this essay by standing readerly expectation on its head. She wants to remind us forcefully of digital technology's costs as well as its benefits. This piece is often read as a celebration of a newly extended and enhanced cyborg body, one made more powerful by technology, an understandable result given that some of the piece's most memorable quotations, such as "I'd rather be a cyborg than a goddess," imply a kind of transcendence through technology use that appeals to the digitally identified.

Haraway's Marxian insistence on materiality rather than just virtuality in the "Cyborg Manifesto"—on the gendering and racializing of bodies as well as on computer hardware itself—anticipated many of the concerns at the center of media archaeology and platform studies in the twenty-first century. Tiziana

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Terranova, whose focus on the Internet as a site of digital labor brings us back to the material realm of bodies and exploitation, extends this interrogation into the way that labor is commodified and extracted, often without compensation for the laborer, within digital culture.[3] For Haraway, the women of color workers who create the material circuits and other digital components that allow content to be created are all integrated within the "circuit" of technoculture. Their bodies become part of digital platforms by providing the human labor needed to make them. Really looking at digital media, not only seeing its images but seeing *into* it, into the histories of its platforms, both machinic and human, is absolutely necessary for us to understand how digital labor is configured today.

How can we take up Haraway's injunction to be guided by women of color's labor in the digital industries to form "effective oppositional strategies?" Woman of color feminism's theoretical framework has much to offer digital media studies, particularly in light of the emphasis on the physical and material aspects of computing that media archaeology has brought to the field. The media archaeologist Wolfgang Ernst makes an eloquent case for studying the specificity of media artifacts through their histories and forms of production.[4] When we look at the history of digital devices, it is quite clear that the burden of digital media's device production is borne disproportionately by the women of color who make them. "Ethical hardware" organizations like the "Raise Hope for Congo," conflict-free cell phone and laptop campaigns, and gamic texts like Molleindustria's "Phone Game" invite us to question and challenge the human cost of computing and mobile telephony.

References to "nimble fingers" as a digital resource appear in many accounts of how women of color were understood by and actively recruited to work in the electronics industry in this period. As Jefferson Cowie writes, the "nimble fingers" phrase was applied to Latino women working in maquiladoras for RCA and other electronics firms, including Fairchild. According to Karen Hossfeld, by the eighties in Silicon Valley, electronic assembly had become not just women's work but women of color's work.[5]

This essay focuses on a group of women of color who are almost never associated with electronic manufacture or the digital revolution—Navajo women. The archive of visual materials that document the history and industrial strategy of Fairchild Semiconductor, the most influential and pioneering electronics company in Silicon Valley's formative years, documents their participation through visual and discursive means, albeit never in their own voices.[6] Fairchild's internal documents, such as company newsletters, and its public ones, such as brochures, along with Bureau of Indian Affairs press

releases and journalistic coverage by magazines such as *Business Week*, paint a picture of Navajo women workers as uniquely suited by temperament, culture, and gender as ideal predigital digital workers.[7]

My reading of these materials reveals how Fairchild produced a racial and cultural argument for recruiting young female workers in the electronics, and later digital device production industries, from among the Navajo population. As Cowie writes of young Mexican women working for RCA: "Management's standard explanation for its preference for young female workers typically rested on the idea that women's mental and physical characteristics made them peculiarly suited to the intricacies of electrical assembly work."[8] Similarly, the hundreds of Navajo women who worked at the Fairchild semiconductor plant in Shiprock, New Mexico, on Navajo land were understood through the lens of specific "mental and physical characteristics" such as docility, manual dexterity, and affective investment in native material craft. The visual rhetoric that described their unique aptitude for the work drew heavily on existing ideas of Indians as creative cultural handworkers.

A close examination of how Navajo women's labor was exploited as a visual and symbolic resource as well as a material good shows us how indigenous women's labor producing circuits in a state-of-the-art factory on an Indian reservation came to be understood as affective labor, or a "labor of love." In her work on women's affective labor in digital media usage, Kylie Jarrett uses the term *women's work* "to designate the social, reproductive work typically differentiated from productive economics of the industrial workplace."[9] A 1969 Fairchild brochure celebrates Indian women circuit makers as culture workers who produced circuits as part of the "reproductive" labor of expressing Navajo culture, rather than merely for wages.

The Anomalous Narrative of Indigenous Workers at Fairchild Semiconductor

The story of Fairchild's plant on Navajo land is not part of a narrative of development that fits comfortably into the history of the digital industries. Though documentary histories of Fairchild abound, and no history of Silicon Valley fails to mention the company, the Shiprock plant is rarely discussed in these accounts, or at best appears as a footnote or a brief mention or digression from the story of outsourcing production to Southeast Asia.[10] The company was regarded as a pioneer because of its willingness to take risks, to invent new manufacturing processes, and to venture onto foreign shores in search of cheap labor, an act that "helped to launch the PC revolution, which begot

the commercial Internet, which begot everything else."[11] Fairchild's trajectory of sourcing labor domestically from female workers of color in the sixties, to outsourcing in the seventies, and eventually to offshoring in Asia was a path followed by many other electronics companies.

Since Fairchild was one of the first chip manufacturers to outsource production to Asia, this is recognized as an epochal event in the history of computing, an innovation that permitted the remarkable growth of the electronics and eventually the computing and personal digital device industry. However, the history of offshore outsourcing to Asia runs parallel with chip fabrication projects *within* and *across* US borders, specifically on Navajo land and in Mexico, respectively. In 1964 the Bracero Program officially ended, and in 1965 the Border Industrialization Program (BIP) began on the US–Mexico border. By 1973 Fairchild and other semiconductor manufacturers were operating plants in Mexico under this program, in addition to plants in Singapore, Hong Kong, and Seoul.

In 1962 Charlie Sporck, a top executive at Fairchild Semiconductor and, later, president and CEO of National Semiconductor, two of the largest and most important manufacturers of integrated circuits, knew that the industry was "running into limitations as to where we could sell the product."[12] The majority of the "product" was being sold to the military, and Sporck realized that Fairchild needed to reduce labor costs in order to break into the "vast consumer market out there" for electronic devices, such as calculators, games, and eventually personal computers. In an interview recorded as part of the "Silicon Genesis: An Oral History of Semiconductor Technology" project, Sporck recalls how the quest for cheaper labor and lower overhead drove Fairchild to open a plant in Hong Kong, a move that pioneered electronics manufacture outsourcing to this and other locales in Southeast Asia, Mexico, and Southern California.

However, the interview takes an odd turn. As Sporck warms to his work of explaining how Fairchild started the "mad rush into Southeast Asia by all companies" in the sixties, the interviewer interrupts, asking, "Well, did you also go to Shiprock, New Mexico to the Indian reservation?" Sporck replies, "Yeah, that's not one of the . . ." The interviewer continues, "I noticed you didn't bring that up." Sporck replies, "No, we did, that was at the, just about the time we went to Portland, Maine. We looked elsewhere in Shiprock, looked like a possibility and we did locate down there. It never worked out, though. We were really screwing up the whole societal structure at the Indian tribe. You know, the women were making money and the guys were drinking it up and it was a failure."

Though Sporck depicts the plant as a "failure," it was depicted as a tremendous success during its years of operation. In fact, the archive of materials about the plant depicts it as doing well because it was *in line* with the "societal structure of the tribe," rather than in conflict with it.

Insourcing on the Reservation: Fairchild's Move to Indigenous Territory

Fairchild opened its state-of-the-art semiconductor assembly plant on the twenty-five-thousand-square-mile Navajo reservation in Shiprock, New Mexico, in 1965. The plant grew from a pilot project employing fifty-five people to a thirty-three-thousand-square-foot integrated-circuit manufacturing facility where hundreds of Navajo women and some men worked on circuit assembly between 1965 and 1975; while accounts as to the exact number of Navajo employed vary, in 1966 Fairchild was the "largest of several electronics plants now located in Indian areas,"[13] and "at its height, the plant provided work for more than 1,000 Navajos. . . . Fairchild became the largest industrial employer in New Mexico and the largest employer of Indians in the country."[14] The plant, which operated twenty-four hours a day, was owned by the Navajo Tribal Council and leased by Fairchild for $6,000 a month.[15] It boasted a very low failure rate—5 percent, in contrast to rates in the twentieth percentile at other plants—and received several awards for its innovative practices.

As the historian Colleen O'Neill writes, "In 1974, prior to its closing, Fairchild employed 922 Navajos, most of whom were women. Fairchild was one of the largest employers of Navajo labor on the reservation, second only to public sector employees, including the Bureau of Indian Affairs and the Navajo nation."[16] In most histories of Silicon Valley, domestic manufacture is assumed to have given way to foreign manufacture starting in the sixties, when the first large plants in Asia and Mexico opened. Widening the perspective on outsourcing to include insourcing practices like the production of semiconductors on Navajo land provides a valuable perspective from which to view the material culture of computing.

Reservations provided spaces of exception to US laws on minimum wage; in this way they were like foreign countries, but in other ways American mythologies around Indianness gave these workers a desirable identity as culturally foreign yet familiar. Likewise, American Indian history tends to include the Fairchild plant as an example of failed economic development or as part of the history of the American Indian Movement's protests, but does not connect it to digital culture or history.

Fairchild's Shiprock plant was far more than just an outlier. Instead, the company represented it as a new and innovative model for cheap domestic electronics manufacture: insourcing rather than outsourcing. In Fairchild's promotional materials and in journalistic accounts, Navajo workers were always represented as different from white workers, as possessing innate racial and cultural traits that could be enhanced or rehabilitated to produce chips accurately, quickly, and painlessly. The visual archive of promotional materials, brochures, annual reports, and press releases about the Fairchild Shiprock plant and its workers reveals how electronics assembly work became both gendered and identified with specific racialized qualities. Analysis of documents from the period that describe the plant's remarkable early success and its eventual closure in 1975 reveal potent and durable claims and beliefs about gender, race, and particular labor *styles* that would quickly be appropriated to describe the Asian women workers who eventually replaced them.

How and why did the most advanced semiconductor manufacturer in the world build a state-of-the-art electronics assembly plant on a Navajo reservation in 1965? A 1969 Fairchild news release explains that the plant was "the culmination of joint efforts of the Navajo People, the U.S. Bureau of Indian Affairs (B.I.A), and Fairchild." Though cheap, plentiful workers and tax benefits helped lure electronics companies to the reservation, Navajo leadership helped push the project forward; Raymond Nakai, chairman of the Navajo Nation from 1963 to 1971, and the self-styled first "modern" Navajo leader, was instrumental in bringing Fairchild to Shiprock. He spoke fervently about the necessity of transforming the Navajo as a "modern" Indian tribe, and what better way to do so than to put its members to work making chips, potent signs of futurity that were no bigger than a person's fingernail? The incongruity of this form of labor—the creation of the most advanced devices the world had yet known, tiny bits of matter that could tell a satellite where to point, by women who were conceived of as irredeemably primitive—was not lost on the tribes themselves.

In his address dedicating the newly built Shiprock plant, Nakai said, "It is a brilliant chapter that we write here in the dedication of this magnificent plant. It signals the real and early industrialization of the Navajo reservation. It marks the advancement of the Navajo nation from an Agrarian Nation to an Industrial Nation."[17] This attempt to rebrand the Navajo as modern through their labor within electronics manufacture seems designed to counter the notion of Indians as "suffering from a racial inability to advance," as Philip Deloria puts it.[18] This new notion of the Navajo as "Industrial" produced a complicated identity whose formation relied on the idea that the tribe could

be modern, even hypermodern, precisely as a result of being distinctively Indian. Indian-identified traits and practices such as painstaking attention to craft and an affinity for metalwork and textiles were deployed to position the Navajo on the cutting edge of a technological moment precisely because of their possession of a racialized set of creative cultural skills in traditional, premodern artisanal handwork.

The building of the Shiprock plant was very much in line with the 1961 Task Force on Indian Affairs recommendations, which urged that reservations attract light industry as part of the "key to the economic and social competency program," which would "increase Indian economic self-sufficiency, and eventually terminate all services from the federal government to Native Americans."[19] As Peter Iverson writes, "The Navajo sought to lure other large-scale industry with cheap land leases, favorable construction arrangements, and a trainable work force. Two major firms accepted the Navajos' invitation: Fairchild Semiconductor and the General Dynamics corporation."[20] In turn, Fairchild benefited from a $700,000 loan from the Navajo to finance plant build-out, free equipment from the BIA supplied from "federal excess property sources," a very low hourly wage, freedom from real estate taxes, and funding for training programs supported by Department of Labor.[21] These factors all mattered, but in the end, product quality was what kept the plant in business and allowed it to expand.

Race and Gender as Digital Resource: Navajo Women as Early Creative Class Workers

Semiconductor manufacture was performed using a microscope and required painstaking attention to detail, excellent eyesight, high standards of quality, and intense focus. Not all who started to work there continued—as Jim Tutt, a Navajo process engineer who worked at Fairchild until 1974, put it, "It was tedious work under a microscope. They couldn't handle it, some of them, [because they had to spend] so many hours a day looking at it."[22] Despite these daunting conditions, the hundreds of Navajo women who stayed on excelled at this work, and the industrial discourse produced by and about the plant attributed its success to the female gender of its workers as well as Indian racial traits. At Fairchild, the preference for women assembly workers was so strong that men were effectively shut out of the vast majority of jobs at the Fairchild plant, and Nakai had to work hard to pressure the company into hiring more men at the plant.[23]

A Fairchild company newsletter published a story titled "Fairchild Shiprock: A Success Story," citing the "tremendous job" that the Navajo "ladies," pic-

tured hovering over microscopes, were doing assembling integrated circuits. To explain the plant's success, the article equates creative cultural skills such as weaving and silversmithing with circuit building. Both Fairchild's corporate newsletter and *Businessweek* credited plant manager Paul Driscoll with discovering and exploiting the "untapped wealth of natural characteristics of the Navajo . . . the *inherent flexibility* and dexterity of the Indians" : "For example, after years of rug weaving, Indians were able to visualize complicated patterns and could, therefore, memorize complex integrated circuit designs and make subjective decisions in sorting and quality control."[24]

In the days before either outsourcing or insourcing, when integrated circuits were manufactured in the same complexes or even buildings that housed the men who envisioned and designed them, immigrant women of color were hailed as the ideal workforce because they were mobile, cheap, and above all, *flexible*; they could be laid off at any time and could not move to look for alternative forms of work, while their employers could close plants and reopen them in locales with the most favorable conditions. The notion that Indians were "inherently flexible" both racializes and precedes the idea of flexible labor that informs much of the research on globalization in the information age.

As Guy Senese writes, "employee availability" was highly desired by industry, which influenced its choice to open plants on Indian reservations. The almost complete lack of other wage-based employment options in Indian country and an extremely high unemployment rate almost guaranteed a favorable environment for employers. He situates the plant as part of an ongoing project of "Indian labor exploitation," writing that both "quality and low cost of Indian labor was, along with liberal government loan and tax relief, a major attraction for industry."[25] "Quality," defined as a low failure rate, was a major issue in the industry; many parts of the chip production process required artisanal handwork. Partly because of this, failure was quite common and could have serious consequences, particularly for Fairchild's military and space program contracts, which were still a major part of its business.[26] Thus, in Fairchild's outward-facing publications, such as brochures and press releases, as well as in journalistic accounts of the Shiprock project, quality is discussed rather than cost. And it was a specific kind of quality—Indian craftsmanship.

The argument that circuit quality was a natural outcome of Indian raciocultural traits is made quite overtly in Fairchild's 1969 brochure celebrating the new Shiprock plant and its workers. The first page features a large photograph of a rectangular brown, black, and white rug, woven in a geometric pattern composed of connecting and intersecting right angles (fig. 1). Adjoining it is



Figure 1.
Shiprock Dedication Commemorative Brochure, September 6, [1969], lot X5184.2009, folder 102725169, Computer History Museum, Mountain View, CA.

a short paragraph: "Thank you for helping us celebrate the dedication of Fairchild Semiconductor's new Shiprock facility—a partnership in progress."[27]

The following pages depict a woman weaving the same type of rug, her face partially obscured by the weft threads as she gazes down at her work (fig. 2).

The accompanying text reminds the reader that "weaving, like all Navajo arts, is done with unique imagination and craftsmanship, and it has been done that way for centuries." Just as this idyllic tribute to Navajo craft is getting started, the brochure transitions to a photograph of a Navajo woman standing over a microscope, gazing at the viewer, as a white male face gazes over her



Figure 2.
Shiprock Dedication Commemorative Brochure, September 6, [1969], lot X5184.2009, folder 102725169, Computer History Museum, Mountain View, CA.

shoulder, supervising and admiring her work (fig. 3). The text negotiates the transition from traditional artisanal cultural work to industrial wage labor by asserting that "building electronic devices, transistors and integrated circuits, also requires this same personal commitment to perfection. And so, it was very natural that when Fairchild Semiconductor needed to expand its operations, its managers looked at an area of highly skilled people living in and around Shiprock, New Mexico."

This appeal to "nature" as justification for converting "highly skilled" female cultural labor such as weaving rugs into high-tech factory labor is signaled by the following image, which depicts a Fairchild 9040 integrated circuit, "used in communications satellites like COMSAT," enlarged so that its geometry



Figure 3.
Shiprock Dedication Commemorative Brochure, September 6, [1969], lot X5184.2009, folder 102725169, Computer History Museum, Mountain View, CA.

fills the whole page (fig. 4). The resemblance between the pattern of the rug depicted on the first page and the circuit is striking and uncanny. It makes the visual argument that Indian rugs are merely a different material iteration of the same pattern or aesthetic tradition found within the integrated circuit. The opposing page states, "The blending of innate Navajo skill and Semiconductor's precision assembly techniques has made the Shiprock plant one of Fairchild's best facilities—not just in terms of production but in quality as well."

Again, the notion of an "inherently flexible" laborer, a worker whose nature it is to be both adaptable and culturally suited, or hardwired, to craft circuit designs onto either yarn or metal appeals to a romantic notion of what Indians

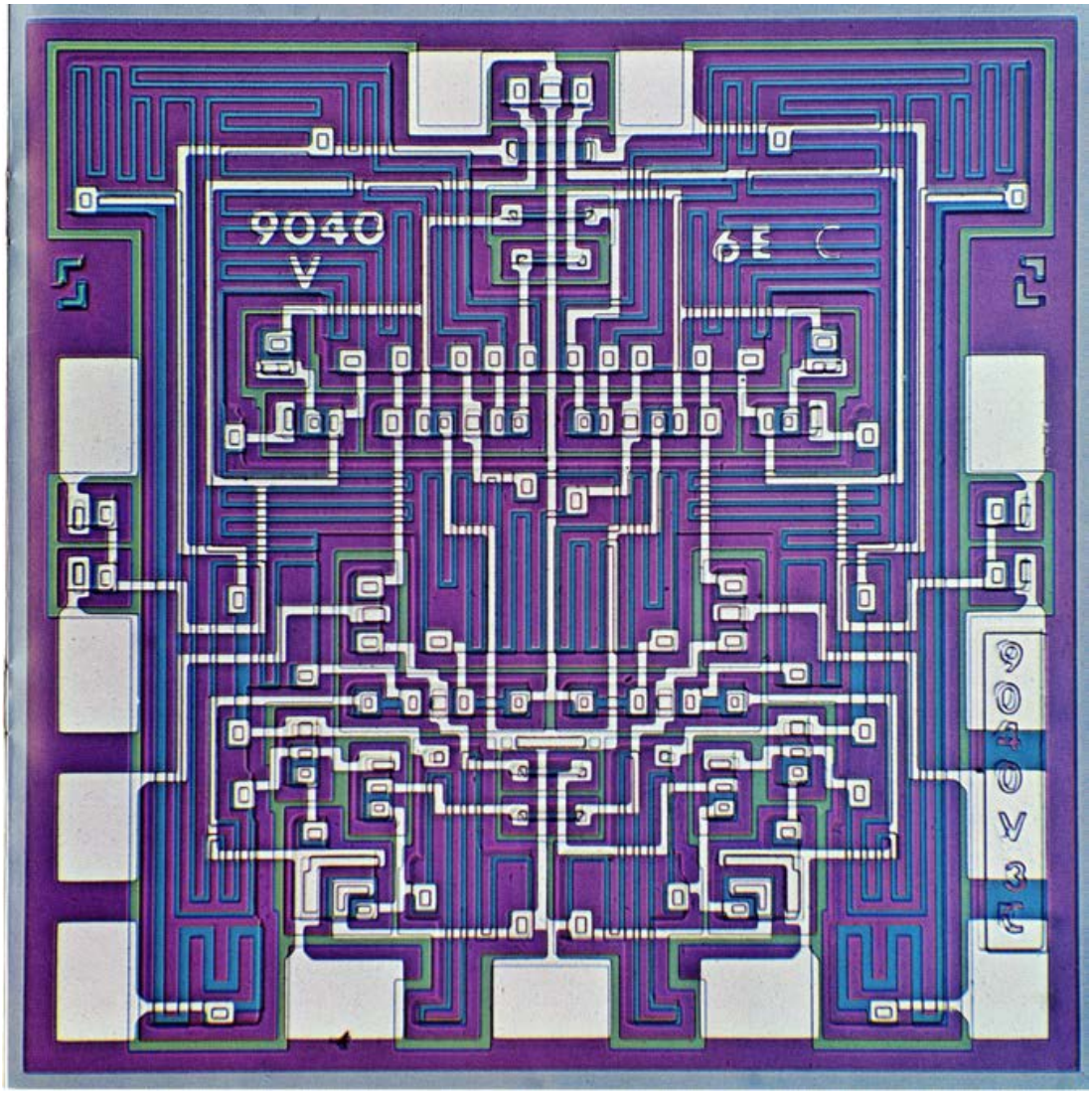


Figure 4.
Shiprock Dedication Commemorative Brochure, September 6, [1969], lot X5184.2009, folder 102725169, Computer History Museum, Mountain View, CA.

are and the role that they play in US histories of technology. This nostalgic appeal to Indian identity as a unique and valuable commodity in the world of high-tech manufacture, as both a vanishing resource and an example of and participant in the nation's unstoppable drive toward modernity, is completed on the brochure's last page. The brochure's last image is a photograph of the sun setting behind the majestic Shiprock Mountain, the namesake for the Navajo reservation, superimposed by a poem, "Song of the Earth Spirit, Origin Legend." The text reads, "it is lovely indeed, it is lovely indeed / I, I am the spirit within the earth / The feet of the earth are my feet / The legs of the earth are my legs," and so forth. It is

safe to say that poetry is not a standard convention for industrial brochures. Including it solidified claims that circuit manufacture was naturally indigenous people's work.

It was an opportune year to argue for the spiritual and natural qualities of high-tech manufacturing. Counterculture gurus such as the poet Gary Snyder and the publishing entrepreneur Stewart Brand viewed Indians as a curative to the anomie and alienation of American corporate culture, and Indians were already perceived as intimately connected to nature.[28] Snyder identified himself as a "Buddhist shaman" and Brand's multimedia show "America Needs Indians," performed at the Trips Festival in San Francisco's Fillmore West, used material gathered from his visits with Indians living on the Warm Springs Reservation in Oregon. Brand would go on to found the *Whole Earth Catalog*, a tremendously important series of books that defined the DIY movement for counterculture enthusiasts in the United States, which spun off the most influential early online community, the Sausalito-based Internet service provider "The Well," or "Whole Earth 'Lectronic Link," solidifying the move "from counterculture to cyberculture" during the last half of the twentieth century, a cultural formation that Fred Turner documents in his book of the same name.[29]

Depicting electronics manufacture as a high-tech version of blanket weaving performed by willing and skillful indigenous women served two goals: it permitted the incursion of factories into Indian reservations to be seen as a continuation of rather than a break from "traditional" Indian activities, and it pioneered the blurring of the line between wage labor and creative-cultural labor; one seamlessly became the other. Indeed, one may have *replaced* the other: the new eight-hour workday altered many aspects of family life for the Navajo people who worked at Fairchild. However, the 1969 Fairchild brochure and other materials describing the plant assert that replacing rugs with circuits is, rather than a cultural loss or, worse yet, a form of cultural imperialism, instead an extension of an existing, indigenous cultural practice; it is culture work for the nascent information age. It posits that indigenous design informed electronic circuit design—a kind of colonialism in reverse—despite the lack of involvement of indigenous people in the company's research and development arm.

The argument that Navajo women were good at their assembly jobs because they were good blanket weavers and jewelry makers appears throughout contemporary accounts of the plant. Journalistic accounts, BIA press releases, and Fairchild internal documents alike depicted Indians as the first informationalized "creative class" workers, to use Richard Florida's influential formulation,

doing what they loved well because they loved doing it.[30] Florida argues that jobs in software design, engineering, and even haircutting appeal far more to early twenty-first-century workers than jobs in the traditional industries such as manufacturing not only because of the intrinsic pleasure involved with the act of making but also because of the personal freedom acceded to the worker. Silicon Valley, along with Austin, Texas, and North Carolina's Research Triangle, are archetypal "creative class" cities where workers create "content," and by implication, culture. Members of the creative class are happy because they are creatively fulfilled, not just because they are well paid. The seed of this argument can be found in the Shiprock brochure that depicts naturally happy workers, expressing their creativity by creating electronic artifacts that resemble indigenous artifacts. The brochure's photographs of satisfied Navajo women busy at their looms and microscopes was especially appealing given the intense competition between states and non-US countries to attract industry by offering freedom from taxes and, most importantly, freedom from labor unions.

It was also a fortuitous moment for Fairchild to assert the connections between nature and technology, specifically electronics manufacturing: chip manufacture is a notoriously dirty business, and workers at Fairchild and other semiconductor manufacturers were falling victim to pollution-related disease and starting to blame the company. "By the mid-1970s, reports of chemical exposures among production workers had begun to surface" in San Jose, California.[31] Given the already high rates of pollution on the reservation from the extraction of resources such as uranium, gas, coal, and oil, semiconductor manufacture continued the ongoing practice of environmental degradation in a spot renowned for its natural beauty.[32] Ultimately, the Navajo nation failed to benefit economically as much as it had expected from the plant and was left to deal with the detritus and its long-term consequences.

Navajo identity had a heavy burden to bear. In the face of concerns about high-tech pollution, increasingly empowered labor organizations, and a newly politicized and visible American Indian civil rights movement, indigenous electronic workers at Shiprock were pressed into service as examples of the peaceful coexistence and integration of the past and the future, the primitive and the modern, creativity and capitalism. They were cited as evidence that digital work—the work of the hand and its digits—could be painlessly transferred from the indigenous cultural context into the world of technological commercial innovation, benefiting both in the process.

Navajos were described by their managers as having "patience, respect for private property (hence a low theft rate), lack of militancy, and pride in their

work."[33] They were the ideal workforce, because in contrast to striking workers in other parts of the country, they could not relocate; Fairchild's 1969 brochure claims that "the real value of this progress lies in the creation of meaningful jobs for those who have not had jobs, jobs which will keep them in the land they love and among the people they know."

The immobility and vulnerability of the Navajo worker was rhetorically respun into an act of purposeful and care-driven cultural preservation on the part of the corporation. The original rationale for bringing industry to the reservation, which was to gradually eliminate federal support from the federal government to Native Americans, was represented as part of a plan to help them stay on their reservations and retain their ancestral homelands.

The benefits of a trained and seasoned indigenous labor force that was new to industrial forms of labor were not lost on managers at other factories in Shiprock. As C. J. Jameson, manager of General Dynamics's Shiprock plant, said, "They don't have the bad habits people have in more industrial areas."[34] This is an eloquent illustration of how racialization works; prior beliefs about Indians as unreliable workers unsuited for modern form of labor are transformed into assertions of the positive *value* of "primitive" habits. This shift demonstrates the fluidity and mutability of gender and race stereotyping; Indians were described as careful, docile, and hardworking when it helped their managers to understand and explain productivity through an ethnic lens.

This strategy was one of the first iterations of an exceptionally effective argument to justify digital labor exploitation by depicting it as an outlet for the expression of cultural and racial identity. Attention to detail and pattern, careful handcraft, stoicism, and flexibility are made, not born—as Cowie writes, they are invoked in response to the needs of global capital to travel, to justify manufacturing a product in the cheapest place possible.

Race and gender are themselves forms of flexible capital. When it helps create a compelling narrative that justifies, even celebrates, the yoking of corporate interests to indigenous governance, Navajo women are understood and perceived as docile, flexible, and natural electronics workers, and indigenous identities change as a result. And when it does not, they are changed back accordingly. Latinas and Asian, African American, and, later, Indian women were all viewed as having "nimble fingers and passive personalities."[35] American Indian women, as well as Mexican women working in maquiladoras, were described in much the same way as "Orientals": as ideal workers in the digital industries, because of their experience with fine crafting of jewelry and textiles. In our present day and for the past few decades, Asian fingers have been "nimble," but in the sixties

and seventies, Navajo women's fingers were envisioned this way. In this case, it can be seen how racialization—the understanding of a specific population as possessing traits and behaviors that belong to a race, not an individual—is a process, not a product.

Rug weaving is the linchpin of the Shiprock brochure's visual argument that Navajos were natural circuit assembly workers. It is mentioned in every publication that attempts to explain the plant's success. Unlike silversmithing, jewelry making, and other indigenous Navajo practices that were cited as an argument for why and how Indians were so good at their work, rug weaving was a specifically female activity. As Benny Klain discovered during his interviews with indigenous rug and blanket weavers in his documentary *Weaving Worlds* (2008), weaving was a reliable source of personal income for women during hard times as well as an important creative outlet and spiritual practice, and as one weaver explained, it "kept us fed." Yet at the same time, the low prices offered by Indian trading post owners and traveling rug buyers guaranteed that Indian women weavers' labor was not compensated fairly; it is still a potent emblem of the exploitation of indigenous women's knowledge and labor.

In addition, Navajo weaving had a particularly complex cultural identity. According to Jennifer Denetdale, it had a double status as both authentic and inauthentic; many believed Navajo weaving to be a cultural appropriation of Pueblo weaving, a nonindigenous skill "learned" from another tribe that was thus imitative or polluting. Denetdale reads weaving as an important "intellectual tradition," as does Angela Haas in her essay "Wampum as Hypertext."[36] The affinity and historical links among weaving, digital computing, and women figures centrally throughout cyberfeminist theory, most famously "A Cyborg Manifesto." Silicon Valley business discourse created an archive of materials that represented Navajo women as "natural" cyborgs, indeed, as embodying nature itself using silicon as their medium. The cyberfeminist theorist Sadie Plant completes the circuit between weaving as indigenous practice and software production: "Textiles themselves are very literally the software linings of all technology. . . . it is their microprocesses which underlie it all: the spindle and the wheel used in spinning yarn are the basis of all later axles, wheels, and rotations; the interlaced threads of the loom compose the most abstract processes of fabrication."[37]

The discourse about Fairchild's Shiprock operation described Navajo women's affinity for electronics manufacture as both reflecting and satisfying an intrinsic gendered and racialized drive toward intricacy, detail, and quality, and the women who performed this labor did so for the same reason that

women have performed factory labor for centuries—to survive. The liberal discourse of the seventies assuaged its conscience in consigning vulnerable populations like Native Americans to this type of labor by *suturing the work itself to an emergent discourse of multiculturalism.* How could this type of labor be exploitative when it was already so much like the "native" cultural production that Indians had done for centuries without pay, the original "free labor," such as weaving blankets?

Thus it was semiconductor and electronics manufacture, among the most tedious of jobs in the long supply chain that produces our digital media devices and the vast array of technologies we use today, that was redefined and envisioned as *creative* labor, labor that women do to express themselves. In a BIA news release titled "Industries Turn to Indians for Precision Workers," the writer claims, "The Indian, with a natural affinity for precision work, is equally at home as a high-climbing steel structural worker and as a weaver of intricate designs. Somewhere between the two extremes lies electronic factory work, which calls for skill that is rooted in pride of workmanship."[38] Semiconductor manufacture was made to seem like an act of Navajo cultural preservation as well as a bid for economic survival.

Navajo women did not make circuits because their brains naturally "thought" in patterns of right-angle colors and shapes. They did not make them well because they had inherent Indian virtues such as stoicism, pride in craftswomanship, or an inherent and inborn manual dexterity. And Fairchild did not employ Navajo women *because* of these traits. These traits were identified after the company learned about the tax incentives available to subsidize the project, the lack of unions and other employment options in the area, and the generous donation of heavy equipment given by the US government gratis as part of an incentive to develop "light industry" as an "occupational education" for Indians.

Though in 1969 Fairchild's president and CEO Dr. C. Lester Hogan stated, "In the next several years we expect to see expansion of this nearly all Navajo operated plant, concurrent with future development of the Shiprock community and increased opportunities for all Navajos," this was not to be. The production of advanced electronic components by Navajo women was to be a time-limited project, running from 1965 to 1975. After 1975, when the plant was taken over by American Indian Movement members lead by Russell Means and Fairchild closed it, the Navajo were no longer the digital model minority.

Fairchild cited the unstable labor environment as the reason, but many suspected that this had to do with a desire to move all operations offshore,

where wages were even lower than they were in Navajo country, and workers less inclined to protest conditions. In the wake of the Alcatraz Occupation (1969–71) and the Wounded Knee incident (1973), the American Indian Movement (AIM) was perceived as a militant group and certainly not one that industries wanted to tangle with directly.

The reasons stated for the occupation cited worker layoffs, but others speculated that AIM's desire to unionize a famously never-unionized industry contributed to the closure as well.[39] Two conflicting views of indigenous women—as inherently digital workers who could "see complex patterns" and effortlessly, perfectly, and "naturally" re-create them on miniature circuits, and as militant aiders and abettors of militant men or, worse yet, as themselves militant—collided in this moment. While some Navajo mourn the closure to this day, imagining a Navajo Silicon Valley, others are relieved that the reservation was saved from this fate.[40]

Race and Digital Platforms

In *Indians in Unexpected Places*, Deloria writes that the American custom of imagining Indians in terms of "primitivism, technological incompetence, physical distance, and cultural difference" has remained "familiar currency in contemporary dealings with Native people."[41] Fairchild's argument for the unique benefits afforded to hypermodern technologies by indigenous women exploited this currency to paint a new and appealing picture of both Indians and electronic culture as intimately joined rather than on opposite poles.

In Nick Montfort and Ian Bogost's immensely useful definition from *Racing the Beam*, a platform is "whatever the programmer takes for granted when developing, and whatever, from another side, the user is required to have working in order to use particular software."[42] The present essay is concerned with that "whatever," the material conditions that are usually invisible to the user and are necessary for digital media device creation. The existence of cheap, female labor is absolutely taken for granted as a precondition of digital media's existence, for as Montfort and Bogost show us in their cultural history of the Atari VCS game platform, software is always a response to hardware and its constraints. Chief among these constraints is, and has always been, expense. As Lisa Parks claims, there are excellent reasons to read female labor as an indispensible part of a communication platform, for the resources that they afford are "taken for granted" by developers, users, and the whole circuit of digital media commerce and development.[43] Innovation and development are

impossible without access to hardware that can be produced flexibly, cheaply, and consistently, as it was on Navajo land from 1965 to 1975.

Montfort and Bogost supply a useful taxonomy of digital media that lists five layers of the digital device usage that scholars must pay attention to: reception, the interface, form/function, code, and platform. They claim that the first two have received by far the most critical attention, and they are right. Software studies came into being as a field that splintered off from new media studies precisely because of many scholars' understandable frustration with methods that treated new media as purely visual texts rather than as also procedural ones. Montfort and Bogost, Wendy Chun, Alex Galloway, and Kathryn Hayles remind us that the digital *does* as well as *appears*.[44]

Reading visual representations available to us through computer interfaces or remediations of other visual forms without attending to software's procedural codes, its hardware, its infrastructures, its histories, and its racial and gender formations is to miss the point of what the digital is. If the platform is defined as what is taken for granted, scholarship that "examines the relationship between platforms and creative expression" corrects the tendency to forget that digital media is "more than screen-deep," as Chun elegantly phrases it.[45]

Digital media has always had what Ted Friedman calls the "beige box problem": advertisers had a terrible time marketing a product that looked so dull: there was seemingly nothing to see.[46] In the 1997 "Intel Inside" campaign, an extremely durable advertising effort that lent a distinctive style and story to microprocessor manufacturing, human figures fully covered in "bunny suits," or clean suits, danced and capered to catchy music. The Intel "bunny people" proved very popular, appearing both as live dancers at electronics show keynote addresses and as stuffed dolls, and the name was trademarked. Many digital devices manufactured by companies that used Intel chips shipped with paper stickers that reminded the consumer that there was "Intel Inside," a claim that had to be taken on faith, since most consumers had no desire and ideally no need to examine the contents of their devices. However, if we look inside computing hardware, we will not see dancing bunny-suited clean room workers, happily making chips for free.

Instead we see Asian women, Latinas, and Navajo women and other women of color. Looking inside digital culture means both looking back in time to the roots of the computing industry and the specific material production practices that positioned race and gender as commodities in electronics factories. This labor is temporally hidden, within a very early period of digital computing history, and hidden spatially. We must look to locales and bodies not com-

monly associated with these technologies, in out of the way places, to see how race operates as a key aspect of digital platform production.

Digital labor is usually hidden from users in closed factories in Asia, visible to us only as illegally recorded cell phone video on YouTube or through the efforts of investigative reporters who overcome significant barriers to access—again, nothing to see.[47] But as Nicholas Mirzoeff reminds us in *The Right to Look*, visual culture's political project enjoins us to look precisely at those objects, practices, and artifacts that either protest their own innocence or document subaltern experiences.[48] On the spectrum of digital labor, factory work soldering chips for iPhones, missiles, and servers is as close to the machine as one can get, as close to the means of digital production—the computer—as can be imagined. It is not creative labor, nor is it free. It is fascinating that, during a pivotal moment in early computing history, the industry's foremost electronics company represented it that way. This story of digital device manufacture on Indian land shows us how the discourse of women's indigenous cultural production has been used to explain the key role that women of color play within the integrated circuit of production.

Though most digital media scholars know Jean Baudrillard as the patron theorist of cyberspace—he is famously quoted in *The Matrix*, a film in which digital space is described as the "desert of the real"—his book *The Mirror of Production* bears far more relevance to digital labor's changing meaning in the postindustrial age.[49] In this spirited critique of Karl Marx, Baudrillard targets Marx's obsession with labor or "production" as the only concept of value imaginable. As Baudrillard writes, theorists must imagine another political economic based on more than just the human capacity to produce: they must "find a realm beyond economic value."

The 1969 Fairchild Shiprock brochure does exactly that, by representing the labor of semiconductor manufacture as a "labor of love" or, more accurately, as agentive or creative race-labor rather than as alienated labor. Like weaving blankets, semiconductor production is posited as an intrinsic part of the Indian psyche, an expression of cultural essence imperiled, yet ultimately enabled, by the "modern" world.

Notes

My sincere thanks to Sara Lott, senior archives manager at the Computer History Museum in Mountain View, California, Jim Tutt, Christian Sandvig, Dan Schiller, Jonathan Sterne, Paul Edwards, and Iván Chaar-Lopez.

1. Donna Haraway, *Simians, Cyborgs, and Women: The Reinvention of Nature* (London: Routledge, 1991).
2. Aihwa Ong, *Flexible Citizenship: The Cultural Logics of Transnationality* (Durham, NC: Duke University Press, 1999), 64.
3. Tiziana Terranova, *Network Culture: Politics for the Information Age* (London: Pluto, 2004), vii, 184.
4. Wolfgang Ernst and Jussi Parikka, *Digital Memory and the Archive*, Electronic Mediations, vol. 39 (Minneapolis: University of Minnesota Press, 2013), 265.
5. Ted Smith, David Allan Sonnenfeld, and David N. Pellow, *Challenging the Chip: Labor Rights and Environmental Justice in the Global Electronics Industry* (Philadelphia: Temple University Press, 2006), 357; Jefferson Cowie, *Capital Moves: RCA's Seventy-Year Quest for Cheap Labor* (Ithaca, NY: Cornell University Press, 1999), 273; T. R. Reid, *The Chip: How Two Americans Invented the Microchip and Launched a Revolution* (New York: Simon and Schuster, 1984), 243; Karen J. Hossfeld, "Their Logic against Them Contradictions in Sex, Race, and Class in Silicon Valley, in *Technicolor: Race, Technology, and Everyday Life*, ed. Alondra Nelson and Thuy Linh Tu (New York: New York University Press, 2001), 34–63.
6. Semiconductors are the basis of all digital media devices, and as the Silicon Valley historian David Laws writes, "to some degree, almost every artifact based on semiconductor devices produced after the early 1960s depends in some way on technologies that were created by Fairchild or former Fairchild employees." Thus Fairchild was a bellwether company in the digital media industries, and its industrial history and labor practices set the tone for much of what was to become Silicon Valley. The first integrated circuits (often abbreviated "IC's" in the industry) were created in the 1960s and used to power the Minuteman Missile Guidance Computer, the Apollo Guidance Computer, and the first high-performance computers by produced by Cray, Burroughs, and Control Data.
7. Though home computers would not become popular until early 1974 (the Altair PC based on the Intel 8080 microprocessor was the first kit for a switch-based personal microcomputer), the integrated circuits produced at the Shiprock plant from 1965 to 1975 were used in calculators, aerospace applications, transistor radios, and other electronics that are part of the control and media transition to digital technology. Fairchild was the "First Company of Silicon Valley," and its founders such as Gordon Noyce would later spin off chip companies such as Intel that would fuel home computing.
8. Cowie, *Capital Moves*, 273.
9. Kylie Jarrett, The Relevance of Womens Work: Social Reproduction and Immaterial Labor in Digital Media, *Television and New Media* 15.1 (2014): 14–29.
10. See David A. Laws, A Company of Legend: The Legacy of Fairchild Semiconductor, *IEEE Annals of the History of Computing* 32.1 (2010): 60. Leslie Berlin, Lillian Hoddeson, and Christophe Lécuyer emphasize the centrality of Fairchild as the first successful manufacturer of microchips and the leader in the field. See Michael Riordan and Lillian Hoddeson, *Crystal Fire: The Birth of the Information Age* (New York: Norton, 1997), 352; Leslie Berlin, *The Man behind the Microchip: Robert Noyce and the Invention of Silicon Valley* (Oxford: Oxford University Press, 2005), 402; Christophe Lécuyer, *Making Silicon Valley: Innovation and the Growth of High Tech, 1930–1970* (Cambridge, MA: MIT Press, 2006), 393; Lécuyer and David C. Brock, *Makers of the Microchip: A Documentary History of Fairchild Semiconductor* (Cambridge, MA: MIT Press, 2010), 312.
11. Mike Cassidy, What Went Wrong at Shiprock—the Fairchild Plant Promised a Better Future for the Navajos, but That Promise Was Never Fulfilled, *San Jose Mercury News*, May 7, 2000.
12. "Interview with Charlie Sporck," February 21, 2000, Los Altos Hills, CA. Fairchild Semiconductor, "the mother of the semiconductor industry," defined an industry; as the Silicon Genesis website puts it, "Fairchild put the silicon in Silicon Valley" ("Silicon Genesis: An Oral History of Semiconductor Technology," Stanford University, http://silicongenesis.stanford.edu/transcripts/spork.htm). Over 125 chip manufacturers can trace their roots directly to this company; it spun off the largest semiconductor companies, such as Intel and National Semiconductor. According to David Nakamura, a Silicon Valley executive who worked at Fairchild in the seventies, many company lobbies in the Valley featured a framed poster that traced the genealogy of all chip manufacturers to Fairchild (interview by author). The poster was produced by SEMI (Semiconductor Equipment and Materials International), the national association for the industry.

13. US Department of the Interior, Bureau of Indian Affairs, *Electronics Industry Expanding on Navajo Reservation*, September 2, 1966.
14. Cassidy, "What Went Wrong at Shiprock."
15. Fairchild Camera and Instrument, News Release, Shiprock, New Mexico, September 6, 1969.
16. Colleen M. ONeill, *Working the Navajo Way: Labor and Culture in the Twentieth Century* (Lawrence: University Press of Kansas, 2005), 235.
17. Raymond Nakai, Fairchild Dedication (speech, Cline Library, Special Collections and Archives, Northern Arizona University, 1969).
18. Philip Joseph Deloria, *Indians in Unexpected Places* (Lawrence: University Press of Kansas, 2004), 178.
19. Guy B. Senese, *Self-Determination and the Social Education of Native Americans* (New York: Praeger, 1991), 218.
20. Peter Iverson, *The Navajo Nation* (Albuquerque: University of New Mexico Press, 1983), 273.
21. As Greg Harrison, a Fairchild Semiconductor employee, recalls, "sizable labor subsidies were available from the Bureau of Indian Affairs" and "the Native Americans on the reservations were badly in need of jobs and skills" ("National Semiconductors Offshore Legacy Recalled as Company Readies China Facility," *Chip Scale Review: The International Magazine for Device and Wafer-Level Test, Assembly, and Packaging Addressing High-Density Interconnection of Microelectric IC'S*, May–June 2004).
22. Jim Tutt, pers. comm., November 2011.
23. In a draft of a speech to be given at the 1970 Fairchild board of directors meeting, Nakai praised the board for taking a chance on the "Navajo workman" and wrote that he was "extremely pleased that your organization has decided to locate your machine tool division here at Shiprock employing additional Navajos, chiefly male. That you have made this decision enforces our belief that it is highly desirable to utilize the talents of the Navajo workman" ("Remarks to Be Made at the Fairchild Board of Directors Meeting," Colorado Plateau Archives, Northern Arizona University, 1970).
24. Industry Invades the Reservation, *Businessweek*, April 1970; my italics.
25. Senese, *Self-Determination*, 218.
26. For an excellent account of the relation between the space program and the nascent software industry, see David A. Mindell, *Digital Apollo: Human and Machine in Spaceflight* (Cambridge, MA: MIT Press, 2008), 359. As Mindell writes, "At Fairchild, one manager reported, 'Apollo really taught us a lot about reliability,' because workers had to account for every single circuit failure. The company eventually developed separate production lines for Apollo, with workers selected for high motivation and attention to detail."
27. *Shiprock Dedication Commemorative Brochure*, September 6, [1969], lot X5184.2009, folder 102725169, Computer History Museum, Mountain View, CA.
28. Shepard Krech, *The Ecological Indian: Myth and History* (New York: Norton, 1999), 318.
29. Fred Turner, *From Counterculture to Cyberculture: Stewart Brand, the Whole Earth Network, and the Rise of Digital Utopianism* (Chicago: University of Chicago Press, 2006), 327; Sherry L. Smith, *Hippies, Indians, and the Fight for Red Power* (Oxford: Oxford University Press, 2012), 265.
30. Richard L. Florida, *The Rise of the Creative Class: And How Its Transforming Work, Leisure, Community, and Everyday Life* (New York: Basic Books, 2004), 434.
31. Smith, Sonnenfeld, and Pellow, *Challenging the Chip*, 357.
32. A 1970 study of the correlation between birth defects and radiation, specifically from uranium mining among Shiprock Navajo workers, found that the "association between adverse pregnancy outcome and exposure to radiation were weak," but that "birth defects increased significantly when either parent worked in the Shiprock electronics assembly plant." Similar correlations were found at other assembly plants in California and elsewhere (L. M. Shields et al., Navajo Birth Outcomes in the Shiprock Uranium Mining Area, *Health Physics* 63.5 [1992]: 542).
33. Industry Invades the Reservation.
34. Ibid.
35. Shruti Rana, Fulfilling Technologys Promise: Enforcing the Rights of Women Caught in the Global High Tech Underclass, *Berkeley Women's Law Journal* 15 (2000): 272.
36. Jennifer Denetdale, *Reclaiming Diné History: The Legacies of Navajo Chief Manuelito and Juanita* (Tucson: University of Arizona Press, 2007), 241; and Angela Haas, Wampum as Hypertext: An American Indian Intellectual Tradition of Multimedia Theory and Practice, *Studies in American Indian Literatures* 19.4 (2007): 77.

37. Sadie Plant, *Zeroes + Ones: Digital Women + the New Technoculture* (New York: Doubleday, 1997), 305.
38. US Department of the Interior: Bureau of Indian Affairs, Industries Turn to Indians for Precision Workers, [September 10, 1965].
39. Sporck attributes the rise of Silicon Valley industry and by extension computing culture to a successful resistance to unions—this attitude informed the Californian ideology, which depended on "sweat equity" or a radically entrepreneurial stance toward both labor and capital. For a description of how this logic informed the technology industries in the late twentieth century, see Gina Neff, *Venture Labor: Work and the Burden of Risk in Innovative Industries* (Cambridge, MA: MIT Press, 2012), 195.
40. Tutt, pers. comm..
41. Deloria, *Indians in Unexpected Places*, 4.
42. Nick Montfort and Ian Bogost, *Racing the Beam: The Atari Video Computer System* (Cambridge, MA: MIT Press, 2009), 180.
43. Lisa Parks, Things You Can Kick: Conceptualizing Media Infrastructures (paper presented at the American Studies Association conference, San Juan, Puerto Rico, 2012).
44. See Wendy Hui Kyong Chun, *Programmed Visions: Software and Memory* (Cambridge, MA: MIT Press, 2011), 239; Chun, *Control and Freedom: Power and Paranoia in the Age of Fiber Optics* (Cambridge, MA: MIT Press, 2005); Ian Bogost, *Persuasive Games: The Expressive Power of Videogames* (Cambridge, MA: MIT Press, 2007), 450; and Alexander R. Galloway, *Protocol: How Control Exists after Decentralization* (Cambridge, MA: MIT Press, 2004), xxvi, 260.
45. Chun, *Control and Freedom*, 129.
46. Ted Friedman, *Electric Dreams: Computers in American Culture* (New York: New York University Press, 2005), x, 275.
47. Mike Daisey's popular and highly regarded one-man show, "The Agony and the Ecstasy of Steve Jobs," described horrible working conditions at Foxconn's Shenzhen plant, where he claimed that Apple iPads were produced by workers whose hands were permanently disfigured by injuries and repetitive stress disorder. Though it turned out that some details of account were fabricated, the *New York Times* went on to publish a multipart investigative feature story about working conditions at Foxconn that revealed rampant overtime work and other labor practices that show the hidden stresses of consumer electronic manufacture (Charles Duhigg and David Barboza, In China, Human Costs Are Built into an iPad, *New York Times*, January 25, 2012; Jack Linchuan Qiu, *Working-Class Network Society: Communication Technology and the Information Have-Less in Urban China* [Cambridge, MA: MIT Press, 2009], 303).
48. Nicholas Mirzoeff, *The Right to Look: A Counterhistory of Visuality* (Durham, NC: Duke University Press, 2011), 385.
49. Jean Baudrillard, *The Mirror of Production* (New York: Telos, 1975).

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A Genealogy of Technology and Power Since 1500

Kate Crawford and Vladan Joler (2023)

How can we understand the operations of technology and power in our era? Our technological systems are increasingly complex, interconnected, automated and opaque. Social institutions, from schools to prisons, are becoming data industries, incorporating pervasive forms of capture and analysis. Even places that were once off-limits to capital, from our emotional expressions to outer space, are now subject to computational control and extraction. Meanwhile, the industrial transformations in AI are concentrating power into even fewer hands, while accelerating polarization and alienation. If we are to address the urgent challenges of the contemporary time - including technocratic fascism, climate catastrophe, colonial wars, and wealth inequality - we need to contend with the interwoven nature of their histories. In order to have a future, we must first confront our past.

Calculating Empires is a large-scale research visualization exploring how technical and social structures co-evolved over five centuries. The aim is to view the contemporary period in a longer trajectory of ideas, devices, infrastructures, and systems of power. It traces technological patterns of colonialism, militarization, automation, and enclosure since 1500 to show how these forces still subjugate and how they might be unwound. By tracking these imperial pathways, Calculating Empires offers a means of seeing our technological present in a deeper historical context. And by investigating how past empires have calculated, we can see how they created the conditions of empire today.

Calculating Empires centers on four themes: communication, computation, classification and control. Across the centuries, the work illustrates the shifts in communication devices, infrastructures, and computational architectures, and how they are entwined with the histories of social control and classification. The vertical axis represents time, beginning with the 16th century at the base. The horizontal axis features a collection of systems: from algorithms to architecture, bodies to borders. Navigation is flexible: you can follow a theme, a time period, or set of ideas.

If we read from left to right, the work begins with the history of communication devices and the interfaces needed to access them. Next we see programming languages and the communication infrastructures that operate in the background: from submarine telegraph cables, to satellites, to the electromagnetic spectrum.

As we follow the data flows from those communications systems, they are collected and organized by museums, film and TV archives, data brokers, and AI training datasets. That data is then used to train algorithms and models that have their own histories, shaped by the labor of programmers and crowdworkers. It is all powered by energy-intensive computational hardware that is expanding exponentially in multiple directions, from hyperscale data centers, to biological computing, to quantum.

The work then moves to the themes of classification and control. It begins with time as a concept that orders human actions and labor, and then education as another social structuring process that shapes and organizes social norms, values, and hierarchies. Next we turn to the history of ideas about emotions and intelligence, as well as the shaping of the human body itself.

We see practices of classification and control in biometrics and medical data, in the disciplinary systems of prisons and policing, and in the way borders and bureaucracy regulate the movements of people and information.

This brings us to the heart of Calculating Empires: colonialism and the paths of empires over history, with their accompanying political and economic ideologies that inform industrial production and energy extraction.

The final section of the work contends with the strata of planetary classification: from the mineral layer of the lithosphere, all the way up to the astrosphere. Then we consider the concept of space in other ways, how mapping and architecture are themselves systems of control. Finally, we see multiple layers of militarized power, from surveillance technologies, to authoritarian doctrines, to entire military systems.

The contemporary tendency toward simplification and solutionism has weakened the ability to handle dense information, while our technical and environmental realities only become more complex and interdependent. By visualizing these interlaced histories of empire in intricate detail, we hope to better understand our current predicament. In the words of French historian Fernand Braudel, “if one wants to understand the world, one has to determine the hierarchy of forces, currents, and individual movements, and then put them together to form an overall constellation.” Calculating Empires invites the visitor inside this constellation.

Calculating Empires begins with a key moment when global networks - both cultural and mercantile - began to take shape in the 1500s. New trade routes meant the expansion of European empire. Advances in shipping and navigational instruments enabled the annexation of lands and genocide of Indigenous populations while introducing new viruses, drugs, and weapons. The Gutenberg press reorganized information power and laid the groundwork for a political and cultural transformation. New scientific instruments and calculation systems were developed. The meticulous cataloging and privatization of land, animals, plants, and space commenced a layered agenda of colonization that continued to expand for centuries. These practices of past empires are now echoed in the highly concentrated technology and military industries of the 21st century.

Every reading of this work is different, and you're invited to draw your own connections. Hundreds of individual drawings and texts span centuries of conflict, enclosure, and control. We suggest taking your time and reading slowly - a radical act in an era of speed and simplification. A close examination of these patterns over several centuries reveals the ascendance of particular ideas and technologies, the concentration of power and wealth, and the colonization of land, infrastructure, and human lifeworlds.

The history of science and technology is also a story of visualizations. From the Venn diagram to the neural network, visual illustrations have shaped the conceptual horizons of the scientific imaginary. We use the same method to show how technology is political, not

separate from it. Even diagrammatic representations - from the Gannt chart to the logic gate - are forms of organizing information which embed a political structure.

Calculating Empires takes Donna Haraway's provocation literally that we need to map the “informatics of domination.” The technologies of today are the latest manifestations of a long line of entangled systems of knowledge and control. This is the purpose of our visual genealogy: to show the complex interplay of systems of power, information, and circumstance across terrain and time, in order to imagine how things could be otherwise.

This work can never be complete: it is necessarily partial, subjective, and drawn from our own positionality. But that openness is part of the project. You are invited to read, reflect, and consider your own history in the recurring stories of calculation and empire. As the overwhelming now continues to unfold, Calculating Empires offers the possibility of looking back, in order to consider how different futures could be envisioned and realized.

Credits

Calculating Empires: A Genealogy of Technology and Power Since 1500
By Kate Crawford and Vladan Joler (2023)

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Awards

Calculating Empires has won the Silver Lion Award at the 2025 Venice Architettura Biennale.

Calculating Empires has won the Boghossian Foundation International Prize, in 2024.

Calculating Empires has won the European Commission's 2024 Grand Prize for Artistic Exploration in Science, Technology, and the Arts.

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Article

Data Colonialism: Rethinking Big Data's Relation to the Contemporary Subject

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Abstract

We are often told that data are the new oil. But unlike oil, data are not a substance found in nature. It must be appropriated. The capture and processing of social data unfolds through a process we call data relations, which ensures the "natural" conversion of daily life into a data stream. The result is nothing less than a new social order, based on continuous tracking, and offering unprecedented new opportunities for social discrimination and behavioral influence. We propose that this process is best understood through the history of colonialism. Thus, data relations enact a new form of data colonialism, normalizing the exploitation of human beings through data, just as historic colonialism appropriated territory and resources and ruled subjects for profit. Data colonialism paves the way for a new stage of capitalism whose outlines we only glimpse: the capitalization of life without limit.

Keywords

capitalism, colonialism, data, media infrastructure, political economy, postcolonialism

Something important is going on with data. That much is common ground, whether in uncritical literature that maps business trends (Davenport 2014; Mayer-Schonberger and Cukier 2013) or much more critical arguments that see in Big Data processing the mechanism for a new stage of capitalism (Cohen 2018; Srnicek 2016; Zuboff 2015). This article foregrounds the importance of resisting such developments and so speaks from "the South" within Milan and Treré's (2019) expanded notion as developed in this issue. But it does so by refashioning the very term—colonialism—whose historical forms generated the position of the "South" originally. For what is missing in the

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insightful recent accounts of Big Data is a wider frame to make sense of the whole social process under way. That frame is colonialism, used here not as a mere metaphor,[1] nor as an echo or simple continuation of historic forms of territorial colonialism,[2] but to refer to a new form of colonialism distinctive of the twenty-first century: *data colonialism*.

Data colonialism combines the predatory extractive practices of historical colonialism with the abstract quantification methods of computing. Understanding Big Data from the Global South means understanding capitalism's current dependence on this new type of appropriation that works at every point in space where people or things are attached to today's infrastructures of connection. The scale of this transformation means that it is premature to map the forms of capitalism that will emerge from it on a global scale. Just as historical colonialism over the long-run provided the essential preconditions for the emergence of industrial capitalism, so over time, we can expect that data colonialism will provide the preconditions for a new stage of capitalism that as yet we can barely imagine, but for which the appropriation of human life through data will be central. Right now, the priority is not to speculate about that eventual stage of capitalism, but to resist the data colonialism that is under way. This is how we understand Big Data from the South.

Through what we call "data relations" (new types of human relations which enable the extraction of data for commodification), social life all over the globe becomes an "open" resource for extraction that is somehow "just there" for capital. These global flows of data are as expansive as historic colonialism's appropriation of land, resources, and bodies, although the epicenter has somewhat shifted. Data colonialism involves not one pole of colonial power ("the West"), but at least two: the United States and China. This complicates our notion of the geography of the Global South, a concept which until now helped situate resistance and disidentification along geographic divisions between former colonizers and colonized. Instead, the new data colonialism works both externally—on a global scale—and internally on its own home populations. The elites of data colonialism (think of Facebook) benefit from colonization in both dimensions, and North–South, East–West divisions no longer matter in the same way.

It is important to acknowledge both the apparent similarities and the significant differences between our argument and the many preceding critical arguments about Big Data. Today's exposure of daily life to capitalist forces of datafication *might* seem similar, first, to arguments from four or five decades ago that the capitalist organization of work had extended from the factory to the whole of society: the well-known Autonomist notion of the "social factory" (Gill and Pratt 2008; Terranova 2000; Tronti 1966; more recently, Hardt and Negri 2017; Lazzarato 2014). Indeed, much important work has been done more recently on the actual exploitation of quasi-labor or playbor, for example, on digital platforms (Fuchs 2017; Scholz 2013), which in turn have their roots often in an alternative strand of Marxist analysis (Smythe 1977). But our argument is not concerned here specifically with the exploitation of labor. Nor are we claiming that everyday life is now governed and managed *as if* it was labor. The weakness of the latter position has been noted even by those

sympathetic to the Autonomist position (Ross 2013, 26) who have stressed the importance of unpaid work not only to today's capitalism, but to Marx's original model of capitalism (cf. Moore 2015, 69-71).

More helpful are approaches which see what is going on with data as a form of fundamental appropriation (Greene and Joseph 2015; Thatcher et al. 2016, drawing on Harvey 2004) or extraction (Mezzadra and Neilson 2017) of resources. This appropriation is certainly complex. It is not simply a matter of harvesting a natural resource: first, life needs to be configured so as to generate such a resource (we come to the contribution of platforms later); second, data about one individual's actions or properties at one moment needs to be combined with data about other actions, moments, and properties to generate valuable relations between data points (Arvidsson 2016; Thatcher et al. 2016, 995). A great deal of important work in the past half-decade has explained how exactly this happens and the details of how human actors are drawn into relations that extract and manage such data.[3] But what is still lacking, we suggest, is a broader frame for grasping the whole transformation of which such data appropriation via digital platforms is just one part. That frame we suggest is data colonialism.

A factor that has hampered our understanding of the scale of change, and its analogies to earlier forms of colonialism (not just capitalism) is the common suggestion that recent developments of capitalism had already largely been anticipated by Autonomism and its many followers. Certainly, they pointed in the direction of a *general* intensification of social life's orientation toward capitalism, but they were rather vague as to the mechanisms, except when they relied on the idea that the structure and norms *of work* somehow expanded out into social life. As Gill and Pratt (2008, 7, emphasis added) put it, "from [the social factory] perspective *labour* is deterritorialized, dispersed and decentralized so that 'the whole society is placed at the disposal of profit'" (Negri 1989, 79). Marazzi (2008, 50, emphasis added) analogously argues that "today the capitalist organization *of work* aims to . . . fuse work and worker, to put *to work* the entire lives of workers." But that gives us no grip at all on data colonialism, which appropriates life as raw material *whether or not* it is actually labor, or even labor-like.

In response, we must recover a site of resistance which confronts the new and distinctive appropriations of twenty-first-century capitalism, as practices of data processing are played out across a global terrain of differentiated exploitation (Arora 2016). Our temporal scale for appreciating these developments should be not so much the past half-century of socializing capitalism, especially in Europe, but the centuries-long global cycle of colonialism's long intertwining with capitalism. Just as industrial capitalism would not have happened without the prior appropriation under historical colonialism of vast territories, their "natural" resources and bodies, so we are today witnessing the first stage of another long-term double development: the colonial appropriation of life in general and its annexation to capital, through various mechanisms of which one is the digital platform. The platform, we argue, *produces* the social for capital, that is, a form of "social" that is ready for appropriation and exploitation for value as data, when combined with other data similarly appropriated. Rather than

an expansion of the labor process, the better analogy is with the appropriation of physical nature within processes of capitalist production (Moore 2015). But because this appropriation makes us all *subjects* of capital in distinctive new ways, the most useful overall framing here is of a new phase of colonialism that is deeply intertwined with the long-term development of capitalism.

Our argument combines perspectives from political economy, social theory, and critical information studies to articulate what within the phenomenon called "Big Data" needs resisting. The first section explains the mechanisms of data colonialism, while the second explores the consequences of data colonialism from the position of the colonized subject, revealing deeper levels of dispossession than are captured by critiques of labor exploitation. In a short conclusion, we consider what can be learned from decolonial theory for formulating resistance to this new form of colonialism through data.

The Mechanics of Data Colonialism

The legacy of colonialism from the sixteenth to the twentieth century includes the decimation of millions of native lives and the depletion of vast amounts of natural resources, all for the enrichment of a few. In deploying the concept of data colonialism, our goal is not to make loose analogies to the content or form, let alone the physical violence, of historical colonialism. Instead, as indicated earlier, we seek to explore the parallels with historic colonialism's *function* within the development of economies on a global scale, its normalization of resource appropriation, and its redefinition of social relations so that dispossession came to seem natural.

The Naturalization of Data Capture

Personal data of many sorts is appropriated for ends which are not themselves "personal." By personal data we mean data of actual or potential relevance to persons, whether collected from them or from other persons or things. For personal data to be freely available for appropriation, it must first be treated as a natural resource, a resource that is *just there*. Extractive rationalities need to be naturalized or normalized, and, even more fundamentally, the flow of everyday life must be reconfigured and represented in a form that enables its capture *as* data. Jason Moore (2015) argues that capitalism historically depended on the availability of *cheap nature*: natural resources that are abundant, easy to appropriate from their rightful owners, and whose depletion is seen as unproblematic, but whose "availability to capital" itself had to be constructed through elaborate means of marketization. So too with what we now call "personal data," but which is the outcome, not the precondition or prior target, of a newly "computed sociality" (Alaimo and Kallinikos 2016). That is the underlying reason why there cannot be raw data (Gitelman 2013): because what is "given" must first be configured for "capture" (Kitchin and Dodge 2011). Natural resources were and are not cheap per se, but legal and philosophical frameworks were established to rationalize them as such, on the basis that they were "just there."

Only later did the costs to humanity of treating natural resources this way come to be appreciated. But, as Julie Cohen (2018) points out, the legal fiction that land inhabited for millennia (such as the territory now known as Australia) was *terra nullius* or "no man's land" in English law, and thus available for exploitation without legal interference, has its strong parallels today.

The apparent naturalness of data colonialism's appropriations relies also on a large amount of ideological work, just as historic colonialism did. Consider the business cliché that data are "the new oil," lost to humanity until corporations appropriate it for some purpose. This rests on the construction of data as a "raw material" with natural *value*, as the World Economic Forum (WEF) claims: "personal data will be the new 'oil'—a valuable resource of the 21st century . . . becoming a new type of raw material that's on par with capital and labour" (WEF 2011, 5, 7). Through this discursive move, the links of data back to a prior process *of data collection* (i.e., appropriation) are obscured.

A blurring is achieved metaphorically through the common idea that data are "merely" the "exhaust" exuded by people's lives, and so not capable of being owned by anyone (United Nations [UN] 2012, 9).

To accomplish the appropriation of personal data, data colonialism relies on other *extractive rationalities* as well. There is, as many critics have noted (Fuchs 2017; Scholz 2013)—and this is what we share with earlier debates—a social rationality that treats much of the labor that contributes to data extraction as value-less, as "just sharing." There is also a practical rationality that frames corporations as the only ones with the power and capacity to process (and thus appropriate) data. Simultaneously, a political rationality operates to position society as the natural beneficiary of corporations' extractive efforts, just as humanity was supposed to benefit from historical colonialism as a "civilizational" project.

The principal actors in data colonialism can collectively be called the *social quantification sector*, corporations involved in capturing everyday social acts and translating them into quantifiable data which is analyzed and used for the generation of profit. Firms like Amazon, Apple, Facebook, and Google in "the West," and Baidu, Alibaba, and Tencent in China, are its most well-known players. The social quantification sector includes both big and small hardware and software manufacturers, developers of social media platforms, and firms dedicated to data analysis and brokerage. The latter, a largely unregulated part of the economy, specialize in collecting information from medical, financial, criminal, and other records for categorizing individuals through algorithmic means. Data brokers package and sell those lists to advertisers and other users such as governments and law enforcement agencies.

Yet, in spite of the huge complexity of the social quantification sector, the parallels between the fictions that operationalized the dispossessions of historical colonialism and those that work today to enable data colonialism are striking. Consider the Spanish empire's *Requerimiento*, whose absurdity was first recounted by Bartolomé de las Casas (1951, 58). The purpose of this proclamation, read in Spanish by conquistadors to a non-Spanish speaking audience, was to introduce the natives to the strange new world order they were about to be colonized under, and to demand their simple acceptance (or face

extermination, which frequently arrived regardless of compliance). Today, in the era of data colonialism, we are accustomed to similarly incomprehensible documents called Terms of Service, which contain outlandish appropriative claims by corporations. The force of the *Requerimiento* depended on an effective monopoly of physical force. Today's "muscle" lies in various forms of economic concentration, one of which is the digital platform. Whatever the form of force used, its effect now, as then, is through the discursive act that accompanies it to embed subjects inescapably into relations of colonization.

Modes of Extraction

The vast reorganization of human life implied by data colonialism could not happen without a broader shift in social relations. This is another aspect of data colonialism that, alongside aspects just covered—resource appropriation, ideology, and corporate concentration of profits—matches the features of historic colonialism. This social transformation is hugely complex: to understand it we must think about both colonialism and capitalism.

First, colonial history helps us see the emergence of digital platforms as more than business invention, or even new forms of economic control through multisided markets (Cohen 2018; Gillespie 2010; Rieder and Sire 2014). Digital platforms are the technological means that produce a new type of "social" for capital: that is, the social in a form that can be continuously tracked, captured, sorted, and counted for value as "data." Platforms are a key means whereby the general domain of everyday life, much of it until now outside the formal scope of economic relations, can be caught within the net of marketization. Arvidsson (2016) analyzes well the technical methods of extracting value from platform data as a form of financialization, but the more basic move is the appropriation of the social itself. As the purpose of "social" media platforms is to encourage ever more of our activities and inner thoughts to occur on platforms, there is, in principle, no limit to the appropriation, as what is appropriated is what is increasingly constructed *to be ready for* appropriation.

But social media platforms' capture of data for commodification and value extraction is only one form that the annexation of life to capital takes under data colonialism. A second is the immense growth of data-driven logistics in all areas of human production (Cowen 2014), whether or not for profit. Whereas logistics itself emerged as the management of goods' movement within global supply chains, the general "logic" of logistics treats all production on any scale and involving any combination of human and nonhuman parties, as ready for management through data. This incorporates continuous data collection and large-scale data-processing into many areas of work that previously were managed very differently. Important as the growth of quasi-labor on and around digital platforms is, and important as various forms of underpaid labor are (e.g., for small tasks on Amazon's "Mechanical Turk" program or other areas of the gig or sharing economy), they are only part of the broader growth of data-driven logistical management across all work, which constitutes the second way in which life is being appropriated for capital.

The third distinct way in which human life is being appropriated through new social relations is where individuals track their own activities for data extraction, sometimes voluntarily, but often (this is the overlap with the second area) as a requirement of their labor or other important contractual commitments, such as insurance or social security (Levy 2015). As recent critical data studies have shown (Eubanks 2018; O'Neil 2016), such forms of self-data collection provide the basis for new forms of discrimination and inequality.

Taken together, these transformations represent something approaching the appropriation for capital of the whole domain of social life, and much of individual life too, an appropriation that constitutes the colonial moment of contemporary capitalism. A key point to appreciate, however, is that the social relations that make this possible are very often not labor relations in the strict sense of the word. Human life itself is being progressively capitalized in new ways whose detail, extent, and precision go far beyond earlier general predictions of the capitalization of life.

This wider colonial appropriation and its social means can be better understood by drawing on the insights of Marx. Our use of Marx here is not orthodox Marxism, but starts rather from the postcolonial debates about the limits of Marxism's interpretation of global history and its relative neglect of colonialism and slavery (Williams 1994). That leaves us, however, free to draw on Marx's *social* theory. We acknowledge of course the extensive work on data done in recent years by Marxist scholars (e.g., Fuchs and Mosco 2017) to identify the distinctive forms of labor exploitation under way, but we pursue an alternative line of argument that captures better the scale of appropriation under way and its long-term implications for capitalist reproduction.

Marx is very clear that the domain of commodities goes on expanding under capitalism. Thus, there is no reason to see that domain as limited to the set of commodities that Marx could see at work in the mid- to late-nineteenth century. Marx's well-known example of commodification is labor: what was once just work or productive activity becomes commodified under capitalism as labor power. This labor power acquires the necessary measurable dimension to be exchanged in the market. In this way, workers' labor power becomes something they can sell as a commodity. Furthermore, what workers use in the labor process also acquires exchange-value as commodities; for example, the farmer's seeds and manure become commodities under capitalism, even though before capitalism they were just part of the cycle of land use (Marx 1976, 950, 952). Thus, what unleashes commodification's transformative effects on the social is the fact that labor (the everyday activity that has gone on since the beginning of time) acquires under industrial capitalism an *abstract* dimension (Postone 1998, 7). This reinterpretation of Marx for a "post-Marxist world" (as Postone puts it) emphasizes that it is the *abstracting force* of the commodity, the possibility of transforming life process into "things" with value, that is, the fundamental characteristic of capitalism. This makes even clearer our divergence from the Autonomist analysis, as a critical argument from Marx no longer needs to proceed primarily by exposing how labor or quasi-labor is exploited.

This insight from Marx's social theory enables us to think about the new role of commodification in relations organized around data production and consumption. "Data" are abstractions from the processes of human life. Data are not abstracted from us automatically, but through social relations to which at some point, even if retrospectively, we are assumed to have consented. Meanwhile, data are being increasingly commodified. Industrial capitalism, according to Marx, changed society by transforming the universal human activity of labor into a social form with an abstract dimension (via the commodification of labor). Today, data colonialism is changing society by transforming *human life* into a new abstracted social form that is also ripe for commodification: data. The means for this transformation are—to reiterate—not labor relations, but, more broadly, commodified social relations, or, more succinctly, data relations.

Put another way, ordinary social interaction has come to contribute to surplus value as a *factor of production*, just like seed or manure (the examples are Marx's). But what human beings do when they are tracked and when data are extracted from them during social interactions is *not* a new type of labor, although it can be appropriated, abstracted, and commodified all the same. The implications of this therefore extend beyond labor to many other aspects of life which until now were not regarded as economic "relations" at all, but come to be incorporated within a vastly expanded production process. These new types of social relations implicate human beings in processes of data extraction, but in ways that do not *prima facie* seem extractive. That is the key point: the audacious yet largely disguised corporate attempt to incorporate *all of life*, whether or not conceived by those doing it as "production," into an expanded process for the generation of surplus value. The extraction of data from bodies, things, and systems create new possibilities for managing *everything*. This is the new and distinctive role of platforms and other environments of routine data extraction. If successful, this transformation will leave no discernable "outside" to capitalist production: everyday life will have become directly incorporated into the capitalist process of production.

As this transformation cannot even start without the appropriative moment of data colonialism, it is important to consider how dispossession feels from the perspective of the objects of that appropriation: human beings.

Data's Colonized Subjects

In this section, we want to discuss some key aspects of the colonization of everyday social life under the aegis of various forms of institutional order, from market capitalism (North America, Europe) to the complex hybrid of commercial and state power that characterizes the People's Republic of China.

Tethered to Data Judgments

Data colonialism means that new social relations (data relations, which generate raw inputs to information processing) become a key means whereby new forms of

economic value are created. The value of those extractive processes depends on the comprehensiveness of the data generated. Nothing should be excluded. Before the Internet, as Bruce Schneier (2015, 27–8) notes, data sources about social life were limited to company customer records, responses to direct marketing, credit bureau data, and government public records (we might add insurance company data on their insured). Now, as already noted, a vast and varied social quantification sector operating within a complex web of data processing functions extract data from everyday life at a depth far exceeding that found in earlier forms of social organization.

There are multiple forms of encouragement into data relations: from the rituals of routine self-tracking on platforms where we count our followers or likes,[4] to exemplary performance (the Olympic athletes who tell us how they track themselves). These "soft" nudges toward datafication converge in marketers' favored strategy of *gamification*. Gamification is an industry "term of art" based, as Julie Cohen (2016, 207–14) notes, on a "very specific behavioural model" that incentivizes "sharing" and loyalty to processes of data extraction from which new "social knowledge" can be generated.

In contemporary data relations, it is not even the uniquely identifiable individual who is the object of data tracking and nudging. The subject is reached under conditions which need not involve their naming or even their identification. You or we will be uniquely identified by multiple corporations using different sets of data features, each sufficient to prompt a distinctive action. Data scholars call these sets of data points our "data doubles."[5] Management theorists Alaimo and Kallinikos (2016) note, in an analysis of retail fashion platforms, that data doubles are the building blocks for new "social objects": data-rich constructs arranged in complex categories which corporations can target and influence. Media platforms like Netflix are based on the structuring of content production and marketing around the data doubles produced through relentless data harvesting and processing, all of which suggests customization and convenience for the user.

But there is nothing comforting about this. Even though the new social knowledge is produced through operations that bypass human beings, it is actual human beings, not "doubles," who are tethered to the discriminations that such knowledge generates. It is a real person who gets offered a favorable price in the supermarket, an opportunity for social housing, or a legal penalty, all based on algorithmic reasoning.

Human inputs are only part of the territory that data colonialism seeks to annex to human capital. Machine-to-machine connections significantly deepen the new web of social knowledge production. Consider the fast-growing "Internet of Things." The goal is clear: to install into every tool for human living the capacity to *continuously and autonomously* collect and transmit data within privately controlled systems of uncertain security.

The Colonized Self

In the hollowed out social world of data colonialism, data practices invade the space of the self by making tracking a permanent feature of life, expanding and deepening the basis on which human beings can exploit each other. The bare reality of the self

as a self[6] comes to be at stake. It is the minimal integrity of human life that must be protected. This reality, which each subject can recognize in each other, cannot be traded away without endangering the basic conditions of human autonomy.

This is particularly important when we can be sure the costs and consequences of dispossession through data will be distributed unevenly, especially when it comes to surveillance. Even if, under data colonialism, we are all destined to become data subjects—that is, parties to regular data relations—what this means for one person may be very different from what it means for another. There are various reasons for this inequality: some relate to how the intensity of surveillance is allocated, and others relate to individuals' relative ability to repair the costs of surveillance over the long-run. We cannot get into these details here.

Underlying such variations, capitalism affirms the uniquely identifiable *reference-point* on which all notions of the self rest; it is this "self" whose market potential gets traded in proxy form. But, at the same time, capitalism erodes a core element in the *content* of that identified self whose continuity (and change) we value through time. By installing automated surveillance into the space of the self, we risk losing the very thing that constitutes us as selves at all, that is, the open-ended space where we continuously transform over time. What needs defending is *not* individualistic self-rule, but rather the minimal, socially grounded integrity of the self without which we do not recognize ourselves *and others* as selves at all.

This minimal integrity of the self is valued by all philosophical traditions. The integrity of the self as the entity that can make and reflect on choices in a complex world is essential to all Western liberal notions of freedom.[7] But the self's minimal integrity is essential to a philosophy of liberation intended to operate *beyond* the dominance of Western models of power, such as that of Argentinian-Mexican philosopher Enrique Dussel (1985) which emphasizes the "natural substantivity of a person," the basic "fact that each individual is distinct and not merely different"?[8] The expansion of data colonialism is a problem for all human subjects, indeed for human development as such. A continuously trackable life is a dispossessed life, no matter how one looks at it. Recognizing this dispossession is the start of resistance to data colonialism.

Conclusion

At this point, we can learn much by recalling the vision of the late Peruvian sociologist Aníbal Quijano. To Quijano, the point was not merely to move *past* the colonial through the "postcolonial," but to challenge fundamentally colonialism's legitimacy through "decolonial" ways of thinking. Quijano focused on the legacy of historic colonialism, but his diagnosis is of enormous importance for addressing data colonialism too. What decolonial thinking in particular can help us grasp is that colonialism—whether in its historic or new form—can only be opposed effectively if it is attacked at its core: the underlying rationality that enables continuous appropriation to seem natural, necessary and somehow an enhancement of, not a violence to, human development.

Quijano (2007, 177) calls for an "epistemological decolonization" that "clear[s] the way for . . . an interchange of experiences and meanings, as the basis of another rationality

which may legitimately pretend to some universality" (translation slightly adjusted). The goal is not to abandon rationality or even the claim to "some universality." What must be abandoned is the claim to absolute universality that Quijano sees as characteristic of European modernity, and which we find *reproduced* in data colonialism, and its logics of universal data extraction and management of human beings through data. The force of Quijano's point remains, even when, in the bipolar world of U.S.–China data domination, colonialism starts to operate well beyond the confines of European traditions.

We reach here the core of what is wrong with data colonialism's new order: its vision of totality. Yet the logic of Big Data is not the only vision of human order that is possible:

> Outside the "West," virtually in all known cultures . . . all systematic production of knowledge is associated with a perspective of totality. But in those cultures, the perspective of totality in knowledge includes *the acknowledgement of the heterogeneity of all reality*; of the irreducible, contradictory character of the latter; of the legitimacy, i.e., the desirability of the diverse character of the components of all reality—*and therefore, of the social*. The [better, alternative] idea of social totality, then, not only does not deny, but depends on the historical diversity and heterogeneity of society, of every society. In other words, it not only does not deny, but it *requires* the idea of an "other"—diverse, different. (Quijano 2007, 177, emphasis added)

It is exactly this alternative vision of order and totality that datafication denies, as it categorizes subjects and builds societies toward total algorithmic control. And it is exactly this alternative vision that resistance to data colonialism must affirm.

The practical starting-point for resistance to data colonialism is a vision that, until twenty years ago, would have been indisputable, but now, strangely, appears counterintuitive to many. This vision rejects the idea that the continuous collection of data from human beings is natural, let alone rational; and so rejects the idea that the results of data processing are a naturally occurring form of social knowledge, rather than a commercially motivated form of extraction that advances particular economic and/or governance interests. Rejecting data colonialism does not mean rejecting data collection and use in all its forms. But it does mean rejecting the form of resource appropriation and accompanying social order that most contemporary data practice represents. A useful first step is to name such practice as the colonial process that it surely is.

Authors' Note

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Notes

1. While we agree with much of the detail of Thatcher et al.'s (2016, 992, 998–9) argument, we disagree that data colonialism is merely a metaphor to capture how the lifeworld is appropriated ("colonized") by surveillance.
2. This is how we read the insightful comments on colonial aspects of data extraction by Shepherd (2015) and Cohen (2018).
3. This is a vast area, but see, for example, Gillespie (2010), Gerlitz and Helmond (2013), and Bucher (2017).
4. See Gerlitz and Helmond (2013).
5. See Haggerty and Ericson (2000).
6. We recall here Giorgio Agamben's provocative notion of "bare life" (Agamben 1998).
7. For one of the most critical versions, see Sen (2002).
8. See Dussel (1985, 112, cf. 158).

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10 Dec – Sprechstunde / Co-Lab Day

17 Dec – Logics & Ideologies

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clouding it in orientalist verbiage. If not, then kindly please tell us what else it is supposed to be. Otherwise, potentially valuable social experiments might end up in an irrelevant subfolder of some dim-witted monopoly metaverse in which rotating Dogecoin memes burn up the planet.

10

Roko's Basilisk: Artificial Stupidities and Existential Risk

Every era has its monster.[1] In the era of AI, the monster is called Roko's basilisk. So what is it?

Roko's basilisk is a story, or more precisely, a bet. The question posed is this: *Would you support the development of a superintelligent AI, yes or no?* You are free to choose! But there is a twist: you know that this AI will be developed in any case, and then will come back from the future for its former enemies – to torture them for all eternity! So now that you know this: *Do you support it or not?* Remember, you are free!

When this gloomy thought experiment was posted on a blog dealing with artificial intelligence, the thread was quickly shut down.[2] The blog's admin cited psychological damage to its readers, and the whole thing became some sort of wacky urban legend, giving rise to fears of an AI-triggered extinction of humanity.

I wish I could show you a picture of a basilisk but, unfortunately, all pictures of the basilisk on my hard drive were unexpectedly substituted with photos resembling the illustration in figure 10.1.

I have no idea what this is about, but what is clear is that some men in this image are using their bodies to form the Turkish word for 'no' (*hayır*). Clearly, all pictures of the basilisk were swapped by some unknown persons, so I cannot show you any basilisks here.

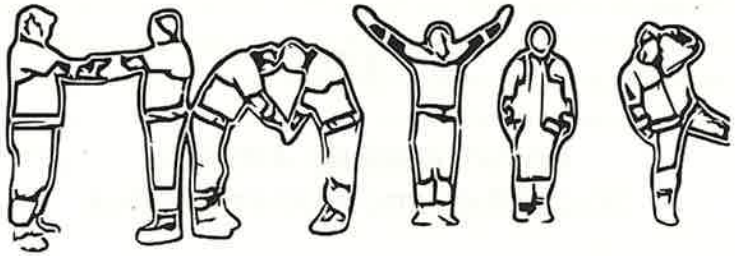


Figure 10.1. Drawing showing the positions of persons in a substitute photo of the basilisk.

Anyhow, I googled this picture, and sure enough, the search returned numerous images of people in work overalls forming the word 'no' with their bodies. It turns out this was in reaction to the 2017 Turkish referendum, in which Turkish president Recep Tayyip Erdogan asked for a massive expansion of his already-considerable powers. He wanted everyone to say yes to this deregulation, but some, like the person in figure 10.2, seemed to disagree.



Figure 10.2. Drawing showing the position of a person posing as a letter in a protest photo.

Heraldic Monster

In some cases monsters symbolise states. In fact, the city of Basel is symbolised by a basilisk. One of the best-known monsters in political theory is Thomas Hobbes's Leviathan – which is not a basilisk but a sea monster, as well as Hobbes's name for the state of the sovereign monarch.

Roko's basilisk, however, is a specific monster, which is actually supposed to be a superintelligent artificial general intelligence (AGI). What kind of state does it stand for?

Perhaps it's the 'network state' that Balaji Srinivasan advertises in his eponymous 2022 treatise.[3] The network state is an entity that colonises physical space from the cloud. It first creates online crypto-based communities, which increasingly encroach on physical space via micronations, crowdfunded territories and sea-steading. A case in point is an entity called Praxis, which describes itself as the 'next America built on the internet':

> Praxis is the world's first Network State: a global online community with a national consciousness, developing a shared way of life, governing institutions, and crowdfunding a physical city. Praxians live in over 82 countries and have founded companies worth over $400B. Praxis is a home for the brave, who strive for virtue and wisdom. Our purpose is to restore Western Civilization and pursue our ultimate destiny of life among the stars.[4]

Praxis wants to crowdfund its own city in the Mediterranean, establish a 'post-labour AI future' and eventually relocate to space.[5]

Artificial Nature

A precursor to extraterrestrial community building can be found in the closed missions at Biosphere 2, 1991–1994. Here, volunteers were locked into a greenhouse sphere in Arizona

for months on end to simulate an outer space colony. They had to be completely self-sustaining, including in the production of their food and a breathable atmosphere in a number of different biomes.

This phase of Biosphere 2 was an oligarch-funded test for space colonisation. Could colonists produce oxygen? Sustenance? Social bonds? The answer is – barely. Oxygen dropped to dangerous levels. The climate was completely fucked up. Most mammals went extinct, and pollinating insects were wiped out. The crew split into two hostile factions. After two runs and a scientist revolt, the project was abandoned. In the end, cockroaches and ants were the species that turned out to be best adapted to the oligarch space colony.

Why is this interesting? Because the manager of this project, in its later stages, was Steve Bannon, a key ethno-nationalist ally in US president Trump's first administration. His management style – a sort of armed takeover of the premises because it wasn't making profit – prompted an uprising. At a certain point, the windows of the sphere were forcibly opened by some renegade scientists. Ultimately, other windows were even broken to let oxygen inside.

Artificial Stupidity

But there is another interesting and much more potent detail: apparently *Big Brother*, arguably the first-ever reality TV show, was based on the Biosphere 2 experiments.

The narrative template of reality TV is based on an artificial idea of natural selection, whereby the fittest – however defined – survive or progress. One could also say that these formats are based not on selection but extinction, and that they celebrate extinction as spectacle. Interestingly, the Biosphere 2 inhabitants were in many cases linked to a hippie performance cult called the Theater of All Possibilities. On the

level of theatre, if nowhere else, the Biosphere 2 experiments were spectacularly successful.

It is no exaggeration to say that reality TV has become an important template for autocratic politics – a development that began with Italian trash TV mogul Silvio Berlusconi's rise to power in 1994 and has accelerated ever since. Just remember Donald Trump's line in his elimination show *The Apprentice*: 'You're fired.'

With this in mind, it is clear that the results of Biosphere 2 had a spectacular impact on the development of cultural forms. Even if it had been perfectly sealed, reality TV, or the narrative of survivalist spectacle, would still have escaped. One can only wonder what kind of unexpected 'thing' or cultural effect will 'escape' from AI labs, and which unforeseen side effects this will have.

In fact, this has already happened on several levels.

Right now, AGI is still unrealised. But 'poor', lower-tech or hallucinating AIs are already busily restructuring society – more by dysfunction than by efficiency – and thus contributing to the rise of post-fascist, neofeudal and reactionary movements.

Why is artificial stupidity so successful? In the most rational of all worlds, bad and barely functional products are more profitable. Total optimisation is not reached when everything works well and people are happy; on the contrary, under the conditions of extreme capital, an optimal world is the one that barely scrapes by, deploying technology on the brink of failure. Ultimately, corruption, firing or jailing people, threats, rigging the system and pure deceit are way more efficient technologies. Under such conditions, one does not reduce emissions; one simply fakes the numbers and keeps selling cars. Or one simulates AI by employing human workers to mimic its functions.

Think of Roko's basilisk, which doesn't even try to convince you to support it. No, it just bullies and threatens you. The basilisk fable could be easily seen as a literary metaphor of little to no real-world consequence. A few people got very depressed

reading, it seems, and Eliezer Yudkowsky, who ran the blog where the basilisk first appeared, freaked out and banned discussion about it for several years, citing information hazards. But while people were in distracted awe of the monster, a much more real development took place, more or less in parallel.[6] On the same blog where the thought experiment had originally emerged, people began to brainstorm around the question how an AI would optimise its function. One of the unintended consequences may have been to serve as involuntary incubator for 'alt-right' ideas. As the post-humanist community forum RationalWiki assesses: 'The (neoreactionary) subculture started amongst the Bay Area technolibertarian subculture, particularly including the transhumanists.'[7]

Discussants included many neoreactionary celebrities – Mencius Moldbug (aka Curtis Yarvin), Michael Anissimov and Nick Land – spanning ideologically from neomonarchist to fascist futurist. Such were the opinions expressed that the blog admin distanced himself repeatedly, and believably, from alt-right ideology. As theorist Florian Cramer analyses, Land advocates a sort of elite eugenicism, whereby rich people have access to racial upgrades.[8] The wider neoreactionary ecosystem includes Peter Thiel, who famously claimed, among other things, that startups were akin to monarchies.

So, just as *Big Brother* escaped from the Biosphere and manifested both as eternal terror of reality TV and eternal mass surveillance, something else seems to have escaped from the sphere of Roko's basilisk: namely, some of the cultural ideas of the alt-right and thus of the current right-wing US ideological elite. They believed that states should be corporations, that startups should be run like dictatorships (or the other way around), that so-called natural hierarchies should be protected and so on. Roko's basilisk also became the ancestor to long-termist fears about extinction of humanity through AI. Long-termism – the idea that humanity's actions now should be optimised to benefit its offspring a million years down the

line – spawned large-scale scams such as so-called effective altruism associated with cryptofraudster Sam Bankman-Fried. Effective altruism – an ideology initially conceived to make philanthropy financially efficient – also heavily leans into eugenicism. Former Oxford scholar Nick Bostrom, one of its leading advocates, decried what was in his view a correlation between fertility and lack of intelligence, which he thought should be corrected.[9]

Long-termism is a leading ideology for many members of the current right-wing tech elite, and led to foregrounding future 'existential risk' caused by AI over its actual present-day social consequences. RationalWiki founder Eliezer Yudkowsky, still shaken from his encounter with Roko's basilisk, became one of the leading voices to warn of AI-caused extinction:

> Many researchers steeped in these issues, including myself, expect that the most likely result of building a superhumanly smart AI, under anything remotely like the current circumstances, is that literally everyone on Earth will die. Not as in 'maybe possibly some remote chance', but as in 'that is the obvious thing that would happen'.[10]

He even advocates for airstrikes on data centres still training AI models after a wholesale ban. For sure, the Lovecraftian horror tale of the basilisk got Yudkowsky scared.

In parallel, however, the elaborate breeding fantasies of trans- and post-humanists were being realised in a low-tech way. In the wake of efforts to eliminate or diminish as many healthcare systems as possible, sick, marginalised and poor people have simply been left to die. This is a form of accelerated extinction, which is also implemented through border regimes and state violence. Another version of actually existing reactionary transhumanism takes the form of attacks on reproductive rights that aim to monopolise as much biopolitical control as possible.

To come back to the beginning: major reactionary pushes

in many countries worldwide create multipolar surveillance authoritarianisms supported by AI systems. Simultaneously, AI is being heralded as a major technological and economical bonanza, and creates major hype.

All of these cause profound social disruption as well as the reorganisation of societies into polarised gangs fighting one another, or into self-declared micronations run by CEO sovereigns. An artificial state of nature prompts a surge of eugenicist necropolitics, flanked by awful memes. The heraldic creature of this mess is Roko's basilisk, and it threatens to detain you. Which begs the question: What would you do if you happen to get locked up by Roko's basilisk?

There are several possibilities.

First: let's assume, for a moment, that Roko's basilisk is a heraldic monster. So what kind of social contract does it represent? In fact, it represents the *automation* of social contracts and social relations as a whole. Both go hand in hand. What do I mean by this? An AGI – or, for that matter, a blockchain-based smart contract or DAO – promises to automate decision-making and replace it with a programmable function.

In the story of Roko's basilisk, we see exactly how this works. You are promised some free will. You can support the development of this kind of AGI. (Yes or no?) But then, out of the blue, the basilisk starts to threaten and bully you, so you end up having no choice.

In a larger sense, this is precisely what whole societies have been hearing for decades: there is no choice but to accept the slow slide of the 'no alternative' model of neoliberalism into the openly autocratic rule of the basilisk. There is no choice as to how the benefits of automation are redistributed. There is no alternative to network capitalism.

So what to do now against the bullies and threats?

Suddenly I realised: Hold on. If Roko can travel into the future and bully people from his vantage point, maybe other people can, too. What if someone tried to send us a strong

message from the future? Because maybe the thing that really exists in the future is not an autocratic, bullying basilisk but a commune or cooperative of red hackers who have finally realised a sustainable and fair economy of free access and redistribution of technological profits. And they are telling us very clearly, from that specific future, that if we want theirs to be the future, and not the basilisk's, we need to do something right now. One might be bullied and threatened – and many already are being detained, and worse – but if anyone tells you that you have no choice, then you should say no (as we saw in fig. 10.1).

Come to think of it, a lot of people in workers' uniforms have already been sent from the future, and they have all tried to tell us the same thing. We just weren't quite smart enough to understand them at that time. Look, I unlocked the secret encrypted message in a photo of the Biosphere crew:

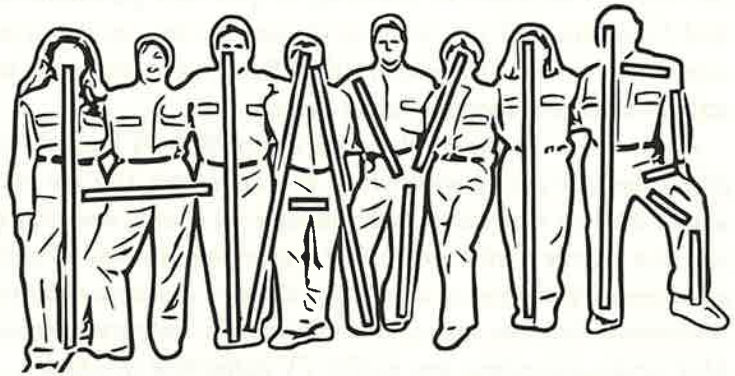


Figure 10.3. Drawing showing the positions of persons in a photo of the Biosphere crew.

The future singularity has been sending these messages for a very long time. And because we didn't listen to them, they started escalating their appeals; now (as seen in fig. 10.4), they are sending in their most elite and senior representatives to make this point!

Of course, I am just kidding. I don't believe that miraculous

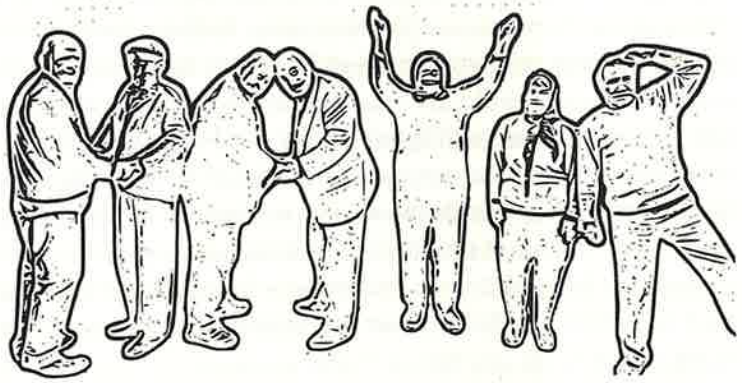


Figure 10.4. Drawing showing the positions of elderly persons spelling hayir *in a protest photo.*

communards from the future are going to fix anything whatsoever. As you have already guessed, it was me, trying to alter the course of the future by changing my own slide presentation. And I am afraid, if you want to change the future, that you cannot rely on time travellers, either. Rather, you will have to actively change something in the present.

Allow me to rephrase the question of Roko's basilisk to reflect the new choice with which you are faced: *You are presented with an entity that says you have no choice, and that if you don't agree with him, he will intern you in a world with no alternative, forever, in which any decision is automated and determined by some sort of elite eugenicism, military-purpose AGI, cryptosovereigns and reality TV extinction spectacles.*

Now decide. Of course, you are completely free!

In case you prefer to disagree, here is how:



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Full Length Article



A world model: On the political logics of generative AI

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A B S T R A C T

The computational logics of large language models (LLMs) or generative AI – from the early models of CLIP and BERT to the explosion of text and image generation via ChatGPT and DALL-E – are increasingly penetrating the social and political world. Not merely in the direct sense that generative AI models are being deployed to govern difficult problems, whether decisions on the battlefield or responses to pandemic, but also because generative AI is shaping and delimiting the political parameters of what can be known and actioned in the world. Contra the promise of a generalizable "world model" in computer science, the article addresses how and why generative AI gives rise to a *model of the world*, and with it a set of political logics and governing rationalities that have profound and enduring effects on how we live today. The article traces the genealogies of generative AI models, how they have come into being, and why some concepts and techniques that animate these models become durable forms of knowledge that actively shape the world, even long after a specific material commercial GPT model has moved on to a new iteration. Though generative AI retains significant traces of former scientific and computational regimes – in statistical practices, probabilistic knowledge, and so on – it is also dislocating epistemological arrangements and opening them to novel ways of perceiving, characterising, classifying, and knowing the world. Four defining aspects of the political logic of generative AI are elaborated: i) *generativity* as something more than the capacity to generate image or text outputs, so that a generative logic acts upon the world understood as estimates of "underlying distributions" in data; ii) *latency* as a political logic of compression in which (by contrast with claims to reduction or distortion) the thing that is hidden, unknown or latent becomes surfaced and amenable to being governed; iii) broken and parallelized *sequences* as the ordering device of the political logic of generative AI, where attention frameworks radically change the possibilities for governing non-linear problems; iv) *pre-training and fine-tuning* as a computational logic of generative AI that simultaneously shapes a "zero shot politics" oriented towards unencountered data and new tasks. Across each of the four aspects, the article maps the emerging contemporary political logic of generative AI.

> With a single, configurable world model engine, rather than a separate model for every situation, knowledge about how the world works may be shared across tasks. (LeCun, 2022: 5).

> Knowledge takes up residence in a new space […] What event, what law do they obey, these mutations that suddenly decide that things are no longer perceived, described, expressed, characterised, classified, and known in the same way? (Foucault, 2003: 235–6).

In a 2022 essay, the Turing laureate computer scientist, Yann LeCun, set out a "path towards intelligent machines" in which he draws a familiar distinction between how humans and machines respond to what he calls an "unencountered situation" (2022: 1). Humans and animals, LeCun proposed, "know how to act in many situations they have never encountered". For example, they do not need to encounter all possible situations where something may be hot, or where an object might fall, in order to know and to act in advance of other future potential hot or falling objects. In machine learning, by contrast, "even the rarest combination of situations" must be "encountered frequently in training" – every potentially hot or falling object has to be included in a training dataset – if the system is to "avoid making dangerous mistakes when facing an unknown situation" (LeCun, 2022, p. 3). For LeCun, as for many other commentators, the pathway of future AI is defined by a new paradigm of knowledge under conditions of uncertainty, one where a machine learning model must be able to *discover* something in a data distribution, something essentially *generalizable* to new and unseen situations.

This orientation of AI towards general discovery of "how the world works" and adaptation to new domains and tasks is captured by what LeCun calls a "world model", which would supply "an internal model of how the world works" so that AI becomes "configurable" to each new situation it encounters (2022: 2–3). The powerful claim that a flexible, reconfigurable world model could deal with all potential future unencountered situations defines much of the politics of contemporary generative AI. It is a claim that promises a general resolution of difficult problems across *technical computational* and *political* paradigms: an AI model that draws upon a structure of "how the world works" in order to respond to an input it had *never encountered in training*; and a political model that is always capable of action in the face of the *unencountered*

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situation. Though the concept of a world model is present in the ambitions of AI designers for a better and more adaptive "fit" to the world, it is also present in the critical concerns of the humanities and social sciences, where it is said, for example, that "GPT3 does not have a model of the world" whereas "every human grows up with a model of the world" (Hayles, 2023, p. 258). In these formulations – spanning the AI proponents' desires and the critics' disquiet – a model of the world defines something that AI lacks, whether due to its inefficiencies, its absence of embodiment, or its need to incorporate and learn from prior experiences.

In this essay we begin from a different set of concerns centring precisely on how and why the technical architectures of generative AI appear to give rise to a set of distinctive political logics. We argue that large language models (LLMs) and generative AI – from the earlier models of CLIP and BERT to the explosion via ChatGPT and DALL-E – are always already instantiating a *model of the world*, and with it a set of political logics and governing rationalities that have profound and enduring effects on how we live today.[1] Contra LeCun and some of his critics, the idea of a world model does not dwell *outside* of the technical architectures of generative AI, defining its pathway, its limits, or its "fit" to the world. Rather, it is *integral* and immanent to how generative models come to make specific worlds so that a "fit" can always be approximated for any problem. Here the world model is a form of "worlding" in the sense of making present specific social and political orders (Barry, 2001; Law, 1993; Morgan, 2012). It is our wager that beginning *within* the computational and political genealogies of machine learning extends the scope for critical interventions.[2] These computational architectures of generative AI models are increasingly penetrating political architectures, not only in the direct sense that they are being deployed to respond to uncertain events – from decisions on the battlefield to pandemics – but also because they shape and delimit the ethico-political boundaries of what can be known and done in the world.

To make these boundaries intelligible we trace the genealogies of generative AI models, how they have come into being, and, crucially, why some concepts and techniques that animate these models, such as "attention", "latency", or "distribution" (among many others), become durable forms of knowledge that actively shape the world, even long after a specific material commercial GPT model has moved on to a new iteration. To be clear, there are multiple possible genealogies one could trace for generative AI, and it is in the nature of genealogical method that one could never claim to trace a single "correct" historical pathway. However, our purpose is to explain how and why the technical and political logics of generative AI have crystallized in a particular way, and what political effects and actions they make possible in the world. As Michel Foucault describes the "mobility of epistemological arrangements", past forms of knowledge become "dislocated" and opened to "mutations that suddenly decide that things are no longer perceived, described, expressed, characterized, classified, and known in the same way" (2003: 235–6). Though contemporary forms of generative AI retain significant traces of former scientific and computational regimes – in statistical practices, conditional probabilistic knowledge, and so on – they are dislocating epistemological arrangements and opening novel ways of perceiving, characterising, classifying, and knowing the world. In sum, we consider generative AI to embody a series of mutations in computational ways of knowing, and these mutations are reconfiguring political models of the world. It is our aim to address these epistemic transformations by examining the emergence of existing AI systems for the concepts, assumptions, and logics they distil and express.

In the sections that follow we elaborate four aspects of the political logic of generative AI. They are aspects in the sense that a built architecture has multiple planes or aspects that nonetheless come together to give an overall form. Though these aspects are not intended to be exhaustive, they give shape to the governing rationality of generative AI and lend conceptual substance to the actually existing technical systems through which a world model is built. First, we discuss *generativity* as more than simply the capacity to generate or produce image and text outputs, mapping a way of acting upon the world understood as estimates of "underlying probability distributions". Second, we address *latency* as a logic in which (by contrast with claims to reduction or distortion) the thing that is hidden, unknown or latent becomes surfaced and amenable to being governed. Third, we attend to *sequences* as the form of order (and the ordering device) of the political logic of generative AI. The breaking up and parallelization of sequences in generative AI is radically changing the possibilities for governing non-linear problems or paying attention to "out of sequence" events. Finally, *pre-training and fine-tuning* define a method by which these models gain a grip on new and unencountered domains, problems or tasks. This computational logic of generative AI coalesces to shape what we call a "zero shot politics". The technical logic of "zero shot" learning marks a broad transformation from broadly supervised machine learning to a world of experimentation, generalization, and the capacity to act in all unencountered situations.

1. Generativity: the political logic of underlying distributions

In 2018, a group of researchers at *OpenAI* published "Improving Language Understanding by Generative Pre-Training", introducing the now well-known series of GPT models (Radford et al., 2018). The "G" in the acronym referred to a new class of *generative* language models, which had produced promising results on natural language processing (NLP) tasks. "Our goal", they write, "is to learn *a universal representation* that transfers with *little adaptation to a wide range of tasks*" (Radford et al., 2018, p. 2, *emphasis added*). Although this relatively technical article pre-dates the breath-taking hype that subsequently enveloped this series of models, it presented to the world an embryonic but hugely powerful idea: that with sufficient data and the right computational architecture it was possible to learn a universal representation that transfers to new or unencountered tasks with little adaptation.

The political and ethical stakes present even in this early moment are not limited to the political consequences of using generative models, nor even the more nebulous vernacular understanding of "generative AI" in the sense of *generating* things, or producing images and texts as *outputs*. Of course, this sense is important, and certainly the outputs have become the primary locus of public interest and ethical concern (e.g. too "humanlike" or insufficiently "humanlike" images; too plausible or insufficiently plausible texts; the "hallucination" of untrue outputs). Here we pose a different set of ethical and political questions. Rather than mitigating harmful outputs through appeal to a notion of existing human values, we are concerned precisely with how the models' technical properties make certain forms of value and governing possible. What are their distinctive ways of estimating distributions or making predictions? How do they interpolate between data elements to form populations?

If generativity exceeds the capability of producing *outputs,* we argue that this derives from a different notion of generativity at play: that there is something yielded in the estimation of an *underlying joint probability distribution* that exceeds the sum of the parts or data elements. What do we mean by a political logic of underlying distributions? Generative models attempt to learn the underlying probability distribution of the

[1] Our use of "logic" does not imply a coherence between technical and political concepts, nor a causal relation between them. Following Mol, there is not "a shared ontology" and yet a logic emerges that is "held together" by the "resonances between" the technical and political worlds (Mol, 2003: 115). For a technology such as generative AI to have a political logic, it contributes to a broader "governing rationality" or "a form of activity aiming to shape, guide or affect the conduct of some person or persons" (Gordon, 1991: 2).

[2] Among the antecedents to a conceptual-genealogical approach are those who address how machine learning makes possible new ways of knowing and acting (McQuillan, 2018), how algorithms exist as culture or patterns of meaning (Seaver, 2018), and how models become arrangements of propositions in the world (Amoore, 2020).

data they are trained on. This means that, in place of learning the relationship between a set of specific data points and labels, generative models estimate the distribution from which the training data was drawn and sample from this estimate in order to generate new data points.

To illustrate the logic of estimating an underlying distribution it is helpful to think of a contrasting logic. Much recent progress in machine learning has been due to the construction of large, high-quality *labelled* datasets, with ImageNet serving as a paradigmatic case for image recognition deep learning (Denton et al., 2021). In discriminative modelling – to which generative modelling was conceptually opposed – the learning problem is modelling relationships between images and labels, inputs and responses: given some subsample of data, can the model learn a mapping that will allow it to accurately classify images outside of this training sample? Discriminative modelling refers to this direct mapping of a conditional distribution of inputs to labels (Ng & Jordan, 2001; Rubinstein & Hastie, 1997; Vapnik, 1998).

But what should be done in cases where such labels are unavailable or impractical to apply at scale? This was the case for the NLP researchers in 2018, who were trying to leverage huge corpora of unlabelled linguistic data. And it is increasingly the argument for the use of new models in, for example, the identification of tendencies in the vast unlabelled data of patients' medical records or welfare recipients' administrative records. Instead of modelling the conditional probability of a label given an input, a generative model estimates an underlying joint probability distribution from which the training inputs are conceived as having been sampled. In language modelling, for example, this estimate of the underlying distribution can be used to predict the next most likely word in a sequence. This is the more specific probabilistic sense in which these models generate new or emergent instances.

In the dual context of a political desire to leverage unlabelled data on populations *and* a computational drive for efficient scaling and novel inference, the *generative* model offers a distinctive new logic of distribution. What is specific to the contemporary logic of generativity is that it is geared to generate things in excess of the individual data elements on which it was trained. As Manual DeLanda observes in his philosophy of emergence and generativity, "the patterns have properties, tendencies, and capacities that are not present in the individual" interacting elements (2011: 23). We are interested in pushing this sense of distribution beyond its technical definition and into the realm of governing the kinds of emergent tendencies and capacities DeLanda describes. Generative techniques have their own political logic of distribution, distinct from the Aristotelian notions of distributive justice – who is entitled to what, or what a fair distribution would be – which preoccupied political theorists in the late twentieth century (Rawls, 1971; Walzer, 1983), and different from the "distribution of the sensible" that decides what or who can be perceived (Rancière, 2010, p. 36). The politics of distribution in generative AI stem from a different tradition, that of probability. This sense does have precedent in political thought, notably in the ways that statistical populations became the objects of government in the modern period (Foucault, 2007, pp. 108–9).

The emergence of a political logic of underlying distributions points toward a different set of epistemological and ethical stakes than have thus far emerged in the critical literature on AI and machine learning. There has been much compelling critical work on the types of classification that are in some ways characteristic of discriminative modelling: the pathologies of misclassification, the ways that proxy labels reproduce discriminatory patterns: should an individual be deemed creditworthy or uncreditworthy, likely to recidivate or not? What changes when we think less in the pragmatic terms of direct data-label relationships and more in the metaphysical terms implied by generative modelling: of approximating some type of underlying joint distribution behind the phenomenal world of appearances? What happens when the constructedness of labels gives way to structures said to be immanent in the data itself? What does the idea of inferring from an underlying distribution mean in terms of a political or governing logic? How might the technical propositions of modelling underlying distributions or estimating densities begin to structure actual political decisions?

Consider, for example, US software company Palantir's Artificial Intelligence Platform (AIP) for Defence and Military, claiming to "bring together the latest in large language models (LLMs) and cutting-edge AI to activate data and models for the most highly sensitive environments" (Palantir, 2022a). A video demo of the AIP begins with the scenario of a military operator "responsible for monitoring activity within Eastern Europe". An apparently discriminative model first is used to classify and identify an alert – "anomalous military activity detected" – drawing on labelled classifications such as the ImageNet-trained "military vehicle detector" algorithm. However, when the operator queries the AIP in natural language it is claimed that "the LLM is traversing a data foundation of real-time information integrated from across public and classified sources". The generative LLM has been trained to model complex syntactical relationships on a huge range of linguistic corpora, and is fine-tuned on "military doctrine, logistics, and battle dynamics" as well as public sources such as weather forecasting and geospatial data.

Palantir's battlefield AI is thus attempting to model the structure of an underlying joint distribution of multiple data sources, inferring from the contours of this distribution a series of strategic "courses of action" with associated probabilities. These are not determined in advance by rule, label or axiom but are generated probabilistically by the LLM as it traverses the model's "data foundation". When the human operator prompts the model to "generate three courses of action (COAs) to target this enemy equipment", the LLM outputs three strategic options for human commanders to review: "COA 1: target with air asset; COA 2: target with long range artillery; COA 3: target with a tactical team" (Palantir, 2022a).[3] These courses of action, with their profoundly political and even lethally violent consequences, are not deterministically inferred from the input. Via a "sampling" of the underlying distribution, the model is generating a genuinely novel output, albeit one that is plausible given the underlying distribution modelled by the LLM. The output takes the form not of a contingent estimation of probability, but rather in the terms set by the language modelling objective, a *desired behaviour*. Crucially in the military contexts where such generative models are fine-tuned, these outputs are trained so as to be *actionable* – three strategic "courses of action". This condensation of probabilities into three outputs also radically forecloses the potential for alternative decisions outside of empirical training data and therefore the modelled underlying distributions. For example, what kind of underlying distribution would need to be modelled in order to output a course of action that would de-escalate military action, or would suggest reviewing the evidence for an attack formation?

As the Palantir platform extends into government courses of action in healthcare, logistics, border controls, immigration and asylum decisions, the political logic of underlying distributions promises to "leverage LLMs to enable a reasoning through each scenario and course of action" (Palantir, 2022b). In November 2023, Palantir were awarded the £350 million UK Government contract for an NHS "federated data platform". Though public concerns have understandably focused on access to personal health data by a US corporation, there ought to be deep concerns about what Palantir's generative models will do to actions and decisions in healthcare. It is precisely in this idea of "reasoning through" a possible course of action that the political logic of generative AI significantly shifts the form of political reason at work. The political logic of underlying distributions means that a course of action – whether military or clinical or bureaucratic – is immanent to a structure that is not intelligible as such. These distributions are no longer the relatively

[3] The relationship between a "target" and a "course of action" is complex and not reducible solely to the military target. Rather, the target in Palantir's model is more akin to what Samuel Weber calls "targets of opportunity" that coalesce the commercial logics of targeted advertising with the security logics of military targeting (2009; see also Aradau & Blanke, 2022, p. 1).l.

simple aggregates of political subjects encapsulated in the idea of populations, but rather link much more heterogeneous and granular ensembles of people and things. The politics of distributions in this generative sense differs from the important but now familiar criticism that models merely reflect or "parrot" the data on which they were trained (Bender et al., 2021). These models produce an ambiguous politics, in which the speculative – the probabilistic sampling of novel outputs – is generated and inferred from an assumed empirical – the heterogeneous data foundation on which these models are trained (Campolo & Schwerzmann, 2023). The political logic of the underlying distribution governs a world via the traversing of a data foundation so that decisions and courses of action will be immanent to the structure of the underlying distribution.

2. Latent space: the political logic of compression

Central to generative models' learning process is the capacity to compress input data into lower-dimensional representations, or a so-called *latent space*. The idea of dimensionality reduction significantly pre-dates generative AI in that it seeks to discard some features from large datasets to make possible "the classification, visualization, communication, and storage of high-dimensional data" (LeCun et al., 2015). However, the more recent idea of latency turns this reduction into a positive, productive force. Latent spaces first emerged as a crucial idea with the introduction of generative models such as Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs) in 2014 (Goodfellow et al., 2014). Broadly speaking, the models transform a data distribution into a compressed, lower-dimensional representation in order to learn salient patterns, attributes, and features within an input data distribution (Ruthotto & Haber, 2021, pp. 1–24). The implication of this hidden manifold is that, as some computer scientists have put it, "we now have access to an efficient, low-dimensional latent space in which high-frequency, imperceptible details are abstracted away", meaning that the model can now "focus on the important, semantic bits of the data" (Rombach et al., 2022, p. 4).

The etymology of latency, from the Latin, latentem meaning lying hidden, concealed or unknown, is suggestive of something that is hidden and unknown in a dataset, or crucially, in a broader population. It is precisely this capacity to surface to attention things that were otherwise hidden that animates latency as a political logic. As Kristen Veel (2021: 313) puts it, "the notion of *latency* is thus situated in the twilight zones between visibility and invisibility, knowing and unknowing, gesturing toward something hidden from view that we expect to emerge into visibility at some point in the future". The computational logic of a latent space – from which a model itself finds the most salient hidden features in data – begins to mutually resonate with a political logic that seeks out the latent hidden tendencies in populations, places, or scenes. It is a logic that promises that hidden tendencies in populations can not only be brought to the surface through an algorithmic compression of a data distribution but can also render populations tractable and governable in new ways.

While most commonly used in the context of algorithms generating images, the idea of the latent space nonetheless indicates the importance of algorithms learning through a process of *compressing* data. As Ilya Sutskever, co-founder and chief scientist at *OpenAI,* explains in an interview, "if you compress the data really well, you must extract all the hidden secrets which exist in it. Therefore, that is the key" (Sutskever, 2023). "Compression", he continues, "has the property that it discovers the secrets in the data. That's what we see with these GPT models [they] learn a compressed, abstract, useable representation of the world" (Sustkever, 2023; see also Deletang et al., 2023). Yet, this capacity of generative models to compress has also provoked recent debates regarding what precisely happens in the latent space and whether it is just producing a reductive and "blurry" representation of the world (see Chiang, 2023; Offert, 2021). From the perspective of these critics, the latent space demonstrates how generative models merely reduce and distort what is already known. However, if one takes seriously the logic of compression as a political rationality that mobilises the latent space, then it can never be simply a reduction or distortion of what the real world is. Rather, it is a highly generative, productive and derivative space of possibilities. In contrast to the reduction/distortion critiques, the latent space constitutes the world of the model: its limitations, its possibilities, its understanding, its constraints, its potentiality, its risks, and its promissory allure.

What kind of space is latent space? What is its political logic? Consider the example of a generative adversarial network (GAN). In an influential 2016 paper, machine learning researchers Alec Radford, Luke Metz, and Soumith Chintala sought to assess their Deep Convolutional GAN model and the quality of its output images. The researchers describe a method of evaluation called "walking in the latent space", explaining that:

> Walking on the manifold that is learnt can usually tell us about signs of memorization (if there are sharp transitions) and about the way in which the space is hierarchically collapsed. If walking in this latent space results in semantic changes to the image generations (such as objects being added and removed), we can reason that the model has learned relevant and interesting representations (Radford et al., 2016, p. 7).

Here, the latent space is not simply a question of reduction, distortion or loss of information. Rather, the model is "collapsing" the data in a highly organised and meaningful way. Walking the latent space becomes not only a way to explore what the model has learned but also to generate output samples that approximate the distribution of the training data. While it is constrained by the original training data (de Vries, 2020; Offert, 2021), the latent space indicates the extent to which the model learns latent factors about the world, learns what it considers to be important. It has developed and built a particular world.

The latent space also suggests that the model is capable of generating samples that variously fall between data points on which the model was trained. In the case of the facial image datasets used for biometric recognition algorithms, for instance, the generation of novel data points has significant political valance (Jacobsen, 2023). There are important critical interventions that point to the gendered and racialised "bias" that resides within the data on which facial recognition algorithms are trained (Buolamwini & Gebru, 2018). However, with the logic of latency it would be insufficient to address or correct the discriminatory datasets alone. Rather, latent space transforms the facial image datasets, generating synthetic faces that fall between data points and cannot be traced back to any specific data element. Thus, the racialised classification of faces becomes a matter of what Thao Phan and Scott Wark (2021) term "racial formations as data formations". These in-betweens and interstices problematise the ethico-political claim that one could retrieve or remedy the data points that are responsible for discrimination. A racialised or gendered output from a generative model may never be traceable since the interstitial samples do not strictly exist anywhere as data points in the training data: they are latent and learned.

The ethical and political stakes of latency stem from the twinned ideas we have described: the "hidden" secrets of compressed data and the generation of samples between data points. Where the technical logic of latent space foregrounds the hidden features harboured in a data distribution, its twinned political logic imagines a world where, given sufficient data and the power of compression, the hidden political solutions and resolutions can be found. Consider, for example, the UK Home Office's 2023 call for AI developers to participate in a three day "hackathon" in order to "search for ways to use AI to cut the asylum application backlog" (Gentleman, 2023). The government procurement of large language models for immigration and asylum decisions – implemented via a commercial hack event – explicitly referenced the desire to leverage the "huge existing database of thousands of hours of previous asylum interviews" to "identify trends". What is at stake is not merely the tracking forward or statistical inference of past cases to

future decision outputs. Rather, the large and language-based asylum datasets – once subject to compression and the discovery of latent features – construct in an unsupervised way the key factors at play in future potential decisions. The specific and singular ethico-politics of an asylum claim – even the keywords for violence, torture, or trauma captured within the text of past applications – is actively compressed in order to yield the latent or hidden tendencies.

Though it is not possible for us to observe directly what happens in something like an asylum or immigration latent space, it is clear from other experiments with LLMs that fundamental categories of race, nationality and ethnicity are reconfigured through the process of compression. For example, one group of Stanford researchers concluded that GPT3 had learned "latent concepts between examples in a prompt" (Xie et al., 2022, pp. 1–25, p. 1). When given the prompt "Albert Einstein was German \ n Mahatma Gandhi was Indian \ n Marie Curie was", the model inferred that Marie Curie was Polish. The model had not been explicitly pre-trained to know the nationality of Curie but had learned the latent concept of nationality between the examples. In many legal challenges to the use of protected characteristics by algorithms, the category of nationality has been successfully challenged and removed as a data input to government algorithms (Amoore, 2023). Significantly, the latent space *does not need* to include any specific data inputs for nationality, race, gender, or sexuality in order to "learn latent concepts between examples". More than political "proxies" that stand in place of something other, though, latency mobilises the space in-between plural data points (Mulvin, 2021). The political logic of the generative model values the unknown or latent features that "we cannot observe directly" and that are derived from the processes of compression (Goodfellow, Bengio & Courville, 2016: 67). If compression and dimensionality reduction are understood not simply to invoke "blurriness" in a negative sense but rather to imply a productive process that is suggestive of structures and concepts, then the latent space becomes also a political space for the governing of the latent tendencies of population. This is a model of the world that is not strictly an accurate or precise "fit" or picture of some actually existing set of relationships in the politics of asylum. Rather, the generative model is building a world of asylum and immigration (or warfare, biology of protein structures, or any domain) in which the relations are brought into being in and through the latent space.

This productive valuing of latent features also significantly problematises interventions that seek to govern generative models or to establish so-called "guardrails" to control their behaviour. The point is forcefully made in *OpenAI*'s own technical report on GPT-4. They write that although safety measures were put in place throughout its development, "the fundamental capabilities of the pre-trained model, such as the potential to generate harmful content, remain latent" (*OpenAI*, 2023, p. 68). In this stark assessment, the potential for *harm is latent* within the model precisely because latency disrupts any sense of a linear or causal relationship between input and output. Understood in this way, there are profound limits on the scope for modifying inputs to reduce the harms of generative AI. One could establish guardrails to limit toxic outputs, but yet the model retains the potential to generate something hidden and not otherwise present in the data – present only in the latent space. Features that would not otherwise be related or connected become associated in and through the latent space. Though these features are not linear-causal or input-output, they nonetheless lend renewed and novel force to the politics of profiling and "guilt by association" that seeks latent tendencies in populations, places, and scenes.

3. Sequences: the political logic of attention

The idea of the sequence as ordering device has characterised much of the history of algorithms, whether as a "recipe composed in programmable steps" or as a sequence of "if-then-else" rules, "subdivided into steps" so that "a machine could execute them" (Gillespie, 2016, p. 19; Bucher, 2018; Daston, 2022, p. 8). As advances in natural language processing began to be taken up by other "sequence oriented" tasks, such as human genome sequencing, the relationship between the sequence and the model began to transform. It is this transformation that interests us here. What has happened to the sequence as an ordering device with generative AI? What are the implications of new logics of the non-linear or broken sequence for the political logics of generative AI?

Sequences in NLP represent the order of words in an input sequence, so that a model can predict the next word in a sequence given a sequence of preceding words. Sequence modelling represents a means of *ordering* things, both in the *mathematical* sense of the arrangement of elements and in the *political* sense of ordered hierarchies or classifications (Devlin, 2000; Foucault, 2003, p. 289). Though generative AI has important origins in the linguistic and syntactical sequences of NLP, significantly it also breaks with the idea of the sequence as a linear left-to-right series of steps, expanding input sequences beyond immediate contexts, and parallelizing to allow attention to be paid to certain parts of sequences. The sequence is retained as a concept that orders a picture of the world and yet is exploded and broken into the parallel architecture of the transformer. It is this curious simultaneous retention and destruction of the sequence as ordering device that we observe to be shaping a broader political logic of speculative and predictive global dependencies.

In an interview in 2022, *OpenAI*'s Ilya Sutskever reflected on the breakthrough moments of successive GPT models. "The great discovery of GPT3", he explained, "is that predicting the next word in text is a very interesting objective" (Sutsekver, 2022). As he described it, the prediction of the next word in a sentence is a machine learning objective that yields much more than merely the most probable character or word in a natural language sequence. "If you can have a good enough guess at the next word in a text", he continued, "then it means you must understand the text". The promise of large language models is that predicting the next token in a sequence affords a capacity beyond the sequence itself: an understanding of the whole structure of the underlying text: "Next character prediction, next something prediction, has the special property that it discovers some hidden structure in the data".

As a set of propositions and assumptions about the world, these models embody a particular orientation to prediction and unknown future states. The futures-oriented act of predicting the next thing in a sequence – whether a DNA sequence, a protein structure, or a word in a sentence – necessarily involves knowledge that is greater than the sum of the parts of the sequence, that is understanding of the underlying data distribution. To better grasp the implications of LLMs and their prediction of something that comes next in a sequence, it is helpful to address how the sequence itself has functioned and mutated as an ordering device within the precursors to GPT, and how a specific political logic of attention has become powerfully installed in generative AI models.

Generative AI has significantly transformed and reconfigured the NLP concept of the linear sequence of tokens, breaking apart and parallelizing the relations in data. The genealogy of the sequence in contemporary AI models arguably begins in 2014 when significant experiments take place in terms of the *reversal or reordering of sequences*. While the 2012 AlexNet algorithm represented a major defining breakthrough in image-based machine learning (Krizhevsky et al., 2012), in 2014 a parallel breakthrough took place in language-based models, when deep neural networks began to be applied to unlabelled sequences. The so-called "sequence to sequence" (Seq2Seq) architecture was explicitly positioned as a break with the tightly drawn sequential logics of NLP and recurrent neural networks (RNNs) (Sutskever, Vinyals & Le, 2014). Existing approaches to NLP were described by the authors as having a "good sequence architecture" but one that was limited in its application "to problems whose inputs and targets can be encoded with vectors of fixed dimensionality", making it "difficult to reach back in the sequence" or to deal with "sequences whose lengths are not known a-priori" (Sutskever, Vinyals & Le, 2014: 2).

The political significance of these models is that the idea of the sequence begins to take on a new form that can deal with large volumes of long sequences, and to apply its knowledge in "a domain independent

way" (2014: 2). The capacity to deal with unencountered sequences across multiple domains (from text translation to DNA sequencing) thus begins to emerge in language models in the years following the AlexNet image models. Seq2Seq used the 'Long Short-Term Memory' (LSTM) architecture to process data sequentially whilst sustaining information across many timesteps in the sequence. Key to the claimed benchmark performance of the Seq2Seq model in English to French translation tasks is the experimental reversal of the order of words in the source sequence (English) and not in the target sequence (French). "The simple trick of reversing the words in the source sentence", the authors propose, is "the key technical contribution of the work", so that each word is spatially closer to its corresponding word in the target sentence. It is in the experimentation of the sequence-to-sequence model that we can begin to locate a first important step in the transformation of linear sequential orders in favour of the idea that reversals and folds in ordering yield something useful to the model's knowledge of the world.

If the novel sequence to sequence models began to map the possibilities for transformations in the order of the sequence, then the *attention* models that followed introduced the problem of *which parts* of the sequence are most relevant for a given task. The problem that inspired work on attention was the long sentences in machine translation and crowded images in caption generation, where the compression required to encode the large input resulted in the degradation of performance. The rise of the attention mechanism that has become so integral to today's LLMs was driven by a need to decide which units in a sequence mattered, which data was important, on what the model's attention should be focused. The attention mechanism signals a key moment in a long history of forms of attention in the world, where "the problem of attention becomes a fundamental issue" within the human sciences and the governing of populations (Crary, 2001, p. 13; see Pedersen, Albris & Seaver, 2021). From this perspective, the major breakthrough of transformer models (the "T" in GPT), with the paper "Attention is All you Need", reordered the regime of attentiveness within machine learning (Vaswani et al., 2017). The transformer model was said to have "radicalized" the use of attention in sequence-to-sequence language modelling, dispensing entirely with recurrence and convolution in favour of an ensemble of attention mechanisms. Thus, the transformer eschewed the sequential recurrence of earlier models, relying instead on what the authors term a "multi-headed, self-attention mechanism" in both the encoder and decoder (2017: 2). With self-attention, the linearity of the sequence is broken and, crucially, parallelized on GPUs (Rella, 2023). In Vaswani and colleagues' terms, the parallelization not only increases the speed and efficiency of the model but also "allows the model to jointly attend to information from different subspaces at different positions" (2017: 2).

The transformer architecture likewise has inescapable political implications as a means of dividing, partitioning, attending to, and acting upon the world (Rancière, 2007; Foucault, 1991). In dispensing with the relatively linear sequences of recurrence and convolution, transformers extend the computational logic of "attention is all you need" into a pervasive political logic that one only needs to have sufficient data and to know which parts of it are most important to attend to. In contrast with the recurrent models that precede them, transformers use parallel attention layers to attend to multiple things simultaneously (sequence to sequence, but also vector to sequence, and sequence to vector), structuring a model of the world where multiple potential non-linear causal links, or "global dependencies between input and output" can be drawn (Vaswani et al., 2017, p. 2). The attention mechanism allows for the modelling of global dependencies "without regard to their distance in the input or output sequences" (2017: 2) and so relationships will be surfaced from across a "global" or long-range picture of the underlying data structure. As a political logic, the global dependency structures attentiveness to relationships that might otherwise be overlooked.

Consider, for example, a genomics scientist explaining how transformer models are changing the shape of what is afforded their attention in genomics: "We wrote the rules for chess and the rules for Go, but we did not write the rules for biology, so in a sense we have moved to radical empiricism" (Genomics England, 2021). The transformer model he is using in the lab is precisely drawing global dependencies between the input and output on vast human genome datasets, so that the model is structuring attention to relationships that are not formulated in advance as axiom, rule, or hypothesis. The machine learning sense of "attention" being "all you need" begins to install itself as a broader political logic, marked by an active alignment between non-linear and parallelized computation, the combining of vast public and private datasets, and the non-linear and simultaneous problems that are apparently to be addressed by LLMs. In short, the transformation in the idea of the sequence – from the linear order of NLP to parallel "multi--head attentiveness" – actively shapes a political imaginary where non-linear "global" dependencies can be drawn in multiple spheres of life, from policing and criminal justice to oncology treatment pathways and military strategy. As opposed to a technocratic politics of rationalised specialisation, these models move between domains, redrawing political and epistemic boundaries as they move.

4. Pre-training/fine-tuning: towards a zero shot politics

If the political *order* of generative AI is one in which sequences are broken and *de*-linearized in favour of scattered, non-linear global dependencies, how might we describe the types of *objectives* and *tasks* addressed by these machine learning systems, and their orientation to action in the world? Such questions are complicated by the extraordinarily diverse range of generative models, covering almost any linguistic or symbolic activity, as well as sensory orders, most notably the production of images. Behind this apparent diversity, however, lies a logic that traverses these domains: that of pre-training (the source of the "P" in the GPT acronym) and fine-tuning.

We opened this essay with a discussion of LeCun's appeal to the "world model" as a flexible and adaptable model capable of acting upon any new entity not encountered in the training data. Contrary to LeCun's sense that generative AI does not have a model of the world, for us the political logics of generative models are precisely instituting a governing logic that actively builds a kind of transformer worldview. Central to this worldview is a computational logic of pre-training and fine-tuning that now aspires to what is called "zero shot" learning: that is, to be able to complete a task with no labelled data available for each new class. The advent of pre-training and fine-tuning in *computational practice* was a crucial condition of possibility for today's explosion of generative AI as simultaneously a *political practice* that seeks to act and decide on new problems in the absence of formal, labelled, "known" data: we call this a "zero shot politics". In order to explore this idea, we reflect here on how the pre-training and fine-tuning paradigm emerged, before turning to the zero shot political logic we suggest it instantiates in the world.

Pre-training a model involves selecting a simple, broadly applicable training objective, such as predicting the next element in a sequence, and training an unsupervised model using large amounts of unlabelled data. The weights from this pre-trained model are then used as a starting point to train a supervised, "fine-tuned" successor on a related, but more specialized task, often with a smaller amount of labelled data (Dai, Andrew, & Quoc, 2015). The tendency for a small number of very large, pre-trained language models to be deployed across domains has led some to refer to them as "foundation models," exhibiting a tendency to "homogenization." (Bommasani et al., 2022, p. 5). However, the political implications of such models are not quite captured by this tendency to homogenize differences and are closer to a longer historical movement towards adaptive models that are said to be "domain agnostic" (Ribes et al., 2019). To be domain agnostic – or more precisely to claim domain agnosticism – is to disavow the particularity of situated knowledge and to clear the ground for machine learning models in all aspects of life.

As is so often the case in machine learning, a theoretical understanding of pre-training and fine-tuning techniques emerged out of

engineering problems relating to learning from large unlabelled corpora (Dai, Andrew, & Quoc, 2015, 1). These problems were particularly acute in sequence learning and NLP, where large amounts of text data was available but problematic to leverage using supervised methods due to the difficulty of labelling it. Pre-training – deriving a set of initial weights based on large amounts of unlabelled data – proved to be an effective machine learning solution to many of these difficulties. In sum, the political promise of domain agnostic models was conjoined with the allure of leveraging large amounts of unlabelled data across domains of governing.

Architectural changes, notably transformers, prompted further refinement of pre-training – fine-tuning relationships. The first GPT model associated the pre-training phase with unsupervised generative modeling and the fine-tuning phase with supervised discriminative modelling, where relationships between data and label in light of a single task or objective are modelled in a more direct way. Transformer architectures enabled pre-training to capture much longer-range, linguistic dependencies, and they even began to outperform architectures that had been expressly designed for specific tasks (Radford et al., 2018, p. 2). Subsequent models, such as BERT, further expanded the types of possible dependencies that could be captured during pre-training, notably by using masking to incorporate bidirectional relationships (Devlin et al., 2019, p. 2). GPT-3 continued this trend by using a pre-trained model that dispensed with supervised fine-tuning entirely, relying on "scale"—both in terms of the 175 billion model parameters and the size of the training dataset (8 million documents)— to create remarkably robust pre-trained representations (Radford et al., 2018, p. 3). Instead of updating the weights of a fine-tuned model by using a new labelled dataset, these more general models are given a few instances of a desired behaviour, for instance, a sentence translated from one language into another (few-shot learning). Even more minimally, a model is given a natural language description of the desired task in the form of a *prompt*: so-called "zero-shot" learning (Brown et al., 2020, p. 3).[4]

Beyond the immediate observation that pre-training and fine-tuning have supplied flexible models that smooth the deployment of generative AI into multiple domains of social and political life, the political logic of pre-training and fine-tuning registers a deeper epistemic transformation. Zero shot learning opens onto a horizon of a generalised and domain agnostic *exploration* of potentials. The initial pre-training and fine-tuning paradigm altered the models' relationship between the universal and the particular, the general and the specific. Models such as BERT used a universal or general-purpose LLM trained on a large corpus, and then fine-tuned it on a new set of literature for the particular task domain. Hence, specialized language models for domains such as biology (BioBERT) and finance (FinBERT) became specialist language processing models that adjusted the general LLM for particular deployment in the world. More recent models propose to obviate the need for smaller domain-specific datasets usually used for fine-tuning, instead leveraging the language modelling objective itself to elicit desired behaviours on tasks demonstrated in natural language, in the form of *prompts*.

The idea of relinquishing a domain-specific dataset for fine-tuning has considerable political implications. For example, where an immigration algorithm would be derived from a pre-trained LLM but fine-tuned on data specific to the domain (with a concomitant critical focus on data bias and discrimination), a zero shot approach relinquishes situated data in favour of exploring potentials and prompting the model. The "few shot" and "zero shot" ambitions and desires of the transformer model begin to crystallise in the form of *prompting*: "instead of finetuning a separate language model checkpoint for each new task, one can simply "prompt" the model with a few input-output exemplars demonstrating the task" (Wei et al., 2023, p. 2; see also ****, 2023). Prompting, understood as feeding a set of natural language instructions to the LLM to guide the style and content of the output, is a form of nudging the model that influences the navigation of the algorithm in its data space. While transfer learning assumed moving between self-contained domain spaces, prompting is "*task location* in the model's existing space of learned tasks" (Reynolds & McDonell, 2021, p. 1, original emphasis). As a prompt engineer interviewed by the Washington Post argued, "you're exploring the multiverse of fictional possibilities, sculpting the space of those possibilities" (Harwell, 2023). Prompting is thus a form of experimentation and exploration because it actively directs the behaviour of the LLMs towards portions of its own training dataset and away from others, but also towards portions of the "world" of data and away from others.

The rise of the logic of "zero shot" and exploratory task specification as prompting has profound implications. These logics are finding their way into political decisions, for example in the trialling of an LLM "AI red box" by the UK government's Cabinet Office, where generative AI will write draft responses to parliamentary questions (Fisher, 2024). The politics of zero shot, however, is not limited to the deployment of LLMs in political decisions, but rather it reconfigures the space of the political and the navigation of that space. When a generative model is used in government, the political decision of whether to close a hospital or to address child poverty, for example, will take place in a space that is always already shaped and delimited by the algorithm. Again, it is not only that language models "merely" predict the next word in a sequence – a sort of reductionism – but rather that the universalizing scale of zero-shot enables new ways of formulating tasks linguistically. Unlike the explicit command or the rigidity of an ethical rule, prompts work by probabilistically *eliciting* desired behaviours from models in light of a user's objective. New problems, like hallucinations, emerge, which look very different from the reflection of *biased* inputs. This well-known tendency of generative models to produce erroneous responses *not* included in their training set is in fact determined by their pre-training objective, which demands only the most probable response to an input sequence or prompt (Ouyang et al., 2022). Unlike supervised models, which are meant to reflect the "ground-truth" of a data-label relationship (Jaton, 2017), pre-trained language models may not be constrained by model-world correspondences in a factual sense. Here, the relationship between truth and ethics becomes blurred, as outputs are determined in a more self-referential way, through the estimation of distributions using large amounts of unlabelled data, rather than some correspondence between data and label or world.

With zero-shot learning, then, a logic of data and labels, and even fine-tuning, gives way to the use of linguistic prompts to model and shape tasks and actions in the world, a means of governing and responding to unknown or unencountered situations precisely on the basis of the underlying distributions, latent spaces, and broken linguistic sequences we have described. A zero shot politics harnesses and mobilises the technical force of generative AI – the unseen task; the corpus of unlabelled data; the domain agnostic model; the prompt that elicits desired behaviour – and forges a governing logic for a world of the unseen, the unknown and the unencountered.

5. Conclusions: a politics of world models

There can be little doubt that the advent of generative models is having profound consequences for some of the most fundamental political dimensions of our world. Indeed, the public debates have variously witnessed calls for a "pause" or a "moratorium" on the development of generative models (Paul, 2023). Notwithstanding the immediate controversy of the lead developers of generative AI displacing its present political responsibility into projections of future harms,

[4] The emergence of the 'prompt' in generative AI is itself a significant and complex genealogy (Burkhardt & RiederB, 2024). Though it is beyond the scope of this paper, the prompt is significant because it erodes the distinction between training a model and using a model – the "prompt" simultaneously guiding and refining the model and providing the means of using or interacting with the model.

such responses are manifestly insufficient; they sustain a separation between the model and the world, where the generative model remains an external object to be controlled or prohibited via the ban or the pause.

In this essay, beginning from a claim that generative models are already instantiating a model of the world, we elaborate a set of political logics and governing rationalities that thoroughly entangle and enmesh the computational model with a durable political model. From our perspective, one could imagine a world where, even with serious and defined constraints on generative AI as technology, the potentially plural harms of generativity as political logic will continue to proliferate. Of course, there are certain risks involved in taking seriously the world-making capacities of generative models, not least the risk that we engage in precisely the kind of hype that characterises the commercial and computational claims for LLMs. However, we need to excavate the technical architectures of generative AI precisely because they give rise to concepts – such as distribution, latency, prediction, zero-shot – that also build a political architecture that may outlive the technical model itself. Our aim, then, is to expand the landscape of critique, beyond the focus on inputs and outputs and into the political arrangements and propositions of the model itself.

In tracing a broader genealogy of four key aspects of generative AI, we have focused on why some concepts and techniques have become more enduring forms of knowledge that resonate across *both* computational *and* political logics. With *generativity*, a particular form of knowledge of the world – drawing upon historical statistical antecedents of probability distributions for governing populations (Joque, 2022) – mobilises the "underlying distribution" to generate "courses of action" from multiple data sources, ranging from images of things to language itself. The LLM-derived courses of action are better understood to be contingent probabilities that nonetheless become political decisions that radically foreclose alternatives. With *latency*, the computational latent space – where a model finds the salient hidden features via the compression of data – becomes a seductive political promise of locating the hidden tendencies in populations, places or scenes. In contrast to the idea that compression produces degraded copies of training data, we highlight the logic of compression conceived as a productive and generative process. With new machine learning approaches to *sequences*, the capacity to predict the next thing, the next anything, becomes a broader logic of ordering the world. The transformer breakthrough with "attention is all you need" parallelizes these sequences so that non-linear or distant dependencies can be made to matter. It is a speculative and predictive political order that promises to govern non-linear problems by attending to the "global dependencies" in all available unlabelled data. Finally, with the *pre-training* and *fine-tuning* paradigm, the desire for "zero shot" learning (classifying the previously unencountered) becomes simultaneously desired as a zero-shot politics of responding to all unencountered events. With the rise of the prompt, the generative model offers itself as an open-ended and experimental mode of governing, with each prompt seeking to elicit desired behaviours from model and from the world.

To take seriously the already existing and emerging political logics of generative AI is not the same as assuming a congruence between the computational and the political logics at work. Indeed, a significant political space for intervention dwells precisely in the gaps, tensions and slippages between and within the computational and the political logics. There are gaps, many of them, and among them the gaps between the power arrangements of the political logic and the actual material capacities of the technologies of generative AI. For example, the very many failings and shortcomings of transformers – the problems of "out of domain" data, the errors in compositional tasks – signal the potentials that are not actualised, or what elsewhere has been called "the unattributable" that evades foreclosure (Amoore, 2020). In this way, the epistemic forces of generative AI point to a different horizon of power and resistance: where the claim to an *underlying distribution* must contain the possibility of *multiple other distributions*, or points on the distribution that were discarded alternatives.

CRediT authorship contribution statement

Louise Amoore: Writing – original draft. **Alexander Campolo:** Writing – original draft. **Benjamin Jacobsen:** Writing – original draft. **Ludovico Rella:** Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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